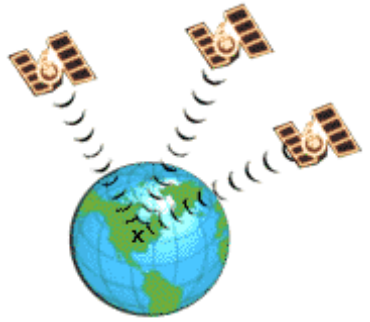
Chapter 14 – Inertial Navigation Systems

14.1 Inertial Measurement Unit
14.2 Attitude Estimation
14.3 Direct Filters for Aided INS
14.4 Indirect Filters for Aided INS



LN-251 IMU with fiber optic gyro (FOG)
https://www.northropgrumman.com/



GNSS

An Inertial Navigation System (INS) consists of the following:

- **Inertial Measurement Unit (IMU)** containing a cluster of sensors. The sensors are rigidly mounted to a common base to maintain the same relative orientations.
- **Navigation Computer** to calculate the gravitational acceleration (not measured by accelerometers) and integrate the acceleration and angular rate measurements to maintain an estimate of the position, velocity and attitude of the craft.

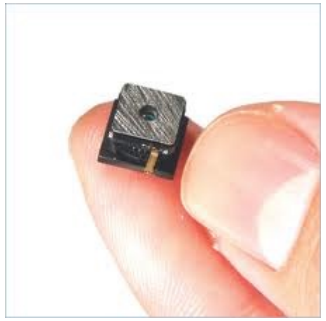
There are two primary types of INS:

- **Gimbal-mounted systems**
- **Strapdown systems**

MEMS-based IMU



Sensonor STIM300 IMU
https://www.sensonor.com



Chapter Goals

Inertial measurement unit (IMU):

- Understand the primary functionality of a MEMS-based IMU. Explain specific force, gyro and ARS.
- Distinguish between 6, 9 and 10 DOFs IMUs
- Be able to write down the IMU measurement equations including the model for white-noise driven biases (drift) terms.
- Be able to transform IMU measurements to different BODY origins using lever arms

Strapdown inertial navigation system (INS):

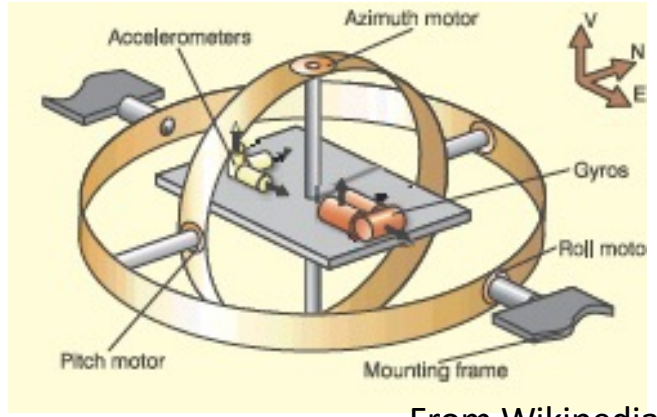
- Understand the principles for strapdown and gimballed-mounted INS.
- Understand what we mean with aided inertial navigation and why it is needed.
- Understand the static relationship between specific force and roll-pitch angles.
- Understand how an AHRS works, from a mathematical point of view.
- Be able to design attitude estimators using EKF and nonlinear observers (reference vectors).
- Be able to design direct extended Kalman filters (EKFs) for strapdown INS using Euler angles.
- Be able to design indirect (error-state) Kalman filters (ESKFs) for strapdown INS.
  Both the Euler angle and the unit quaternion representation (MEKF).
- Understand the differences of bias models in direct and indirect Kalman filters.
- Understand the problems/limitations of ARS and specific force bias estimation, and how observability is related to vehicle maneuverability.

Chapter 14 – Inertial Navigation Systems

- **Gimbal-mounted systems:** An IMU platform (isolated from the craft's rotations through a set of gimbals) which maintains a fixed orientation in space. Three gimbals would technically suffice, but most systems use four in order to avoid what is called *gimbal-lock*, which happens when the axis of two of the gimbals are driven in the same direction, disabling their isolation capabilities. Mechanized systems use angular rate sensors, feedback control and an actuated platform to keep the IMU in a fixed orientation in space.

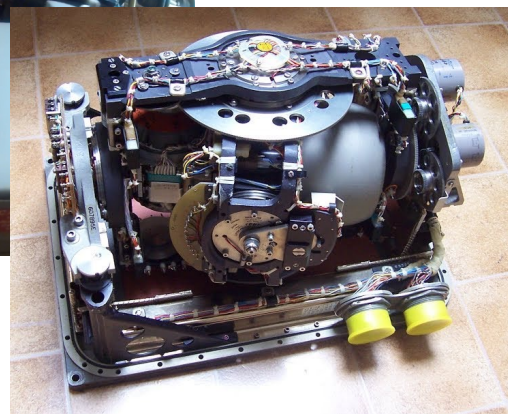
  Gimbal-mounted systems are expensive compared to strapdown INS. However, they are more accurate.

- **Strapdown systems:** An IMU which is strapped to the craft. Hence, the IMU will pick up the motions of the craft. This implies that the strapdown navigation equations must be integrated online in a state estimator to accurately describe the motions of the IMU and the marine craft in order to separate these.



From Wikipedia





Litton LN3 on an F-104 Starfighter
https://www.youtube.com/watch?v=U2JvtNbGVZc

We only consider strapdown INS in the textbook

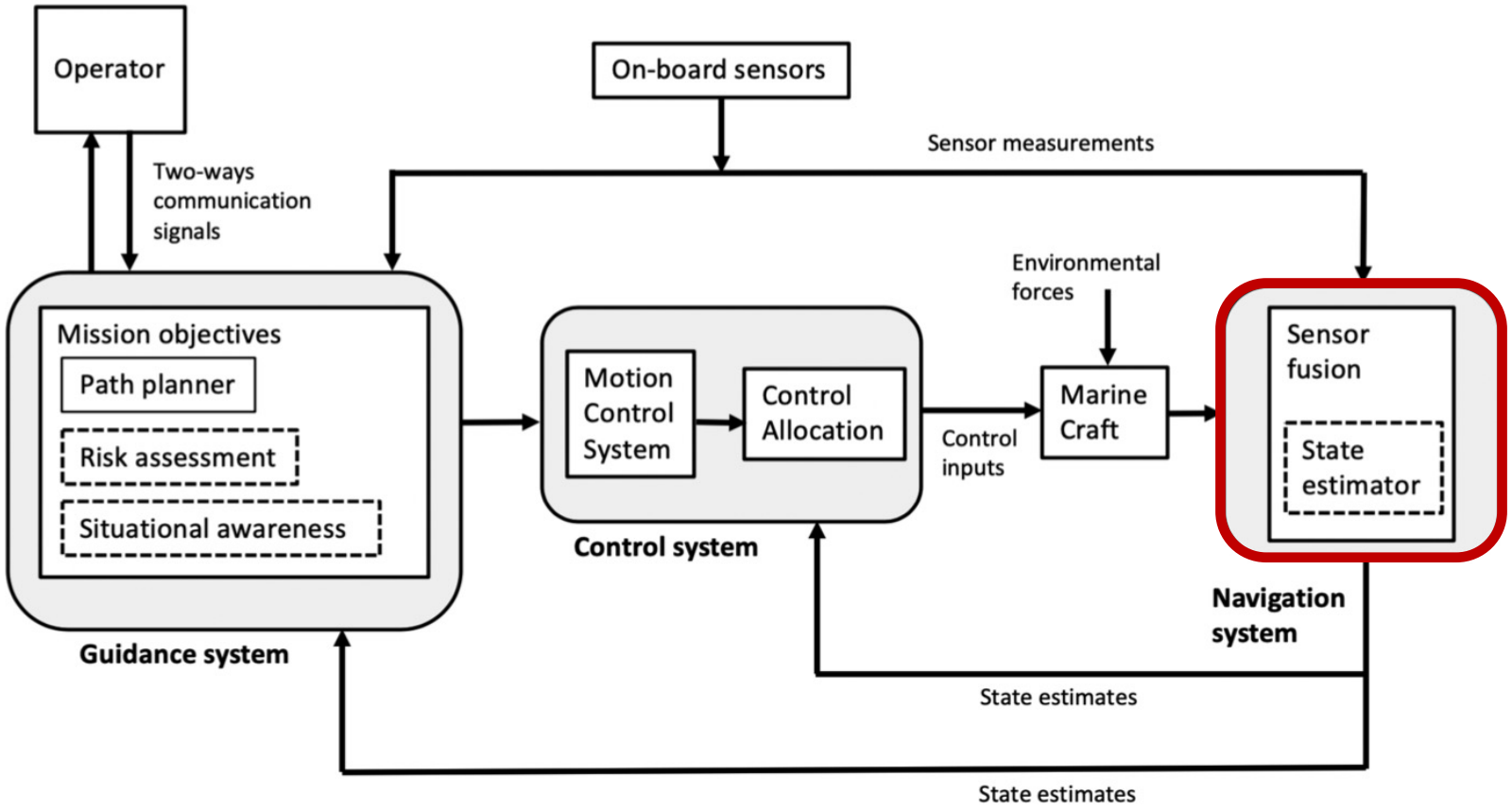


Xsense strapdown INS
https://www.xsense.com



SBG strapdown INS
https://www.sbg-systems.com/products/ellipse-series

Chapter 14 – Inertial Navigation Systems



The sensor and navigation system is usually implemented as an optimal state estimator (Kalman filter) using position measurements combined with motion sensors such as accelerometers and attitude rate sensors (ARS).

Navigation is the science of directing a craft by determining its position/attitude, course and distance traveled. In some cases, velocity and acceleration are determined as well. Navigation is derived from the Latin navis, "ship", and agere, "to drive".
It originally denoted the art of ship driving, including steering and setting the sails.

Guidance, Navigation and Control (GNC)

GNC is a branch of engineering dealing with the design of systems to control the movement of vehicles, especially, automobiles, drones, marine craft, aircraft, and spacecraft.

- **Guidance** focuses on the methods and systems used to follow a route or path to a destination once it has been identified. It's about steering, direction, and instruction on moving from one point to another.
  - How do I get there? is fundamentally a question of guidance.

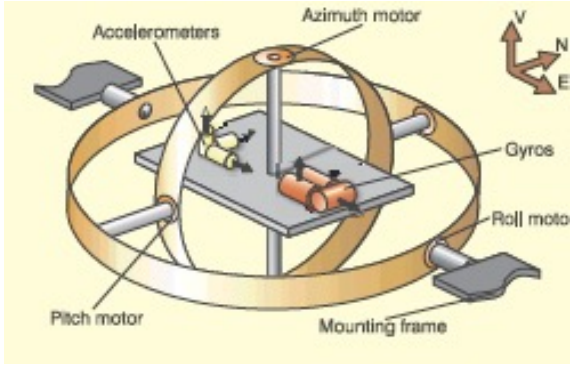
Once you know your current location and your destination (navigation), guidance provides the step-by-step instructions or actions required to reach the destination.

- **Navigation** refers to the process of determining one's current position and planning a route to a desired destination.
  - Where am I? identifies your current location. This is the primary navigation question.
  - Where do I want to go? decides your destination
  - Which way should I go? plans the route to get there, considering various factors such as distance, time, obstacles, and preferences

- **Control** refers to the manipulation of the forces, by way of steering controls, thrusters, etc., needed to execute guidance commands whilst maintaining vehicle stability.

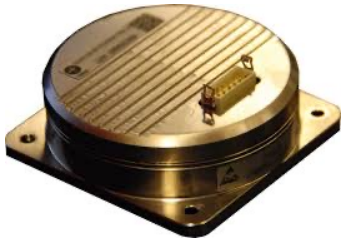
Chapter 14 – Inertial Navigation Systems

Today gimbaled systems are replaced by strapdown INS



From Wikipedia

- **Before 1970:** Early gimbaled INS (e.g., in aircraft and spacecraft) used mechanical gyros mounted on stabilized platforms to maintain a fixed reference orientation.
- **1971 – Bortz formulation:** J. Bortz introduces the *"New Mathematical Formulation for Strapdown Inertial Navigation"*, defining a rigorous method for integrating angular rates directly in body coordinates — a foundation for modern strapdown algorithms and later for the Error-State Kalman Filter (ESKF).
- **1970s – Algorithmic foundations:** Savage, Schmidt, and others develop strapdown mechanization equations and error-state models for linearization and filter design. The concept of *coning and sculling corrections* is formalized.
- **1980s – Practical realization:** The emergence of ring-laser gyros (RLGs) and digital computers (2 kHz update rates) enables fully strapdown INS to replace gimbaled systems in aircraft and ships.
- **Mid-1980s –** Fiber Optic Gyros (FOGs): Development of FOG technology provides a solid-state alternative to RLGs with no moving parts and long-term bias stability.
- **1990s – Miniaturization and MEMS:** Low-cost MEMS rate gyros and accelerometers become available, operating at 1–2 kHz and reducing coning/sculling errors. Strapdown INS becomes feasible for automotive, UAV, and consumer devices.



- **2000–2005 – Quaternion filtering era**: F. L. Markley (2002) formalizes the Multiplicative Extended Kalman Filter (MEKF) for quaternion-based attitude estimation, avoiding normalization issues in traditional EKFs.
- **2008 – Nonlinear observers:** Mahony, Hamel, and Pflimlin (2008) propose a nonlinear complementary attitude observer with proven stability — a simpler alternative to EKF/MEKF.
- **2013 – Stability and bias estimation:** Grip, Fossen, and Johansen (2013) prove USGES (Uniformly Semi-Globally Exponentially Stable) convergence for the Mahony observer and extend it with gyro bias estimation for time-varying reference vectors.



- **2017 – Nonlinear observer–Kalman fusion:** Johansen and Fossen (2017) introduce the eXogenous Kalman Filter (XKF) framework, blending nonlinear observers and Kalman filtering for robust INS and GNSS/IMU integration.
- **Today – Unified frameworks:** Strapdown INS employs tightly coupled sensor fusion, combining IMU, GNSS, magnetometer, and vision data within frameworks such as error-state, invariant, and nonlinear complementary filters, often implemented on embedded GPUs or FPGAs.

Navigation-Grade Inertial Measurement Technology

Today inertial measurement technology is available for commercial users thanks to a significant reduction in price the last decade. Because of this, low-cost inertial sensors can be integrated with satellite navigation system using a conventional Kalman filter or a nonlinear state observer.



LN-200 (*Courtesy Litton)*





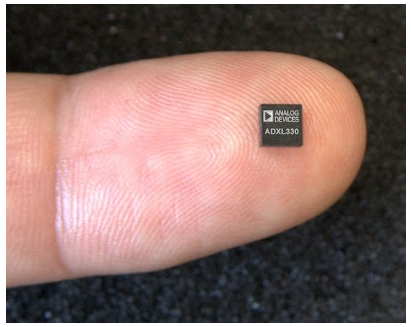
ISA (Inertial Sensor Assembly) - cluster of three gyros and three accelerometers that measure angular velocity and specific force, respectively.

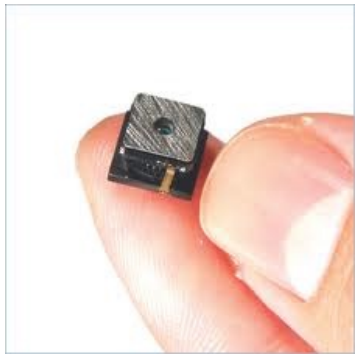
IMU (Inertial Measurement Unit) - consists of an ISA, hardware to interface the ISA, and low-level software that performs down-sampling, temperature calibration, and vibration (sculling and coning) compensation

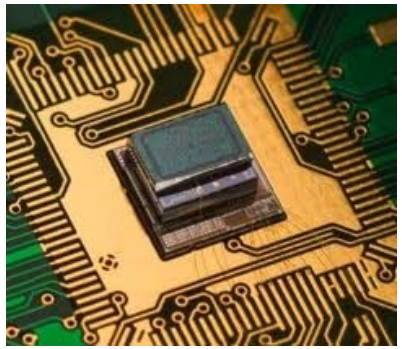
- **Navigation-Grade** IMUs offer the highest accuracy with very low bias drift, enabling long-duration navigation without external aiding (ITAR-controlled, i.e., export restricted).
- **Tactical-Grade** IMUs provide moderate accuracy suitable for aided navigation (sometimes ITAR-controlled, depending on performance).
- **MEMS-Grade** IMUs are low-cost sensors with higher noise and drift, typically used in small autonomous systems (not ITAR-controlled and considered commercial off-the-shelf (COTS) components).

MEMS-Grade Inertial Measurement Technology

Microelectromechanical systems (MEMS) is the technology of very small mechanical devices driven by electricity
It merges at the nano-scale into nanoelectromechanical systems (NEMS) and nanotechnology.







Today inertial measurement technology is available for commercial users thanks to a MEMS technology and a significant reduction in price the last decades.

Low-cost inertial sensors can be integrated with a global navigation satellite or other positioning systems using a conventional Kalman filter or a nonlinear state observer.





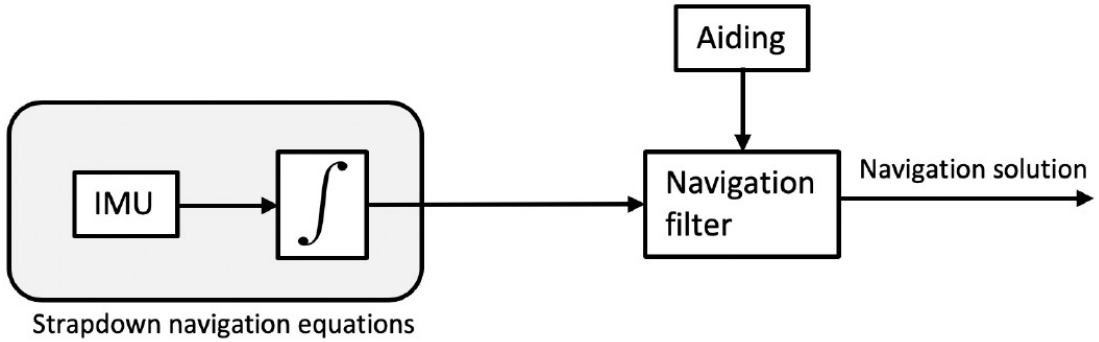




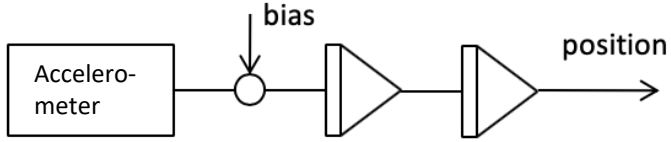


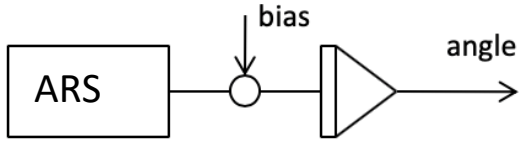


Numerical Integration of Inertial Measurements: From Angular Rates and Accelerations to Attitude, Velocity, and Position



The model of the craft is avoided by using **accelerometers** and **angular rate sensor (ARS)** measurements as inputs. Rotations of the body frame with respect to the inertial frame is compensated by integrating the strapdown navigation equations, which are based on kinematics.





An IMU can be integrated with a satellite navigation system in a state estimator to obtain estimates of *position, velocity* and *attitude* in 6 DOFs.

A stand-alone IMU solution, where

- **Specific force measurements** are integrated twice to obtain positions
- **ARS measurements** are integrated once to obtain attitude

will drift due to sensors biases, misalignments, temperature variations etc.

An INS state estimator aided by position measurements can be used to estimate the accelerometer and ARS bias terms and thus give accurate estimates of position, velocity and attitude.

Specific force: Non-gravitational acceleration measured by the IMU (body acceleration excluding gravity).



Coning and Sculling Effects in Strapdown IMUs — Errors from Discrete Numerical Integration

Coning (Small Rotational Oscillations)

- Happens due to rapid translational accelerations, which affect velocity and position updates.
- Coupling between rotation and translation due to the rotation matrix: In strapdown systems, rotation affects how accelerations are measured. Improper handling of orientation changes during rapid translational motion can cause integration errors in velocity and position.

Sculling (Rapid Translational Accelerations)

- Happens due to rapid translational accelerations, which affect velocity and position updates.
- Coupling between rotation and translation due to the rotation matrix: In strapdown systems, rotation affects how accelerations are measured. Improper handling of orientation changes during rapid translational motion can cause integration errors in velocity and position.

Why High-Frequency Integration (1000–2000 Hz) Helps

1. Capturing rotational-translational coupling: At lower frequencies, changes in orientation and their influence on translational accelerations are poorly captured, leading to cumulative errors. High-frequency sampling accurately captures the interaction between rotation (from the gyroscope) and acceleration (from the accelerometer), ensuring accurate velocity and position integration.
2. Improved handling of fast dynamics: High-frequency updates are crucial for properly compensating for coning and sculling, ensuring that small, high-frequency changes are integrated correctly.

14.1 Inertial Measurement Unit (IMU)

14.1 Inertial Measurement Unit (IMU)

An INS uses a computer, accelerometers and attitude rate sensors (ARS) to continuously calculate by dead reckoning the position, velocity and attitude of a moving craft without the need for external reference signals.

The key sensory component is the IMU, which is composed of the following high-rate sensors (1000-2000 Hz):

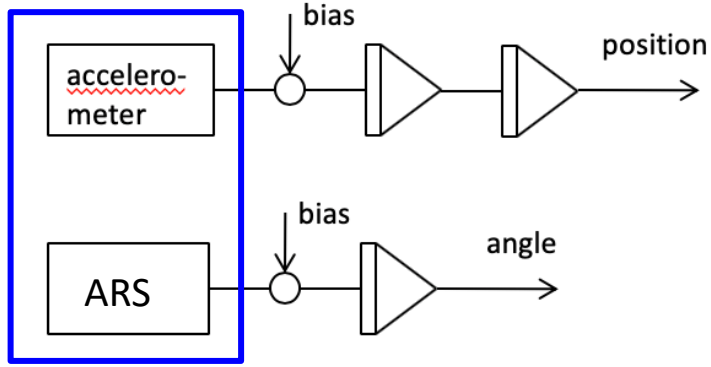


- **Three-axis ARS (MEMS-based or fiber-optic gyros)**
- **Three-axis accelerometer (MEMS or capacity type)**

This is referred to as a 6-DOF IMU. Some MEMS-based IMUs have additional measurements such as:

- **Three-axis magnetometer**
- **One-axis barometric pressure (altitude) sensor**

The complete sensor suite will then be 10 DOFs. In addition to this, it is common to include a temperature sensor that can be used for temperature compensation. This is usually implemented as a temperature-indexed lookup table with offset values.



Aided INS: The integrated signals will drift. Hence, the system must be aided by GNSS, hydroacoustic positioning reference (HPR) systems or other reference systems.

Dead-reckoning: In case of GNSS drop-outs the strapdown equations are integrated without corrections of the Kalman filter. This is referred to as *dead-reckoning*.

14.1 Inertial Measurement Unit

IMU Measurement Frame $\{m_I\}$

The IMU is mounted onboard the craft in a measurement frame $\{m_I\}$ with coordinate origin CM_I located at

$$\boldsymbol{r}^b_{bm_I} = [x_{m_I}, y_{m_I}, z_{m_I}]^\top$$

with respect to the $\{b\}$ frame coordinate origin CO. It is assumed that the axes of $\{m_I\}$ and $\{b\}$ point in the same directions.

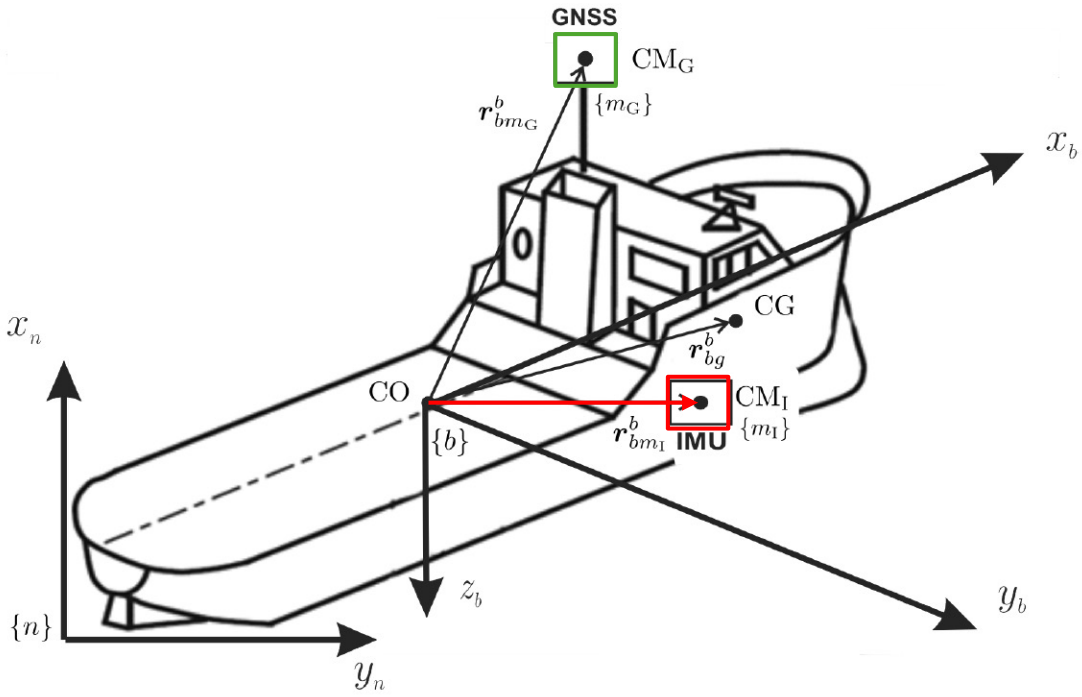
The state estimator is formulated in the CM_I and the estimated states are transformed to the CO using:

$$\begin{bmatrix} \hat{\boldsymbol{v}}^b_{nb} \\ \hat{\boldsymbol{\omega}}^b_{nb} \end{bmatrix} = \boldsymbol{H}^{-1}(\boldsymbol{r}^b_{bm_I}) \begin{bmatrix} \hat{\boldsymbol{v}}^b_{nm_I} \\ \hat{\boldsymbol{\omega}}^b_{nm_I} \end{bmatrix}$$

$$\Updownarrow$$

$$\hat{\boldsymbol{\nu}} = \boldsymbol{H}^{-1}(\boldsymbol{r}^b_{bm_I})\hat{\boldsymbol{\nu}}_{m_I}$$

(Appendix C)



$$\boldsymbol{r}^b_{bg} = [x_g, y_g, z_g]^\top \qquad \boldsymbol{r}^b_{bg} = \boldsymbol{r}^b_{bm_I} + \boldsymbol{r}^b_{m_Ig}$$

$$\boldsymbol{r}^b_{m_Ig} = \boldsymbol{r}^b_{bg} - \boldsymbol{r}^b_{bm_I} = \begin{bmatrix} x_g - x_{m_I} \\ y_g - y_{m_I} \\ z_g - z_{m_I} \end{bmatrix}$$

It is standard practice to express the Kalman filter in the CM_I since specific force $\boldsymbol{f}^b_{nm_I}$ is measured in the same frame. If another reference point is chosen, such as the CG, the specific force measurement must be transformed to the CG. This involves numerical differentiation of the angular rate measurement $\boldsymbol{\omega}^b_{nb}$ since

$$\boldsymbol{f}^b_{ng} = \boldsymbol{f}^b_{nm_I} + \dot{\boldsymbol{\omega}}^b_{nb} \times \boldsymbol{r}^b_{m_Ig} + \boldsymbol{\omega}^b_{nb} \times (\boldsymbol{\omega}^b_{nb} \times \boldsymbol{r}^b_{m_Ig})$$

14.1.1 Attitude Rate Sensors (ARS)

The classic attitude rate sensor (ARS) is a *gyro*, that is a spinning wheel that utilizes conservation of momentum to detect rotation.

For strapdown applications, optical gyros such as ring-laser gyros (RLG) and fiber-optic gyros (FOG) have been used for some time and are also expected to be the standard for high-accuracy strapdown INS for the foreseeable future. For low- and medium-cost applications, ARS based on MEMS are dominating.

IMU measurement equation for a three-axis ARS

$$\boldsymbol{\omega}^b_{\text{imu}} = \boldsymbol{\omega}^b_{nb} + \boldsymbol{b}^b_{\text{ars}} + \boldsymbol{w}^b_{\text{ars}}$$
$$\dot{\boldsymbol{b}}^b_{\text{ars}} = \boldsymbol{w}^b_{b,\,\text{ars}}$$

$\boldsymbol{b}^b_{\text{ars}}$ ARS bias vector
$\boldsymbol{w}^b_{b,\,\text{ars}}$ Gaussian white bias noise
$\boldsymbol{w}^b_{\text{ars}}$ Gaussian white measurement noise



It is necessary to estimate the ARS bias online since it will grow over time.

The ARS measurement is only valid for low-speed applications such as a marine craft moving on the surface of the Earth since it assumes that $\{n\}$ is nonrotating, that is

$$\boldsymbol{\omega}^b_{ib} \approx \boldsymbol{\omega}^b_{nb}$$

For terrestrial navigation, the Earth rotation will affect the results, and it is necessary to use the inertial frame $\{i\}$ instead of the approximate frame $\{n\}$.

14.1.2 Accelerometers

There are several different types of accelerometer. Two of these are mechanical and vibratory accelerometers. The mechanical accelerometer can be a pendulum, which in its simplest form is based on Newton's second law of motion. The vibratory accelerometers are usually based on measurement of frequency shifts due to increased or decreased tension in a string.

The IMU accelerometer is a device that measure three-axis specific force:

$$\begin{aligned} \boldsymbol{f}^b_{\text{imu}} &:= \frac{1}{m} \sum \boldsymbol{f}^b_{\text{non-gravitational}} \\ &= \frac{1}{m} \left(\boldsymbol{f}^b_{\text{total}} - \sum \boldsymbol{f}^b_{\text{gravitational}} \right) \end{aligned}$$

$$\boldsymbol{f}^b_{\text{imu}} = \boldsymbol{a}^b_{nm_{\text{I}}} - \boldsymbol{g}^b \qquad \boldsymbol{g}^b = (1/m)\boldsymbol{f}^b_g$$

The specific force measured by an IMU is the **non-gravitational acceleration** acting on the sensor, equal to the vehicle's acceleration in the body frame minus the local gravity vector.

Marine craft experiencing hydrodynamic, buoyancy, gravitational and thrust forces

$$\begin{aligned} \boldsymbol{f}^b_{\text{imu}} &= \frac{1}{m} \left(\boldsymbol{f}^b_{\text{total}} - \boldsymbol{f}^b_g \right) \\ &= \frac{1}{m} \left(\left(\boldsymbol{f}^b_h + \boldsymbol{f}^b_b + \boldsymbol{f}^b_g + \boldsymbol{f}^b_t \right) - \boldsymbol{f}^b_g \right) \\ &= \frac{1}{m} \left(\boldsymbol{f}^b_h + \boldsymbol{f}^b_t + \boldsymbol{f}^b_b \right) \end{aligned}$$

Newton's second law

$$m\boldsymbol{a}^b_{nm_{\text{I}}} = \boldsymbol{f}^b_{\text{total}}$$

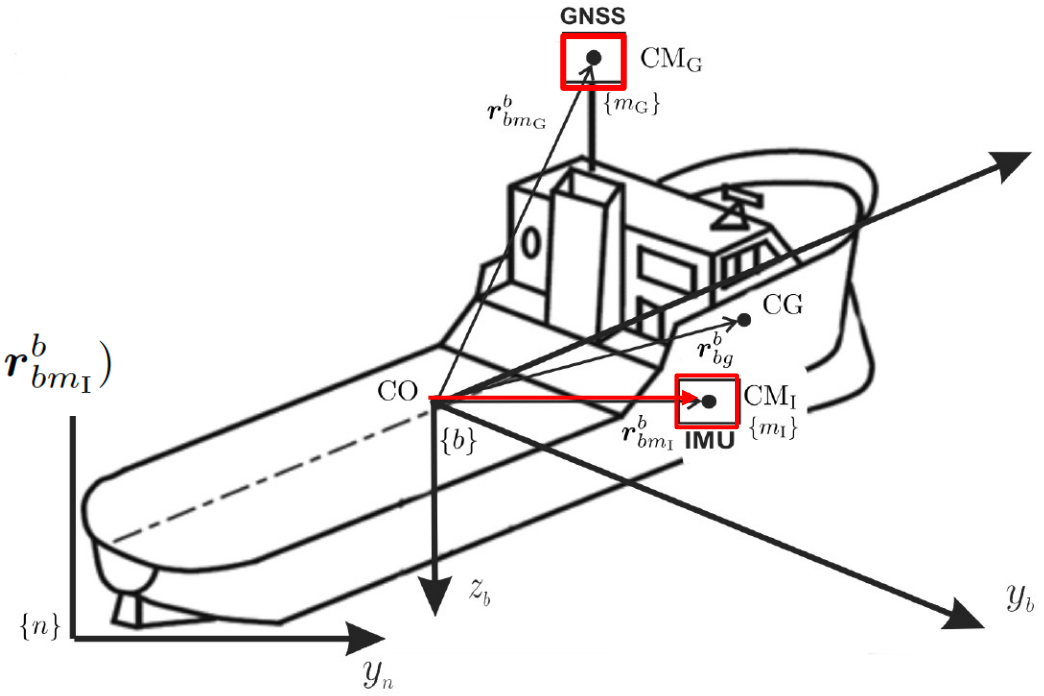
14.1.2 Accelerometers

Linear velocity and acceleration

$$\boldsymbol{v}^b_{nm_\mathrm{I}} = \boldsymbol{v}^b_{nb} + \boldsymbol{\omega}^b_{nb} \times \boldsymbol{r}^b_{bm_\mathrm{I}}$$

$$\boldsymbol{a}^b_{nm_\mathrm{I}} = \dot{\boldsymbol{v}}^b_{nb} + \boldsymbol{\omega}^b_{nb} \times \boldsymbol{v}^b_{nb} + \dot{\boldsymbol{\omega}}^b_{nb} \times \boldsymbol{r}^b_{bm_\mathrm{I}} + \boldsymbol{\omega}^b_{nb} \times (\boldsymbol{\omega}^b_{nb} \times \boldsymbol{r}^b_{bm_\mathrm{I}})$$

$$\boldsymbol{f}^b_{\mathrm{imu}} = \boldsymbol{a}^b_{nm_\mathrm{I}} - \boldsymbol{g}^b \qquad \boldsymbol{g}^b = \boldsymbol{R}^\top(\boldsymbol{\Theta}_{nb})\boldsymbol{g}^n$$



$$\begin{aligned}
f_x &= \dot{u} - vr + wq - x_{m_\mathrm{I}}(q^2 + r^2) + y_{m_\mathrm{I}}(pq - \dot{r}) + z_{m_\mathrm{I}}(pr + \dot{q}) + g\sin(\theta) \\
f_y &= \dot{v} - wp + ur - y_{m_\mathrm{I}}(r^2 + p^2) + z_{m_\mathrm{I}}(qr - \dot{p}) + x_{m_\mathrm{I}}(qp + \dot{r}) - g\cos(\theta)\sin(\phi) \\
f_z &= \dot{w} - uq + vp - z_{m_\mathrm{I}}(p^2 + q^2) + x_{m_\mathrm{I}}(rp - \dot{q}) + y_{m_\mathrm{I}}(rq + \dot{p}) - g\cos(\theta)\cos(\phi)
\end{aligned}$$

IMU measurement equation for a three-axis accelerometer (specific force)

$$\boldsymbol{f}^b_{\mathrm{imu}} = \boldsymbol{R}^\top(\boldsymbol{\Theta}_{nb})(\boldsymbol{a}^n_{nm_\mathrm{I}} - \boldsymbol{g}^n) + \boldsymbol{b}^b_{\mathrm{acc}} + \boldsymbol{w}^b_{\mathrm{acc}}$$

$$\dot{\boldsymbol{b}}^b_{\mathrm{acc}} = \boldsymbol{w}^b_{b,\,\mathrm{acc}}$$

$\boldsymbol{b}^b_{\mathrm{acc}}$ Accelerometer bias vector

$\boldsymbol{w}^b_{b,\,\mathrm{acc}}$ Gaussian white bias noise

$\boldsymbol{w}^b_{\mathrm{acc}}$ Gaussian white measurement noise

$\boldsymbol{g}^n = [0,\, 0,\, g]^\top$

14.1.2 Accelerometers

What will an accelerometer measure on a levelled table?

$$\boldsymbol{f}_{\text{imu}}^{b} = \boldsymbol{a}_{nm_{\text{I}}}^{b} - \boldsymbol{g}^{b}$$

$$\begin{aligned}
f_x &= \dot{u} - vr + wq - x_{m_{\text{I}}}(q^2 + r^2) + y_{m_{\text{I}}}(pq - \dot{r}) + z_{m_{\text{I}}}(pr + \dot{q}) + g\sin(\theta) \\
f_y &= \dot{v} - wp + ur - y_{m_{\text{I}}}(r^2 + p^2) + z_{m_{\text{I}}}(qr - \dot{p}) + x_{m_{\text{I}}}(qp + \dot{r}) - g\cos(\theta)\sin(\phi) \\
f_z &= \dot{w} - uq + vp - z_{m_{\text{I}}}(p^2 + q^2) + x_{m_{\text{I}}}(rp - \dot{q}) + y_{m_{\text{I}}}(rq + \dot{p}) - g\cos(\theta)\cos(\phi)
\end{aligned}$$



Answer:

Note that the accelerometers if leveled ($\phi = \theta = 0$) and at rest $\boldsymbol{a}_{nm_{\text{I}}}^{n} = \boldsymbol{0}$ under the assumptions that the bias is zero and there are no measurement noise, will measure

$$\boldsymbol{f}_{\text{imu}}^{b} = -\boldsymbol{g}^{n}$$

$$\boldsymbol{g}^{n} = [0,\, 0,\, g]^{\top}$$

14.1.2 Accelerometers

Earth-Gravity Model

The gravity of Earth is modelled as a constant vector

$$\boldsymbol{g}^n = [0,\, 0,\, g(\mu)]^\top$$

World Geodetic System (1984) ellipsoidal gravity formula

$$g(\mu) = \mathbb{G}_e \frac{1 + k \sin^2(\mu)}{\sqrt{1 - e^2 \sin^2(\mu)}}$$

$$k = \frac{r_p \mathbb{G}_p - r_e \mathbb{G}_e}{r_e \mathbb{G}_e}, \qquad e^2 = 1 - (r_p / r_e)^2$$

where $g(\mu)$ is the acceleration of gravity as a function of latitude μ. Gravity increases from $\mathbb{G}_e = 9.780\,325\,335\,9\ \mathrm{ms}^{-2}$ at the equator to $\mathbb{G}_p = 9.832\,184\,937\,8\ \mathrm{ms}^{-2}$ at the poles. The nominal "average" value at the surface of the Earth, known as standard gravity, is, by definition of the ISO/IEC 8000, $\mathbb{G}_s = 9.806\,65\ \mathrm{ms}^{-2}$.

```
function g = gravity(mu)
% g = gravity(mu) computes the acceleration of gravity (m/s^2) as a function
% of lattitude mu (rad) using the WGS-84 ellipsoid parameters.
%
% Input:  mu  lattitude (rad)
%
% Author:    Thor I. Fossen
% Date:      19th February 2020
% Revisions:

g = 9.7803253359 * ( 1 + 0.001931850400 * sin(mu)^2 ) /...
    sqrt( 1 - 0.006694384442 * sin(mu)^2 );
```

Trondheim, Norway:

```
>> gn = gravity(63.4 * pi/180)
gn =
    9.8218
```

Online gravity calculator:
https://www.sensorsone.com/local-gravity-calculator/

Parameters	Comments
$r_e = 6\,378\,137.0$ m	Equatorial radius of ellipsoid (semi-major axis)
$r_p = 6\,356\,752.314\,245$ m	Polar axis radius of ellipsoid (semi-minor axis)
$\omega_{ie} = 7.292115 \times 10^{-5}$ rad/s	Angular velocity of the Earth
$e = 0.0818$	Eccentricity of ellipsoid

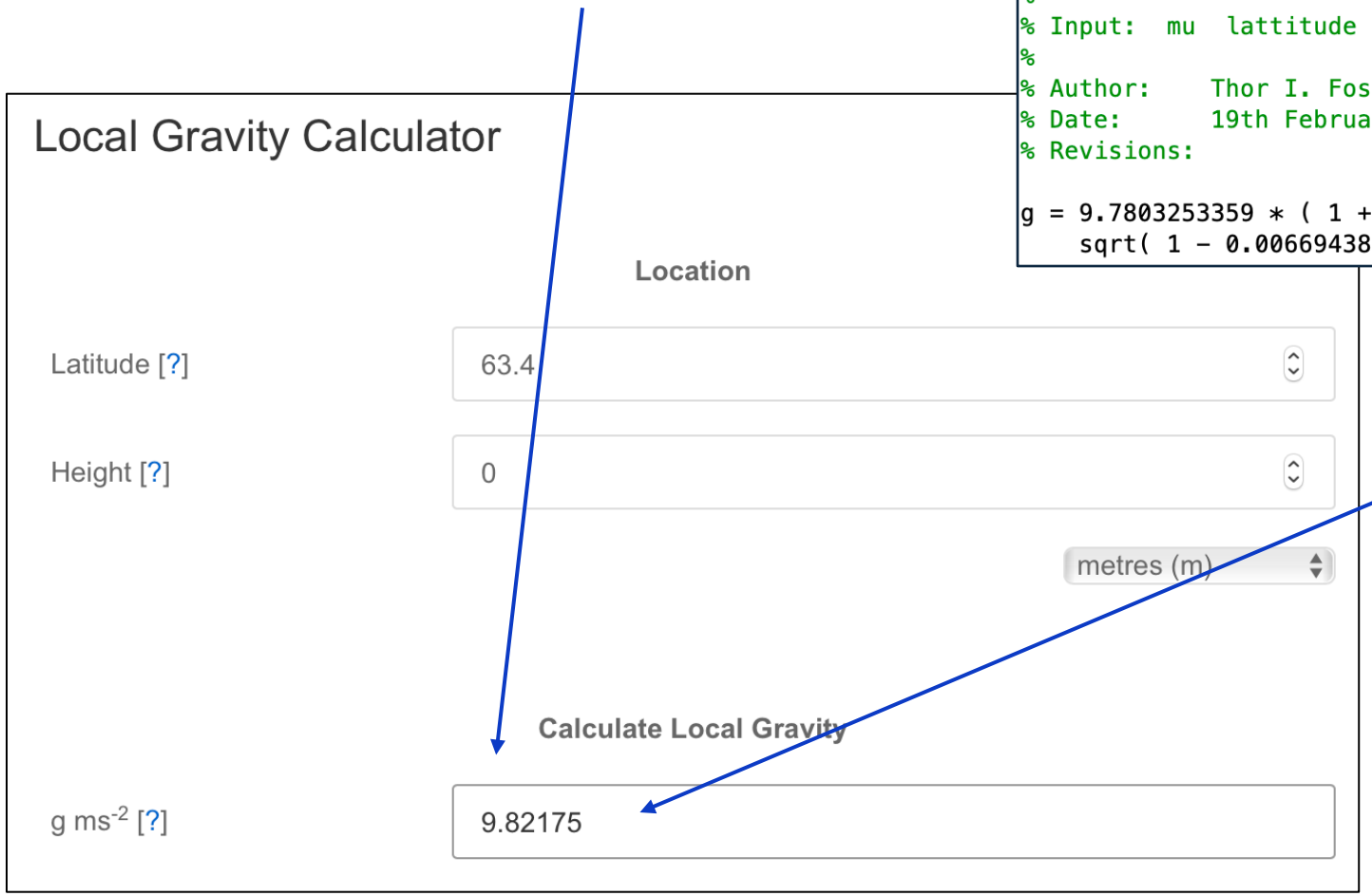
14.1.2 Accelerometers

Online gravity calculator:
https://www.sensorsone.com/local-gravity-calculator/

MSS toolbox

```
function g = gravity(mu)
% g = gravity(mu) computes the acceleration of gravity (m/s^2) as a function
% of lattitude mu (rad) using the WGS-84 ellipsoid parameters.
%
% Input:  mu  lattitude (rad)
%
% Author:    Thor I. Fossen
% Date:      19th February 2020
% Revisions:

g = 9.7803253359 * ( 1 + 0.001931850400 * sin(mu)^2 ) /...
    sqrt( 1 - 0.006694384442 * sin(mu)^2 );
```



Trondheim, Norway:

```
>> gn = gravity(deg2rad(63.4))
gn =
    9.8218
```

14.1.3 Magnetometer

The magnetic field of the Earth is like a simple bar magnet. The magnetic field is a magnetic dipole that has its field lines originating at a point near the South Pole and terminating at a point near the North Pole. The field lines vary in both strength and direction about the face of the Earth. At each location on the Earth, the field lines intersect the Earth's surface at a specific angle of inclination. Near the equator, the field lines are approximately parallel to the Earth's surface and thus the inclination angle δ in this region is 0 degrees.

Magnetic field strength can be measured by a three-axis magnetometer, usually included in the sensor suite of commercially available IMUs

$$\boldsymbol{m}^b_{\text{mag}} = \boldsymbol{R}^\top(\boldsymbol{\Theta}_{nb})\boldsymbol{m}^n + \boldsymbol{b}^b_{\text{mag}} + \boldsymbol{w}^b_{\text{mag}}$$

$\boldsymbol{b}^b_{\text{mag}}$ Magnetometer bias vector

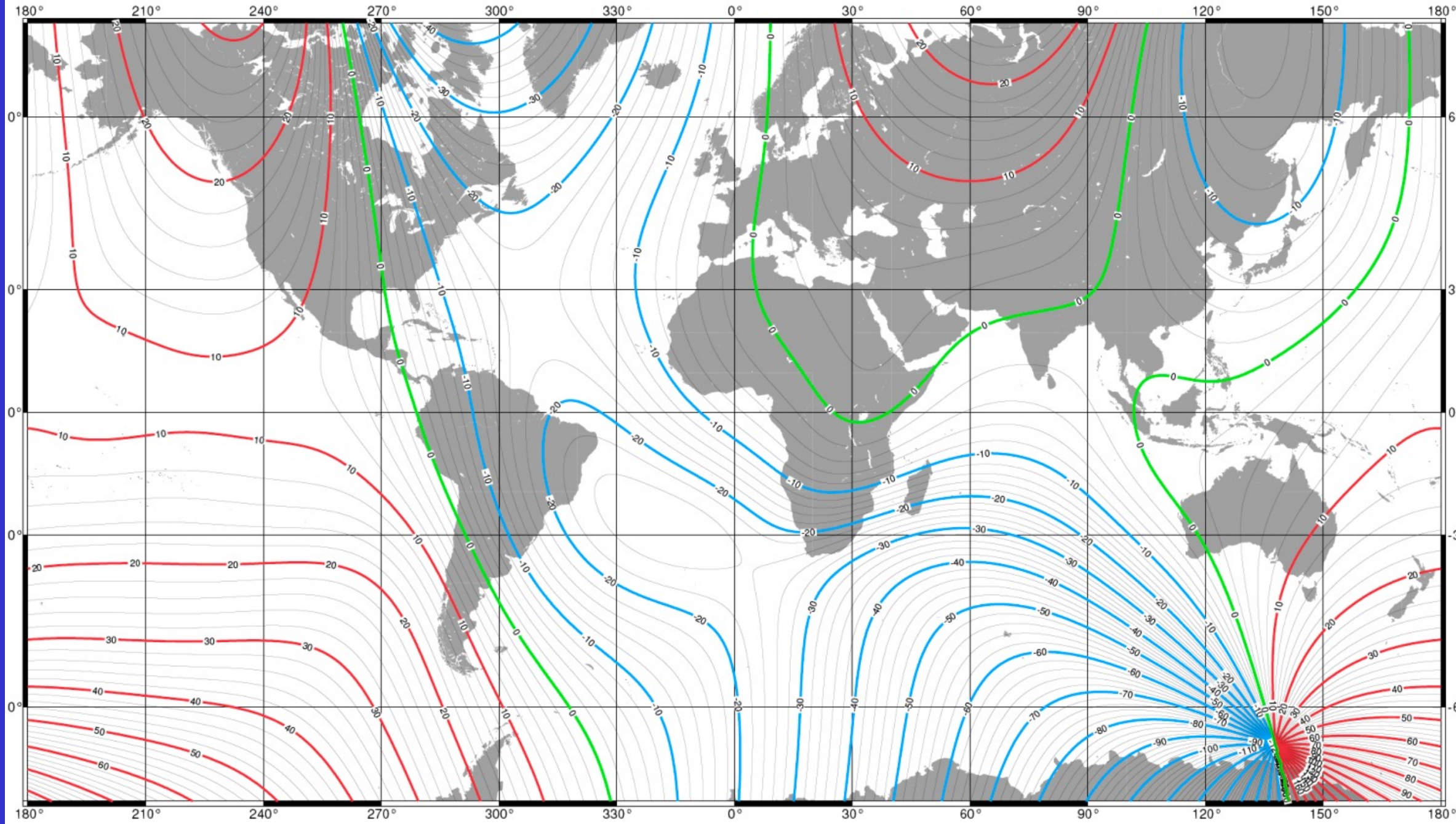
$\boldsymbol{w}^b_{\text{mag}}$ Gaussian white measurement noise

$\boldsymbol{m}^n$ is the strength and direction of the Earth's magnetic field expressed in {*n*}, which will change over time.

The measurement is affected by a largely time-invariant bias, induced by local magnetic disturbances.
It is necessary to perform a hard-iron magnetometer calibration to derive an estimate for $\boldsymbol{b}^b_{\text{mag}}$ which is then subtracted from all future measurements.

However, care must be taken since magnetometers also are prone to time-varying electric currents from motors and transmission wires. As of result of this, ships use gyrocompasses for safe navigation.

Magnetic field declination δ according to the US/UK World Magnetic Model (WMM)



14.1.3 Magnetometer

Magnetic heading from magnetometer measurements

The heading angle is the sum of the the magnetic heading measurement ψ_m and the declination angle δ

$$\psi = \psi_m + \delta$$

The declination angle for a given longitude l and latitude μ can be calculated using the *World Magnetic Model* (WMM), which is a joint project by the United States' National Geospatial-Intelligence Agency (NGA) and the United Kingdom's Defence Geographic Centre (DGC).

If the roll and pitch angles are known the magnetometer measurements can be transformed to the horizontal plane:

$$\begin{bmatrix} h_x \\ h_y \\ h_z \end{bmatrix} = \boldsymbol{R}_{y,\theta}\boldsymbol{R}_{x,\phi} \begin{bmatrix} m_x \\ m_y \\ m_z \end{bmatrix}$$

$$\begin{bmatrix} h_x \\ h_y \\ h_z \end{bmatrix} = \begin{bmatrix} \cos(\theta) & 0 & \sin(\theta) \\ 0 & 1 & 0 \\ -\sin(\theta) & 0 & \cos(\theta) \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\phi) & -\sin(\phi) \\ 0 & \sin(\phi) & \cos(\phi) \end{bmatrix} \begin{bmatrix} m_x \\ m_y \\ m_z \end{bmatrix}$$

The magnetic compass heading is deduced from the horizontal components according to:

$$\psi_m = \text{atan2}(h_y, h_x)$$



Wikimedia Commons
https://commons.wikimedia.org/wiki/File:Kompas_Sofia.jpg

14.1 Attitude Estimation

14.2 Attitude Estimation

A complete navigation solution will estimate positions, velocities and attitude (PVA). This section, however, presents static and dynamic solutions to the attitude estimation problem without using information about the position.

Commercial systems implementing these algorithms are usually called Attitude Heading Reference Systems (AHRS).







14.2.1 Static Mapping from Specific Force to Roll and Pitch Angles

The core algorithm for mapping the three-axis specific force from an IMU to roll and pitch angles is based on the principle that the angle between the acceleration and gravity vectors can be computed using trigonometry. This is a static mapping that suffers from inaccuracies when performing high-acceleration maneuvers:

$$\boldsymbol{f}^b_{\text{imu}} = \boldsymbol{R}^\top(\boldsymbol{\Theta}_{nb})(\boldsymbol{a}^n_{nm_{\text{I}}} - \boldsymbol{g}^n) + \boldsymbol{b}^b_{\text{acc}} + \boldsymbol{w}^b_{\text{acc}}$$

From this formula we can compute the static roll and pitch angles by noticing that for three orthogonal accelerometers onboard a craft at rest:

$$\boldsymbol{a}^n_{nm_{\text{I}}} = \boldsymbol{0}$$

The initial accelerometer biases are usually removed by calibrating the accelerometer in a laboratory for varying temperatures. This can be implemented as a look-up table in combination with a temperature sensor. It is also necessary to remove the dynamic drift, for instance by recalibrating the sensor when the craft is at rest. In addition, the measurement noise should be removed by low-pass filtering such that

$$\boldsymbol{f}^b := \boldsymbol{f}^b_{\text{imu}} - \boldsymbol{b}^b_{\text{acc}} \approx -\boldsymbol{R}^\top(\boldsymbol{\Theta}_{nb})\boldsymbol{g}^n$$

14.2.1 Static Mapping from Specific Force to Roll and Pitch Angles

$$\boldsymbol{f}^b := \boldsymbol{f}^b_{\text{imu}} - \boldsymbol{b}^b_{\text{acc}} \approx -\boldsymbol{R}^\top(\boldsymbol{\Theta}_{nb})\boldsymbol{g}^n$$

Here it is assumed that $\boldsymbol{b}^b_{acc}$ is compensated after an initial calibration (warm start).

$$\begin{bmatrix} f_x \\ f_y \\ f_z \end{bmatrix} \approx -\boldsymbol{R}^\top(\boldsymbol{\Theta}_{nb}) \begin{bmatrix} 0 \\ 0 \\ g \end{bmatrix} = \begin{bmatrix} g\sin(\theta) \\ -g\cos(\theta)\sin(\phi) \\ -g\cos(\theta)\cos(\phi) \end{bmatrix}$$

Ratios

$$\frac{f_y}{f_z} \approx \tan(\phi), \quad \frac{f_x}{g} \approx \sin(\theta), \quad \frac{f_y^2 + f_z^2}{g^2} \approx \cos^2(\theta)$$

Static roll and pitch angles as a function of specific force

$$\phi \approx \tan^{-1}\left(\frac{f_y}{f_z}\right)$$

$$\theta \approx \tan^{-1}\left(\frac{f_x}{\sqrt{f_y^2 + f_z^2}}\right) \qquad \theta = \pm 90°$$

Static RPY Angles

When combined with a 3-D magnetometer (or compass) for the yaw angle ψ, the attitude vector

$$\boldsymbol{\Theta}_{nb} = [\phi, \theta, \psi]^\top$$

is completely determined. Recall that

$$\psi = \psi_m + \delta \qquad \psi_m = \text{atan2}(h_y, h_x)$$

where

$$\begin{bmatrix} h_x \\ h_y \\ h_z \end{bmatrix} = \begin{bmatrix} \cos(\theta) & 0 & \sin(\theta) \\ 0 & 1 & 0 \\ -\sin(\theta) & 0 & \cos(\theta) \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\phi) & -\sin(\phi) \\ 0 & \sin(\phi) & \cos(\phi) \end{bmatrix} \begin{bmatrix} m_x \\ m_y \\ m_z \end{bmatrix}$$

Coordinate Frames and Origins

Symbol	Frame	Description	Typical Use
{b}	**Body-Fixed Frame**	Control Origin (CO): The point about which the control system acts, with Forward–Starboard–Down (x–y–z) axes fixed to the vehicle. The frame moves and rotates with the craft.	The marine craft EoM can be integrated either about the CO or the CG to obtain body-fixed velocities $\boldsymbol{\nu}$. When the EoM are expressed in the CG, the velocities are transformed to the control origin (CO) using: $\boldsymbol{\nu} = \mathbf{H}^{-1}\left(\mathbf{r}_{\mathrm{bG}}^{b}\right)\boldsymbol{\nu}^{\mathrm{CG}}$ where $\boldsymbol{r}_{bG}^{b}$ is the position vector from the CO to the CG.
{n}	**North-East-Down (NED) Frame**	Earth-fixed (local tangent plane) reference frame.	The kinematic equations expresses global position, orientation, and NED velocity.
{i}	**Inertial Frame**	A non-rotating reference frame in which Newton's 2nd law is valid.	Usually, approximated by {n} or another local tangent frame by neglecting the Earth rotation.
{m}	**Sensor Frame**	Local IMU frame with origin at CM_I (possibly misaligned from {b}).	Numerical integration of the strapdown inertial navigation system (INS) equations.

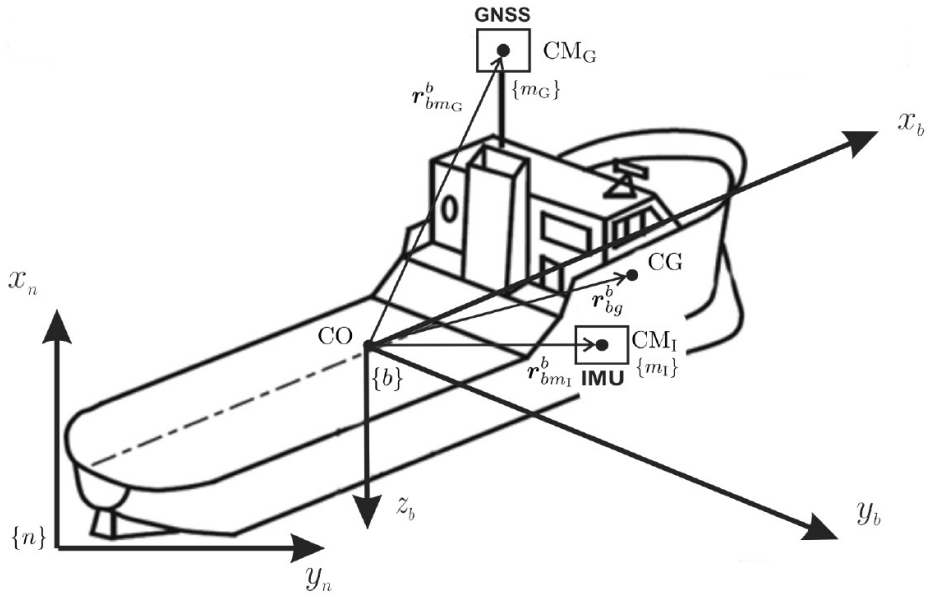
Notes:

- Using $\{n\} \approx \{i\}$ is standard for marine and short-duration applications.
- The distinction between {m} and {b} is important when the IMU is not perfectly aligned with the body axes.

Body-Fixed Points Defined in {b}

Symbol	Name	Description	Typical Use
CO	**Control Origin (body origin)**	The fixed geometric origin of the body frame {b}, which the control system acts. Usually at midships or another convenient reference point such as the CG.	Reference point for the 6-DOF equations of motion and for defining the rigid-body and hydrodynamic matrices $\mathbf{M}_{RB}, \mathbf{M}_A, \mathbf{D}, \mathbf{G}$, etc.,. as well as actuator lever arms and control implementation.
CG	**Center of Gravity**	The craft's mass center. The CO may coincide with the CG or be offset for modeling convenience, for instance when the CG is time varying.	When the EoM are integrated at this point, coupling terms between translation and rotation are minimized. Practical if $\mathbf{D}$ is chosen as a diagonal matrix.
VRP	**Virtual Reference Point**	A virtual point located $\mathbf{p}_{vrp}$ ahead of CO along the x-axis. Used only in guidance and path-following systems to define the lookahead (track) point.	Guidance law tracking errors (LOS, ILOS, ALOS) are computed in this point.
CM_I	**IMU Reference Point**	Physical location of the IMU (strapdown mechanization point). Offset from CO and CG by a fixed lever arm.	Strapdown INS equations, accelerations and angular rates are numerically integrated in this point.
CM_G	**Global Measurement Point**	Point where aided navigation sensors (e.g., GNSS, USBL) measure position in the global frame. Often offset from CM_I by a fixed lever arm.	The aiding updates for the INS (position and velocity) are transformed from the CM_G to the CM_I.

$$\boldsymbol{p}^n_{\text{gnss}} := \boldsymbol{p}^n_{nm_G} = [x^n, y^n, z^n]^\top$$
$$\boldsymbol{v}^n_{\text{gnss}} := \boldsymbol{v}^n_{nm_G} = [\dot{x}^n, \dot{y}^n, \dot{z}^n]^\top$$



$$\begin{aligned}\boldsymbol{p}^n_{nm_I} &= \boldsymbol{p}^n_{nb} + \boldsymbol{R}(\boldsymbol{\Theta}_{nb})\boldsymbol{r}^b_{bm_I} \\ &= \boldsymbol{p}^n_{nm_G} - \boldsymbol{R}(\boldsymbol{\Theta}_{nb})\boldsymbol{r}^b_{bm_G} + \boldsymbol{R}(\boldsymbol{\Theta}_{nb})\boldsymbol{r}^b_{bm_I} \\ &= \boldsymbol{p}^n_{\text{gnss}} + \boldsymbol{R}(\boldsymbol{\Theta}_{nb})(\boldsymbol{r}^b_{bm_I} - \boldsymbol{r}^b_{bm_G})\end{aligned}$$

$$\boldsymbol{v}^n_{nm_I} = \boldsymbol{v}^n_{\text{gnss}} + \boldsymbol{R}(\boldsymbol{\Theta}_{nb})\boldsymbol{S}(\boldsymbol{\omega}^b_{nb})(\boldsymbol{r}^b_{bm_I} - \boldsymbol{r}^b_{bm_G})$$

14.2.2 Vertical Reference Unit (VRU) Transformations

The hardware and software needed to compute ϕ and θ, and sometimes the heave position z^n from inertial sensors are referred to as a *Vertical Reference Unit* (VRU).

The roll and pitch angles can be computed using the static solution (specific force) or dynamically in a state estimator, while the heave position is computed using the vertical accelerometer.

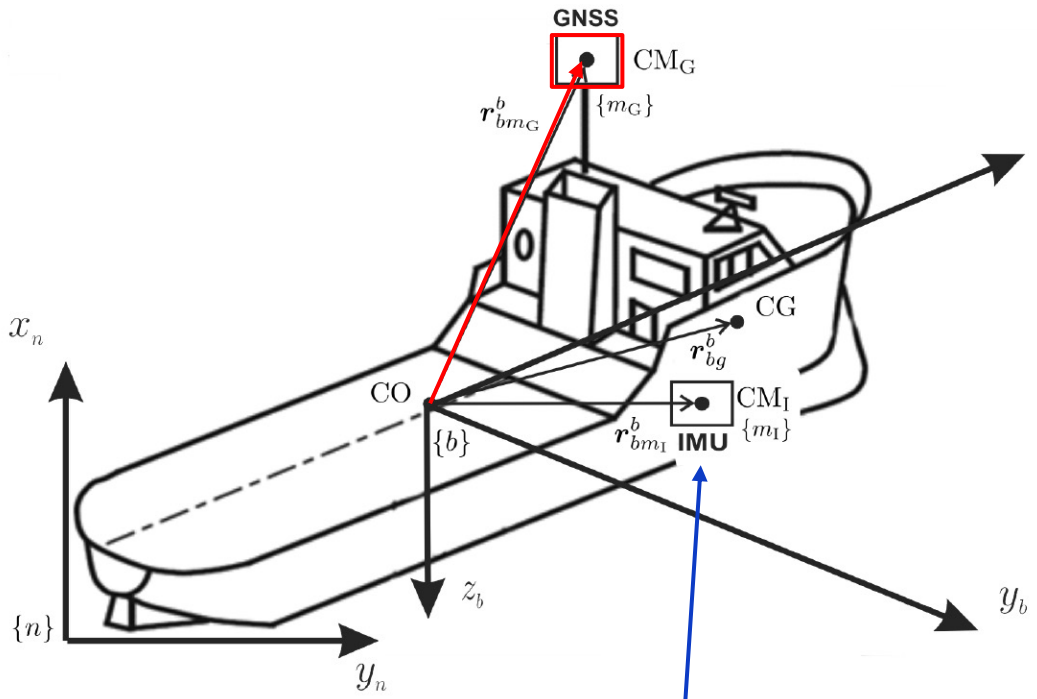
A VRU can be used to transform the GNSS position and velocity measurements

$$\boldsymbol{p}^n_{\text{gnss}} := \boldsymbol{p}^n_{nm_\text{G}} = [x^n, y^n, z^n]^\top$$
$$\boldsymbol{v}^n_{\text{gnss}} := \boldsymbol{v}^n_{nm_\text{G}} = [\dot{x}^n, \dot{y}^n, \dot{z}^n]^\top$$

from the GNSS measurement frame $\{m_\text{G}\}$ to the IMU measurement frame $\{m_\text{I}\}$ according to

$$\begin{aligned}\boldsymbol{p}^n_{nm_\text{I}} &= \boldsymbol{p}^n_{nb} + \boldsymbol{R}(\boldsymbol{\Theta}_{nb})\boldsymbol{r}^b_{bm_\text{I}} \\ &= \boldsymbol{p}^n_{nm_\text{G}} - \boldsymbol{R}(\boldsymbol{\Theta}_{nb})\boldsymbol{r}^b_{bm_\text{G}} + \boldsymbol{R}(\boldsymbol{\Theta}_{nb})\boldsymbol{r}^b_{bm_\text{I}} \\ &= \boldsymbol{p}^n_{\text{gnss}} + \boldsymbol{R}(\boldsymbol{\Theta}_{nb})(\boldsymbol{r}^b_{bm_\text{I}} - \boldsymbol{r}^b_{bm_\text{G}})\end{aligned}$$

$$\boldsymbol{v}^n_{nm_\text{I}} = \boldsymbol{v}^n_{\text{gnss}} + \boldsymbol{R}(\boldsymbol{\Theta}_{nb})\boldsymbol{S}(\boldsymbol{\omega}^b_{nb})(\boldsymbol{r}^b_{bm_\text{I}} - \boldsymbol{r}^b_{bm_\text{G}})$$



$$\boldsymbol{r}^b_{bm_\text{I}} = [x_{m_\text{I}}, y_{m_\text{I}}, z_{m_\text{I}}]^\top \qquad \dot{\boldsymbol{r}}^b_{bm_\text{I}} = \boldsymbol{0}$$
$$\boldsymbol{r}^b_{bm_\text{G}} = [x_{m_\text{G}}, y_{m_\text{G}}, z_{m_\text{G}}]^\top \qquad \dot{\boldsymbol{r}}^b_{bm_\text{G}} = \boldsymbol{0}$$

The navigation computer computes the INS estimates in the origin $\{\text{m}_\text{I}\}$. Hence, the GNSS measurements must be transformed from $\{\text{m}_\text{G}\}$ to $\{\text{m}_\text{I}\}$ using a lever-arm transformation.

14.2.3 Nonlinear Attitude Observer using Reference Vectors

The **static RPY** angles from an IMU gives only instantaneous angles and is easily disturbed during motion. The Mahony et al. (2008) observer improves this by using ARS feedback and reference vector correction for stable and drift-free attitude estimation.

Attitude Observer with ARS Bias Correction (Mahony et al. 2008)
The nonlinear attitude observer of Mahony et al. (2008) is less computational demanding than commercial algorithms such as the the Multiplicative Extended Kalman Filter (MEKF) in Section 14.4.3. The attitude observer renders the equilibrium point of the error dynamics semiglobally exponentially stable.

The observer-based **Attitude Heading Reference System (AHRS)** uses accelerometer and magnetometer measurements directly to update the states. This is achieved by constructing a nonlinear injection term $\boldsymbol{\omega}_{mes}^{b}$ that compares the directions of the measurement vectors in {b} to reference vectors in {n}, which is assumed to be an approximative inertial frame.



$$\boldsymbol{\omega}_{\text{mes}}^{b} = -\text{vex}\left(\sum_{i=1}^{n}\frac{k_i}{2}\left(\boldsymbol{v}_i^b(\hat{\boldsymbol{v}}_i^b)^\top - \hat{\boldsymbol{v}}_i^b(\boldsymbol{v}_i^b)^\top\right)\right)$$

$$\dot{\hat{\boldsymbol{q}}}_b^n = \boldsymbol{T}(\hat{\boldsymbol{q}}_b^n)\left(\boldsymbol{\omega}_{\text{imu}}^b - \hat{\boldsymbol{b}}_{\text{ars}}^b + \boldsymbol{K}_p\,\boldsymbol{\omega}_{\text{mes}}^b\right)$$

$$\dot{\hat{\boldsymbol{b}}}_{\text{ars}}^b = -\boldsymbol{K}_i\,\boldsymbol{\omega}_{\text{mes}}^b$$

Injection term based on reference vectors

Unit quaternion estimate

ARS bias vector estimate

$$\boldsymbol{a}\times\boldsymbol{b} = \boldsymbol{S}(\boldsymbol{a})\boldsymbol{b}$$

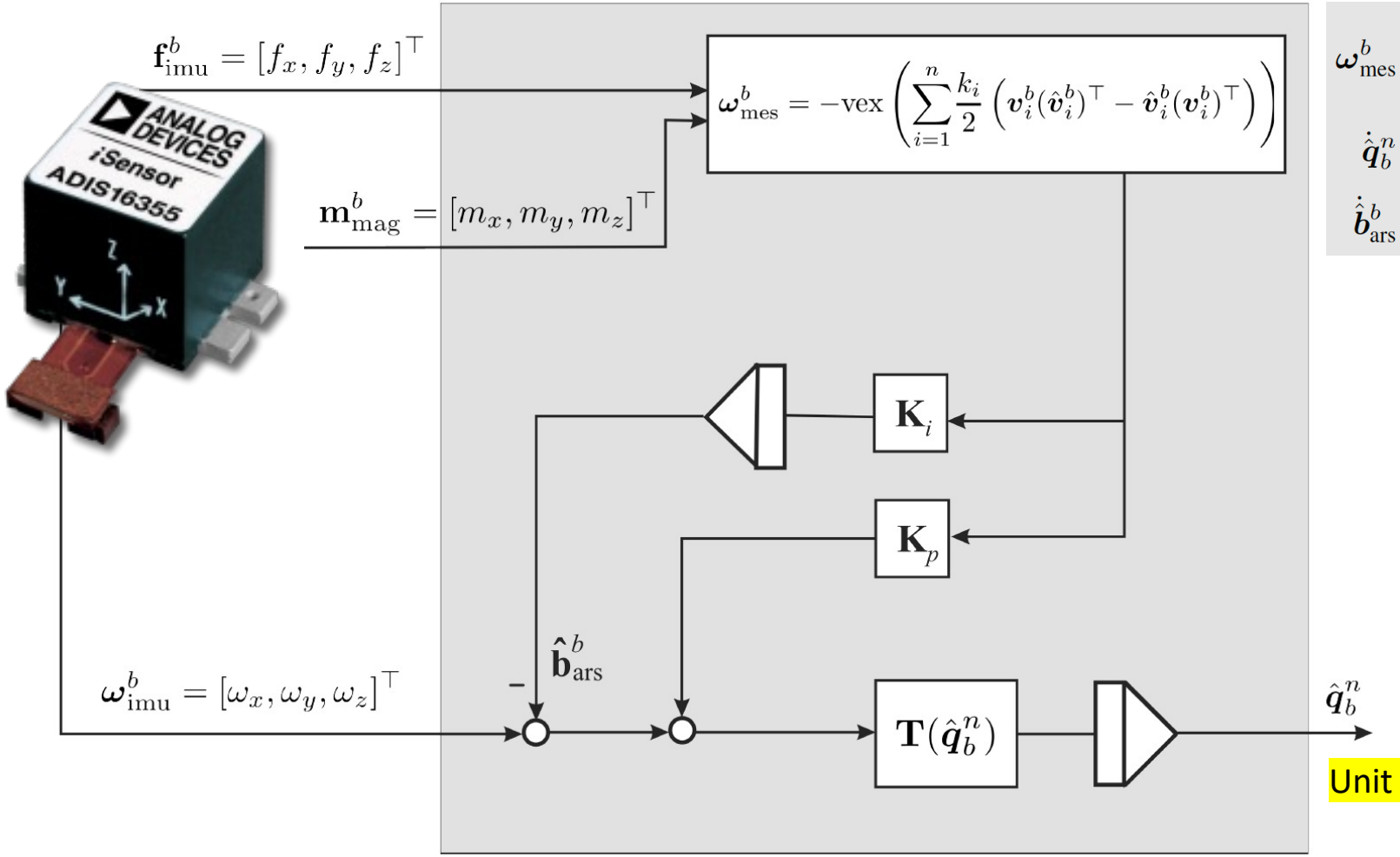
$$\text{vex}(\boldsymbol{S}(\boldsymbol{a})) = \boldsymbol{a}$$

where $\boldsymbol{v}_i^b$ and $\hat{\boldsymbol{v}}_i^b$ $(i = 1,\ldots,n)$ are sets of n true and estimated reference vectors

$$\text{vex}\left(\boldsymbol{a}\boldsymbol{b}^\top - \boldsymbol{b}\boldsymbol{a}^\top\right) = \begin{bmatrix} a_3b_2 - a_2b_3 \\ a_1b_3 - a_3b_1 \\ a_2b_1 - a_1b_2 \end{bmatrix}$$

R. Mahony, T. Hamel and J.-M. Pflimlin (2008). Nonlinear Complementary Filters on the Special Orthogonal Group. IEEE Trans. on Aut. Control 53(5)

14.2.3 Nonlinear Attitude Observer using Reference Vectors



$$\boldsymbol{\omega}^b_{\text{mes}} = -\text{vex}\left(\sum_{i=1}^{n} \frac{k_i}{2}\left(\boldsymbol{v}^b_i(\hat{\boldsymbol{v}}^b_i)^\top - \hat{\boldsymbol{v}}^b_i(\boldsymbol{v}^b_i)^\top\right)\right)$$

$$\dot{\hat{\boldsymbol{q}}}^n_b = \boldsymbol{T}(\hat{\boldsymbol{q}}^n_b)\left(\boldsymbol{\omega}^b_{\text{imu}} - \hat{\boldsymbol{b}}^b_{\text{ars}} + \boldsymbol{K}_p\,\boldsymbol{\omega}^b_{\text{mes}}\right)$$

$$\dot{\hat{\boldsymbol{b}}}^b_{\text{ars}} = -\boldsymbol{K}_i\,\boldsymbol{\omega}^b_{\text{mes}}$$

Nonlinear attitude observer with ARS bias correction (Mahony et al. 2008)

R. Mahony, T. Hamel and J.-M. Pflimlin (2008). Nonlinear Complementary Filters on the Special Orthogonal Group. IEEE Trans. on Aut. Control 53(5)

14.2.3 Nonlinear Attitude Observer using Reference Vectors

The MSS toolbox version of the Mahony et al. (2008) observer is a modified version using the **cross-product representation** of Grip et al. (2013).

$$\boldsymbol{\sigma} = \sum_{i=1}^{n} k_i \boldsymbol{v}_i^b \times \boldsymbol{R}^\top(\hat{\boldsymbol{q}}_b^n)\boldsymbol{v}_{0i}^n$$

$$\dot{\hat{\boldsymbol{q}}}_b^n = \boldsymbol{T}(\hat{\boldsymbol{q}}_b^n)\left(\boldsymbol{\omega}_{\text{imu}}^b - \hat{\boldsymbol{b}}_{\text{ars}}^b + \boldsymbol{\sigma}\right)$$

$$\dot{\hat{\boldsymbol{b}}}_{\text{ars}}^b = -\boldsymbol{K}_i\,\boldsymbol{\sigma}$$

Grip et al. (2013) shows how the nonlinear observer can be extended to terrestrial navigation by using reference vectors in the Earth-fixed Earth-centered reference frame {e}. Hence, the rotation and curvature of the Earth as well as the gravity vector are considered. This is not presented here.

where v^n_{0i} (i = 1, …, n) denote a set of n known inertial directions, approximated by $\{n\}$.

The reference vectors $\boldsymbol{v}_{0i}^n$ are expressed in {b} using the rotation matrix $\boldsymbol{R}(\boldsymbol{q}_b^n)$:

$$\boldsymbol{v}_i^b = \boldsymbol{R}^\top(\boldsymbol{q}_b^n)\boldsymbol{v}_{0i}^n \quad \text{True}$$

$$\hat{\boldsymbol{v}}_i^b = \boldsymbol{R}^\top(\hat{\boldsymbol{q}}_b^n)\boldsymbol{v}_{0i}^n \quad \text{Estimate}$$

We will now discuss four “reference vectors” (i = 1, 2, 3, 4) based on IMU and compass measurements.



Grip, H. F., T. I. Fossen, T. A. Johansen and A. Saberi (2013). Nonlinear Observer for GNSS-Aided Inertial Navigation with Quaternion-Based Attitude Estimation. American Control Conference (ACC), Washington DC, USA, IEEE Xplore, pp. 272-279.

14.2.3 Nonlinear Attitude Observer using Reference Vectors

Reference Vector #1 – IMU Specific Force Measurements

For the specific force measurements:

$$\boldsymbol{f}^b_{\text{imu}} = \boldsymbol{R}^\top(\boldsymbol{\Theta}_{nb})(\boldsymbol{a}^n_{nm_{\text{I}}} - \boldsymbol{g}^n) + \boldsymbol{b}^b_{\text{acc}} + \boldsymbol{w}^b_{\text{acc}}$$

we will assume that $\boldsymbol{a}^n_{nm_I}$ is small and that bias $\boldsymbol{b}^b_{acc}$ is compensated such that:

$$\boldsymbol{f}^b := \boldsymbol{f}^b_{\text{imu}} - \hat{\boldsymbol{b}}^b_{\text{acc}} = -\boldsymbol{R}^\top(\boldsymbol{\Theta}_{nb})\boldsymbol{g}^n$$



Reference vector (gravity) expressed in $\{n\}$

$$\boldsymbol{v}^n_{01} = [0,\, 0,\, -1]^\top$$

Normalized specific force vector expressed in $\{b\}$

$$\boldsymbol{v}^b_1 = \frac{\boldsymbol{f}^b}{\left\|\boldsymbol{f}^b\right\|}, \qquad \hat{\boldsymbol{v}}^b_1 = \boldsymbol{R}^\top(\hat{\boldsymbol{q}}^n_b)\begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}$$

14.2.3 Nonlinear Attitude Observer using Reference Vectors

Reference vector #2 – IMU Magnetic Field Measurements

For the magnetic field measurements

$$\boldsymbol{m}^b_{\text{mag}} = \boldsymbol{R}^\top(\boldsymbol{\Theta}_{nb})\boldsymbol{m}^n + \boldsymbol{b}^b_{\text{mag}} + \boldsymbol{w}^b_{\text{mag}}$$

We will assume that bias is compensated such that

$$\boldsymbol{m}^b := \boldsymbol{m}^b_{\text{mag}} - \hat{\boldsymbol{b}}^b_{\text{mag}}$$



Magnetometer measurements can experience bias from hard iron distortions, caused by permanent magnetic fields from nearby materials, and soft iron distortions, which result from materials that bend the Earth's magnetic field. Other sources of bias include electromagnetic interference (EMI) from electronic devices and temperature changes, which can affect the sensor's accuracy. Proper calibration and sensor fusion techniques are necessary to correct these biases and ensure reliable measurements.

Reference vector (magnetic field) expressed in {n}

$$\boldsymbol{v}^n_{02} = \boldsymbol{m}^n$$

The strength and direction of the magnetic field $\boldsymbol{m}^n$ can be assumed to be constant for a given location and period. The numerical values can be found by averaging the magnetometer measurements for a given period while keeping the roll and pitch angles close to zero.

Normalized magnetic field vector expressed in {b}

$$\boldsymbol{v}^b_2 = \frac{\boldsymbol{m}^b}{\|\boldsymbol{m}^b\|}, \qquad \hat{\boldsymbol{v}}^b_2 = \boldsymbol{R}^\top(\hat{\boldsymbol{q}}^n_b)\frac{\boldsymbol{m}^n}{\|\boldsymbol{m}^n\|}$$

14.2.3 Nonlinear Attitude Observer using Reference Vectors

Reference Vector #3 – Compass Measurements (Small Roll-Pitch Angle Assumption)

For marine craft, an alternative to the three-axis magnetometer is using a gyrocompass or a magnetic compass as reference vector. If the roll and pitch angles are small it follows that $\boldsymbol{R}_b^n \approx \boldsymbol{R}_{z,\psi}$. Hence, the Northwards unit vector:

$$\boldsymbol{e}_3^n = [1,\, 0,\, 0]^\top$$

expressed in BODY becomes:

$$\boldsymbol{e}_3^b \approx \boldsymbol{R}_{z,\psi}^\top \boldsymbol{e}_3^n = \begin{bmatrix} \cos(\psi) \\ -\sin(\psi) \\ 0 \end{bmatrix}$$

The assumption $\boldsymbol{R}_b^n \approx \boldsymbol{R}_{z,\psi}$ only holds for small roll and pitch angles. Hence, it is recommended to use reference vector #4 on the next slide for compass measurements.

$$\psi = \psi_m + \delta$$



Reference vector (compass) expressed in $\{n\}$

$$\boldsymbol{v}_{03}^n = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Normalized compass measurement expressed in $\{b\}$

$$\boldsymbol{v}_3^b = \begin{bmatrix} \cos(\psi) \\ -\sin(\psi) \\ 0 \end{bmatrix}, \qquad \hat{\boldsymbol{v}}_3^b = \boldsymbol{R}^\top(\hat{\boldsymbol{q}}_b^n) \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

14.2.3 Nonlinear Attitude Observer using Reference Vectors

Reference Vector #4 – Compass Measurements (BODY-Forward Direction in NED)

The roll-pitch small angle assumption of reference vector #3 can be removed by using ψ to define the BODY-forward direction in NED, and then compare it with the trivial forward axis in BODY

Reference vector (compass) expressed in $\{n\}$

$$\mathbf{v}_{04}^{n} = \begin{bmatrix} \cos\psi \\ \sin\psi \\ 0 \end{bmatrix}$$

Normalized measurement expressed in $\{b\}$

$$\mathbf{v}_{4}^{b} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

By using ψ to define the **BODY-forward direction in NED** the reference vector is valid for all roll-pitch-yaw angles

$$\psi = \psi_m + \delta$$



NB! Other reference vectors can be deduced from star sensors, lidars, radars, optical flow etc. This will not be discussed.

MSS Toolbox: High-Rate Nonlinear Quaternion-Based Attitude Observer Implementation

```
function [quat, b_ars] = quatObserver(quat, b_ars, h, Ki, k1, k2, m_ref, imu_meas)
```

```
% Transposed unit quaternion rotation matrix: R_transposed[k]
R_transposed = Rquat(quat)';

% High-rate IMU specific force and ARS measurements: f_imu[k] and w_imu[k]
f_imu = imu_meas(1:3)';
w_imu = imu_meas(4:6)';

% Nonlinear injection terms (low rate)
if length(imu_meas) == 9
    % IMU measuremenst with 3-axis magnetometer measurement: [f_imu' w_imu' m_imu']
    m_imu = imu_meas(7:9)'; % Magnetic field IMU measurements: m_imu[k] in BODY
    v01 = [0 0 -1]'; % Normalized gravity reference vector in NED (measuring -g at rest
    v1 = f_imu / norm(f_imu); % Normalized specific force measurement
    v02 = m_ref / norm(m_ref); % Normalized magnetic field reference vector in NED
    v2 =  m_imu / norm(m_imu); % Normalized magnetic field measurement in BODY
    sigma1 = k1 * cross(v1, R_transposed * v01);
    sigma2 = k2 * cross(v2, R_transposed * v02);
    sigma = sigma1 + sigma2;

elseif length(imu_meas) == 7
    % IMU measurements with scalar compass measurement: [f_imu' w_imu' psi]
    psi = imu_meas(7);  % Compass measurement: psi[k]
    v01 = [0 0 -1]'; % Normalized gravity reference vector in NED (measuring -g at rest
    v1 = f_imu / norm(f_imu); % Normalized specific force measurement

    v02 = [cos(psi);  sin(psi); 0]; % Normalized heading reference vector in NED
    v2  = [1; 0; 0]; % Normalized heading measurement vector in BODY
    sigma1 = k1 * cross(v1, R_transposed * v01);
    sigma2 = k2 * cross(v2, R_transposed * v02);
    sigma = sigma1 + sigma2;

else
    % No correction
    sigma = zeros(3,1);
end
```

```
9-DOF measurements:
   [quat, b_ars] = quatObserver(quat, b_ars, h, Ki, k1, k2, m_ref, ...
       [f_imu', w_imu', m_imu'])
7-DOF measurements:
   [quat, b_ars] = quatObserver(quat, b_ars, h, Ki,pen k1, k2, m_ref, ...
       [f_imu', w_imu', psi])
6-DOF measurements (no magnetometer/compass measurements):
   [quat, b_ars] = quatObserver(quat, b_ars, h, Ki, k1, k2, m_ref, ...
       [f_imu', w_imu'])
```

```
% State propagation: quat[k+1] is computed using the matrix exponential,
% which serves as the exponential map for matrix Lie groups, ensuring exact
% discretization with 1st-order hold of the quaternion differential equation:
%   quat_dot = Tquat(w_imu - b_ars + sigma) * quat
% You can replace the built-in Matlab function expm.m with the custom-made
% MSS functions expm_taylor.m or expm_squaresPade.m for this computation.

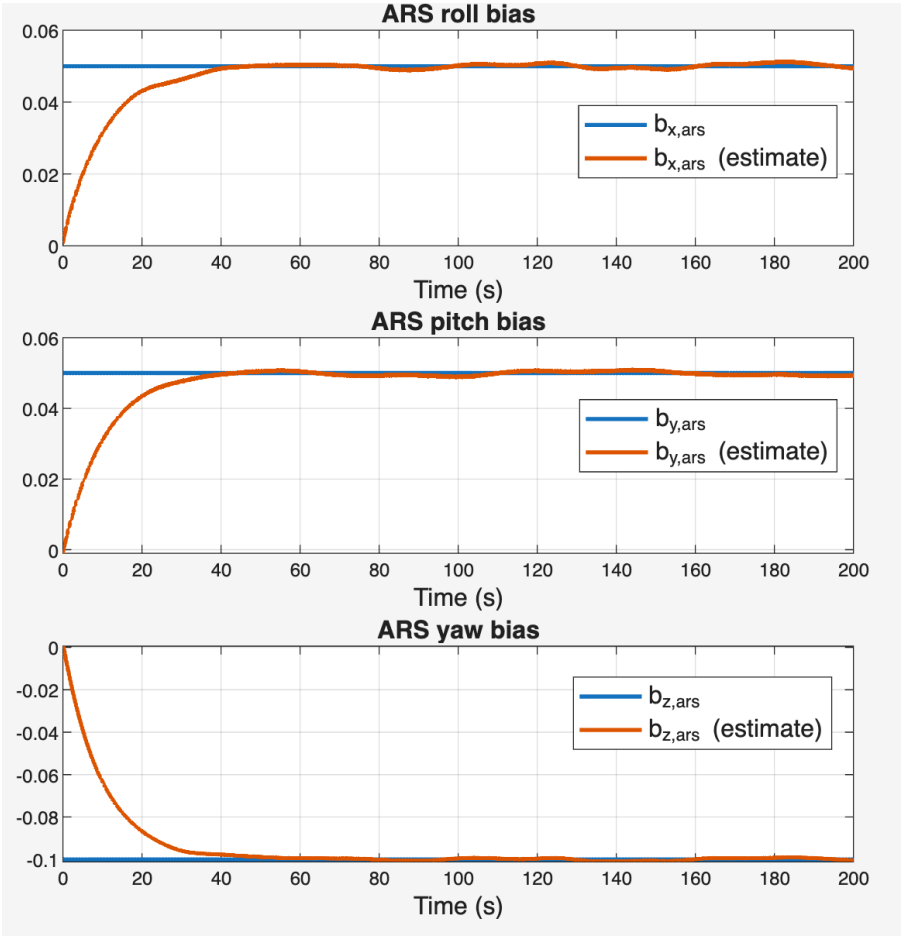
% Angular velocity with bias compensation 'b_ars' and injection term 'sigma'
w_estimated = w_imu - b_ars + sigma;

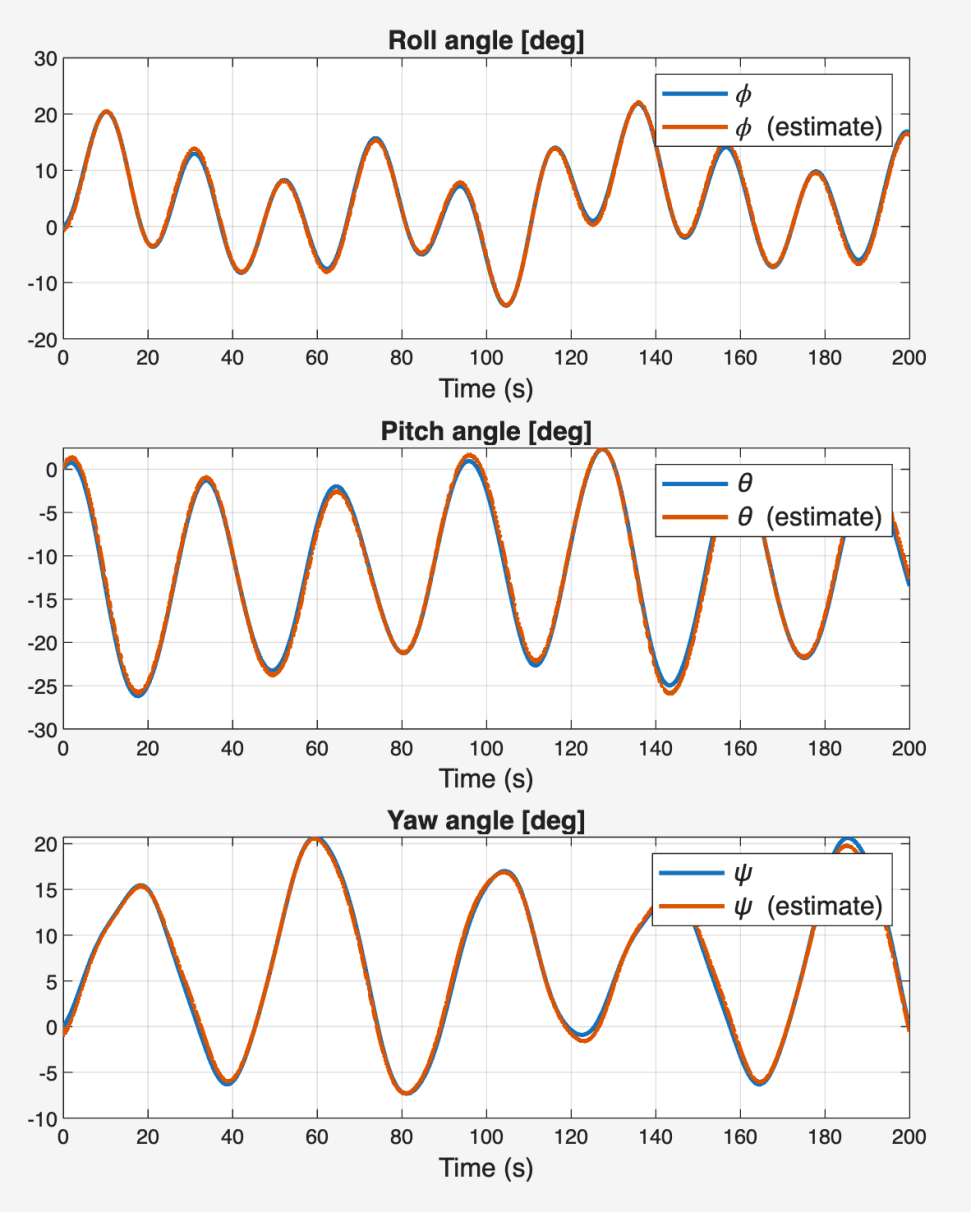
% Exponential map with a 1st-order hold on w (exact if w constant on the step)
quat = expm( Tquat(w_estimated) * h ) * quat;
quat = quat / norm(quat);  % Normalization

% State propagation: b_ars[k+1]
b_ars = b_ars - h * Ki * sigma; % Attitude rate sensor (ARS) bias
```

MSS Toolbox: SIMquatObserver.m

```
>> SIMquatObserver
-------------------------------------------------------------------------
MSS toolbox: Nonlinear quaternion-based attitude observer
IMU inertial measurements (specific force and ARS) at 1000 Hz
IMU magnetic field/compass measurements at 100 Hz
Magnetic field reference vector for Tokyo (>> type magneticField)
-------------------------------------------------------------------------
Simulating...
```





14.2.3 Nonlinear Attitude Observer using Reference Vectors

Experimental Testing







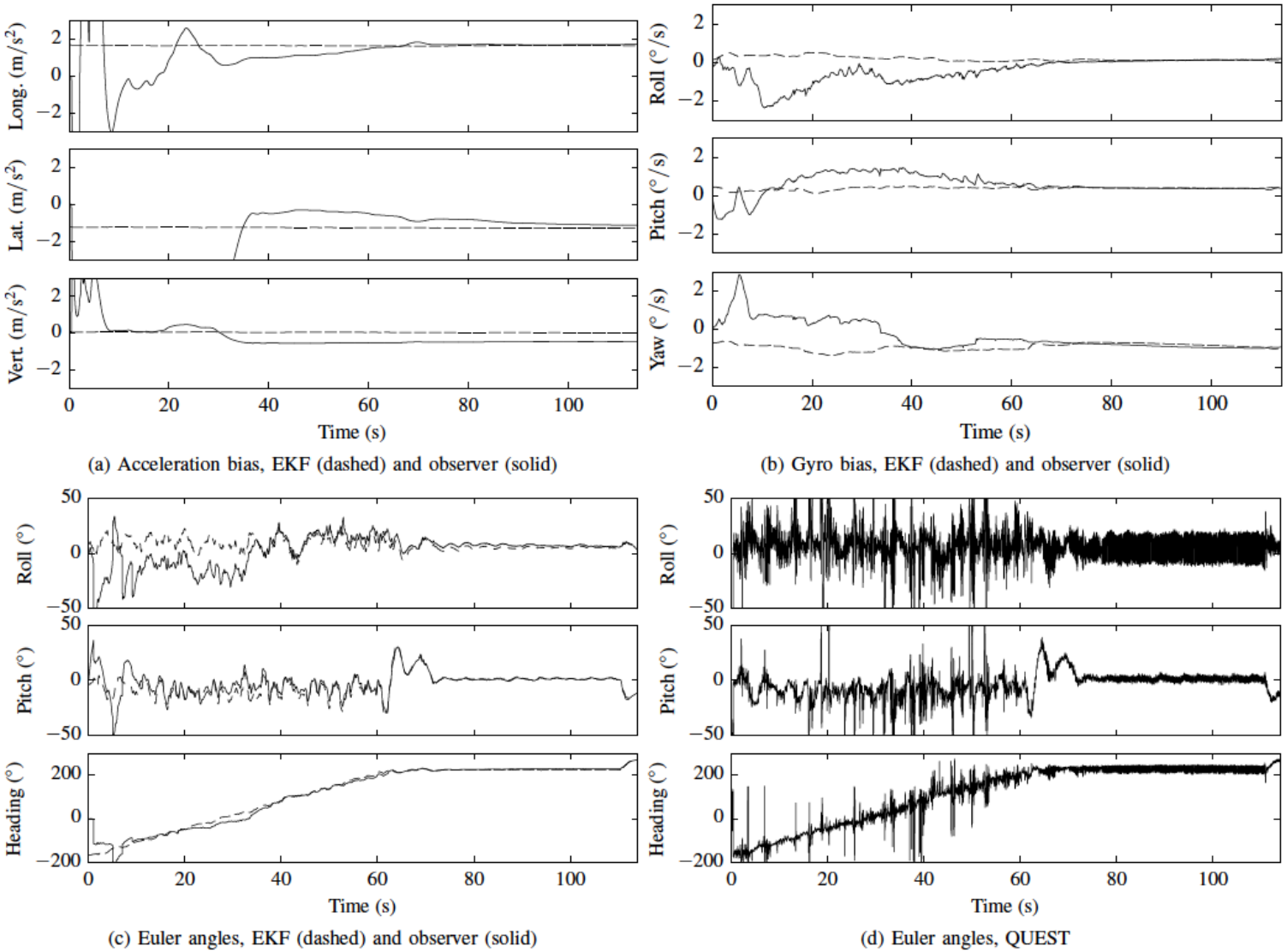
The *Adaptive Flight Hornet Inc*. mini helicopter equipped with the FCS-20 autopilot system. Autopilot system includes IMU, GPS and EKF for navigation.

IMU: accelerometers and gyros (100 Hz)
magnetometers (10 Hz)
GPS: position and velocity (5 Hz**)**

Acceleration reference vector is found by differentiating the GPS position twice (only for demonstration)



14.2.3 Nonlinear Attitude Observer using Reference Vectors



(a) Acceleration bias, EKF (dashed) and observer (solid)

(b) Gyro bias, EKF (dashed) and observer (solid)

(c) Euler angles, EKF (dashed) and observer (solid)

(d) Euler angles, QUEST

H. F. Grip, T. I. Fossen, T. A. Johansen and A. Saberi (2012). Attitude estimation using biased gyro and vector measurements with time-varying reference vectors. IEEE Transactions Automatic Control.

14.2.3 Nonlinear Attitude Observer using Reference Vectors

Experimental testing using the Piper Cherokee 140 light fixed-wing aircraft





Håvard – the pilot

The Piper Cherokee is equipped with:

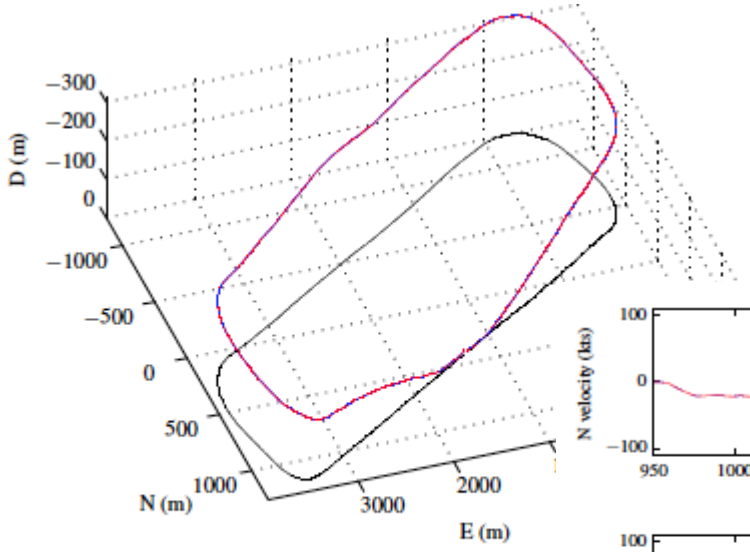
- **IMU:** XSens Mti, accelrometers, gyros and magnetometers (100 Hz)
- **GNSS:** u-Blox LEA-6H, position and Doppler-based velocity measurements (5 Hz**)**





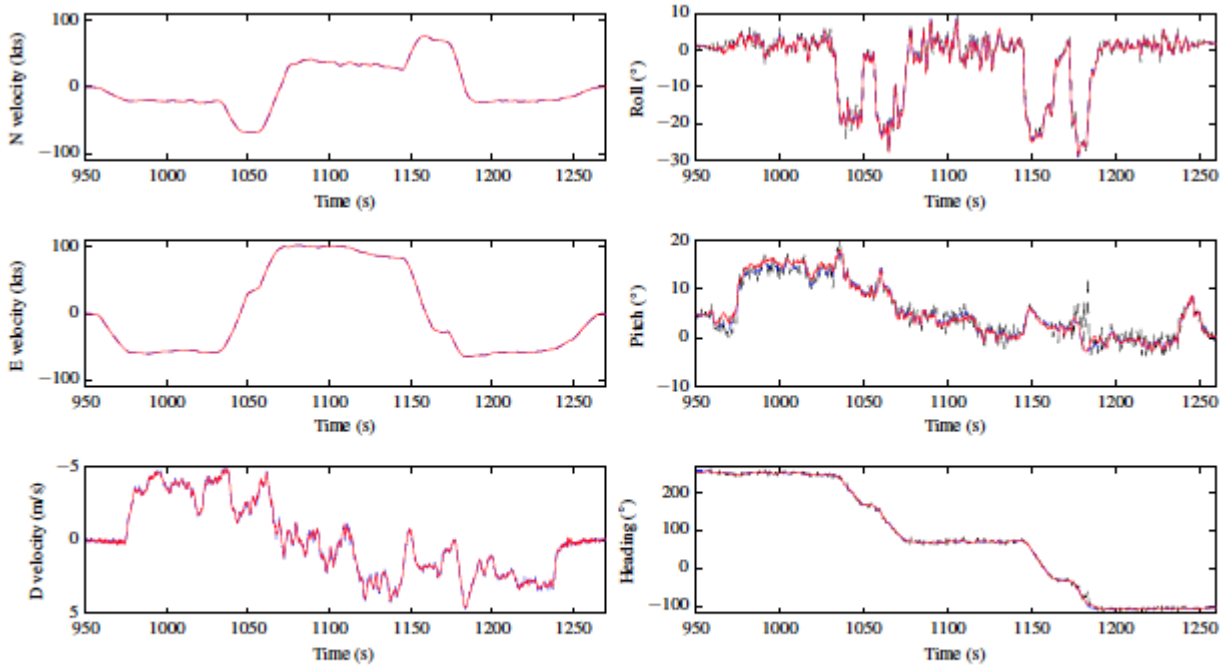
Mounted on a bulkhead within the tail of the aircraft. The GNSS antenna is mounted on top of the instrument panel.

14.2.3 Nonlinear Attitude Observer using Reference Vectors



Left-handed traffic pattern flown from takeoff to landing

Traffic Pattern Maneuver



(b) Measured (blue, dashed) and estimated (red, solid) velocity (c) QUEST (black, dashed), MEKF (blue, dashed), and observer (red, solid) attitude estimates

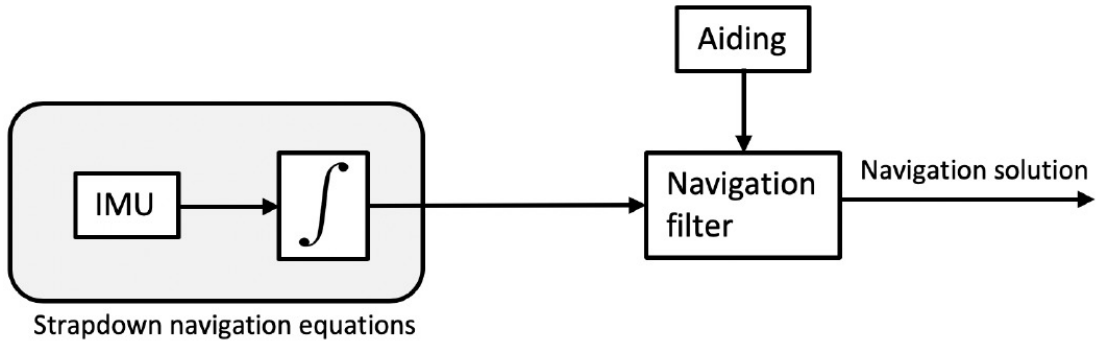
H. F. Grip, T. I. Fossen, T. A. Johansen and A. Saberi (2015). Globally Exponentially Stable Attitude and Gyro Bias Estimation with Application to GNSS/INS Integration, *Automatica.* Volume 51, January 2015, Pages 158–166.

14.3 Direct Filters for Aided INS

Direct filter: The filter acts directly on raw navigation outputs — for example, position or velocity estimates from the INS are smoothed by filtering the measured signals themselves.
Tools: basic Kalman Filter (KF) or Extended Kalman Filter (EKF) applied to raw states.

Indirect filter: The filter operates on the errors between the INS and aiding sensors (GNSS, magnetometer, etc.) rather than on the raw outputs. It estimates and corrects INS state errors and biases without re-integrating the full navigation equations.
Tool: Error-State Kalman Filter (ESKF) applied to the error states

14.3 Direct Filters for Aided INS



In a direct filter design, the marine craft position and velocity are states in the estimator, while specific force and angular rate measurements are used as inputs to the strapdown navigation equations.

It is advantageous to use a signal-based KF based on the **strapdown navigation equations** expressed in $\{n\}$, which is an approximative inertial frame. Kinematic equations are not prone to parameter uncertainty since they are geometrical descriptions (perfect models).

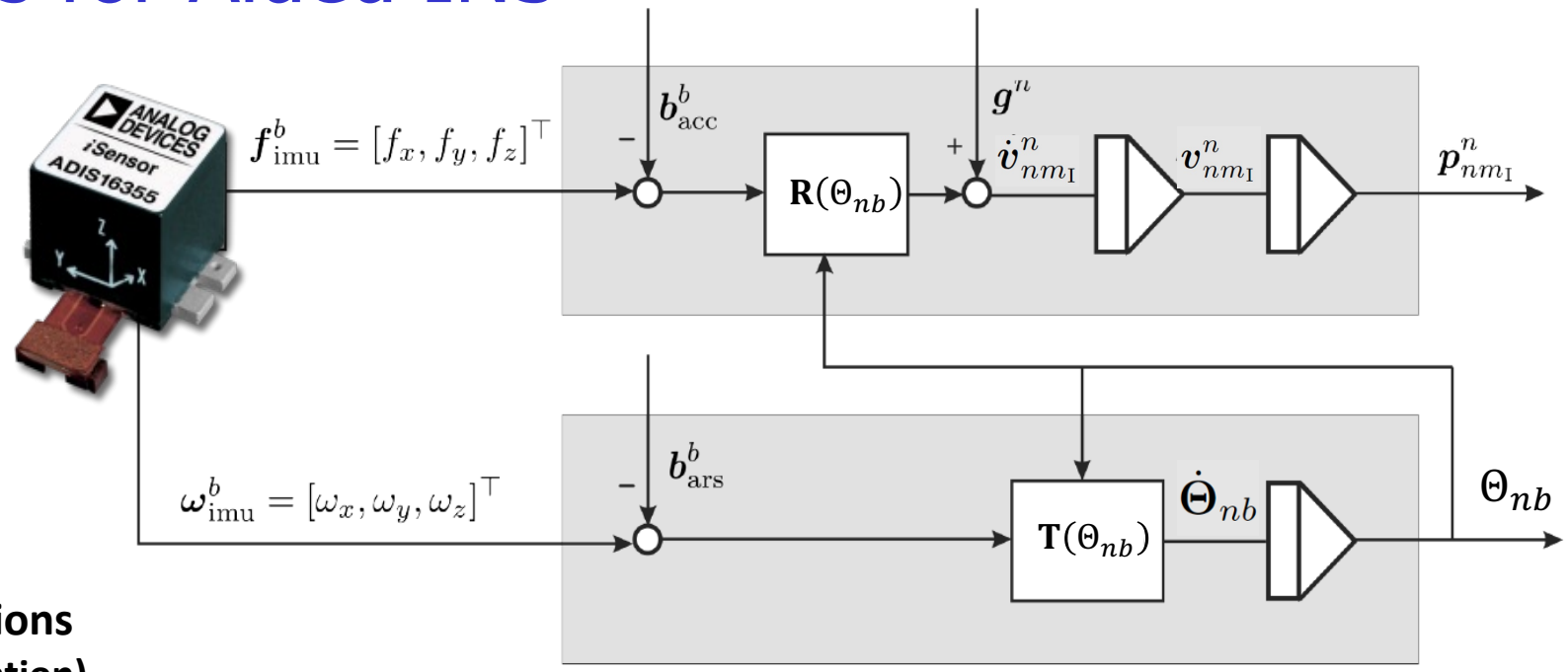
$$
\begin{aligned}
\dot{\boldsymbol{p}}^n_{nm_\mathrm{I}} &= \boldsymbol{v}^n_{nm_\mathrm{I}} \\
\dot{\boldsymbol{v}}^n_{nm_\mathrm{I}} &= \boldsymbol{R}(\boldsymbol{\Theta}_{nb})(\boldsymbol{f}^b_{\mathrm{imu}} - \boldsymbol{b}^b_{\mathrm{acc}} - \boldsymbol{w}^b_{\mathrm{acc}}) + \boldsymbol{g}^n \\
\dot{\boldsymbol{b}}^b_{\mathrm{acc}} &= \boldsymbol{w}^b_{b,\,\mathrm{acc}} \\
\dot{\boldsymbol{\Theta}}_{nb} &= \boldsymbol{T}(\boldsymbol{\Theta}_{nb})(\boldsymbol{\omega}^b_{\mathrm{imu}} - \boldsymbol{b}^b_{\mathrm{ars}} - \boldsymbol{w}^b_{\mathrm{ars}}) \\
\dot{\boldsymbol{b}}^b_{\mathrm{ars}} &= \boldsymbol{w}^b_{b,\,\mathrm{ars}}
\end{aligned}
$$

The inputs to the "Strapdown Navigation Equations" are the specific force and angular rate vectors

$$\boldsymbol{f}^b_{\mathrm{imu}} := \boldsymbol{f}^b_{nm_\mathrm{I}} \qquad \boldsymbol{\omega}^b_{\mathrm{imu}} := \boldsymbol{\omega}^b_{nb}$$

The IMU is assumed mounted onboard the craft in a measurement frame $\{m_\mathrm{I}\}$ with distance vector $\boldsymbol{r}^b_{bm_\mathrm{I}} = [x_{m_\mathrm{I}},\, y_{m_\mathrm{I}},\, z_{m_\mathrm{I}}]^\top$ w.r.t. the CO.

14.3 Direct Filters for Aided INS



Strapdown Navigation Equations
(15-state Euler-Angle Representation)

$$
\begin{aligned}
\dot{\boldsymbol{p}}^n_{nm_\text{I}} &= \boldsymbol{v}^n_{nm_\text{I}} \\
\dot{\boldsymbol{v}}^n_{nm_\text{I}} &= \boldsymbol{R}(\boldsymbol{\Theta}_{nb})(\boldsymbol{f}^b_{\text{imu}} - \boldsymbol{b}^b_{\text{acc}} - \boldsymbol{w}^b_{\text{acc}}) + \boldsymbol{g}^n \\
\dot{\boldsymbol{b}}^b_{\text{acc}} &= \boldsymbol{w}^b_{b,\,\text{acc}} \\
\dot{\boldsymbol{\Theta}}_{nb} &= \boldsymbol{T}(\boldsymbol{\Theta}_{nb})(\boldsymbol{\omega}^b_{\text{imu}} - \boldsymbol{b}^b_{\text{ars}} - \boldsymbol{w}^b_{\text{ars}}) \\
\dot{\boldsymbol{b}}^b_{\text{ars}} &= \boldsymbol{w}^b_{b,\,\text{ars}}
\end{aligned}
$$

The figure shows the nonlinear coupling between the translational and rotational dynamics.

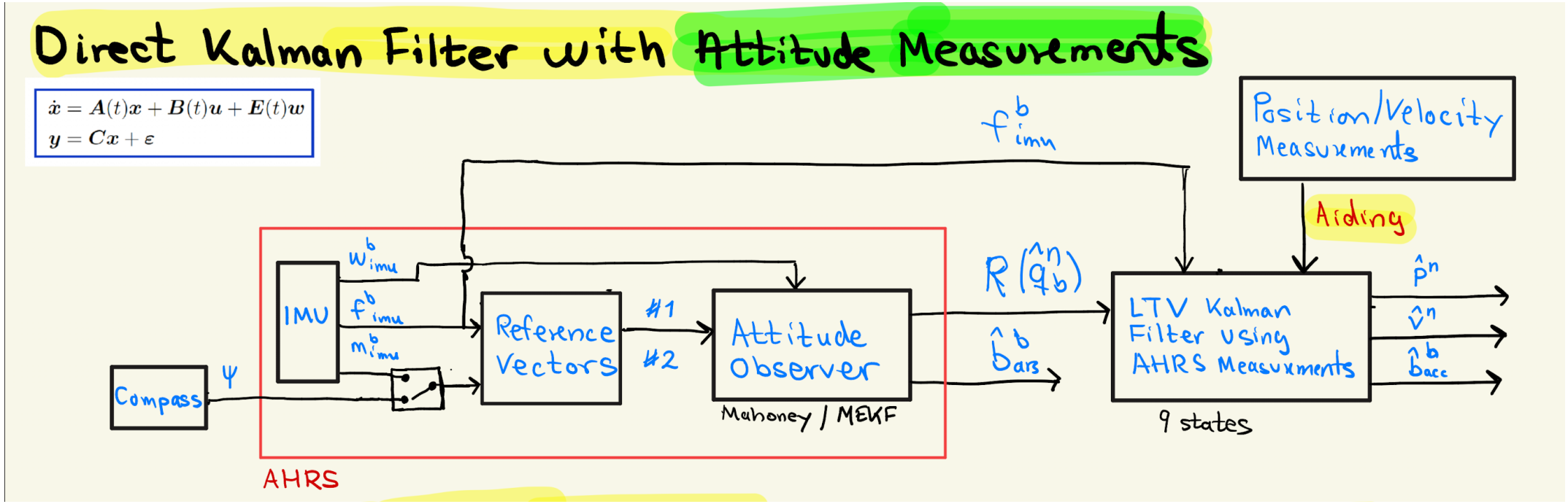
NB! If the transformations matrices:

$$\boldsymbol{R}(\boldsymbol{\Theta}_{nb}) := \boldsymbol{R}^n_b(t), \qquad \boldsymbol{T}(\boldsymbol{\Theta}_{nb}) := \boldsymbol{T}^n_b(t)$$

are measured using an AHRS, the dynamics is decoupled, and the resulting state-space model is an LTV system.

The use of attitude measurements simplifies the estimation problem; this assumption will be relaxed later.

Direct Kalman Filters/Fixed-Gain Observers with Attitude Measurements



14.3.1 Fixed-Gain Observer with Attitude Measurements

Assume that an AHRS measures:

$$\boldsymbol{R}(\boldsymbol{\Theta}_{nb}) := \boldsymbol{R}_b^n(t), \qquad \boldsymbol{T}(\boldsymbol{\Theta}_{nb}) := \boldsymbol{T}_b^n(t)$$



AHRS

Hence, the attitude is known, and the 9-state translational motion with respect to the IMU mounting point CM_I becomes:

$$\begin{aligned}
\dot{\boldsymbol{p}}_{nm_\mathrm{I}}^n &= \boldsymbol{v}_{nm_\mathrm{I}}^n \\
\dot{\boldsymbol{v}}_{nm_\mathrm{I}}^n &= \boldsymbol{R}_b^n(t)(\boldsymbol{f}_{\mathrm{imu}}^b - \boldsymbol{b}_{\mathrm{acc}}^b) + \boldsymbol{g}^n \\
\dot{\boldsymbol{b}}_{\mathrm{acc}}^b &= \boldsymbol{0} \\
\boldsymbol{y}_1 &= \boldsymbol{p}_{nm_\mathrm{I}}^n
\end{aligned}$$

Lever arm compensated GNSS position measurement

$$\boldsymbol{p}_{nm_\mathrm{I}}^n = \boldsymbol{p}_{\mathrm{gnss}}^n + \boldsymbol{R}_b^n(t)(\boldsymbol{r}_{bm_\mathrm{I}}^b - \boldsymbol{r}_{bm_\mathrm{G}}^b)$$

Fixed-gain observer copying the system dynamics

$$\begin{aligned}
\dot{\hat{\boldsymbol{p}}}_{nm_\mathrm{I}}^n &= \hat{\boldsymbol{v}}_{nm_\mathrm{I}}^n + \boldsymbol{K}_1\tilde{\boldsymbol{y}}_1 \\
\dot{\hat{\boldsymbol{v}}}_{nm_\mathrm{I}}^n &= \boldsymbol{R}_b^n(t)(\boldsymbol{f}_{\mathrm{imu}}^b - \hat{\boldsymbol{b}}_{\mathrm{acc}}^b) + \boldsymbol{g}^n + \boldsymbol{K}_2\tilde{\boldsymbol{y}}_1 \\
\dot{\hat{\boldsymbol{b}}}_{\mathrm{acc}}^b &= \boldsymbol{K}_3\boldsymbol{R}_b^n(t)^\top\tilde{\boldsymbol{y}}_1 \\
\hat{\boldsymbol{y}}_1 &= \hat{\boldsymbol{p}}_{nm_\mathrm{I}}^n
\end{aligned}$$

Transform the estimates from the CM_I to the CO

$$\begin{aligned}
\hat{\boldsymbol{p}}_{nb}^n &= \hat{\boldsymbol{p}}_{nm_\mathrm{I}}^n + \boldsymbol{r}_{m_\mathrm{I}b}^n \\
&= \hat{\boldsymbol{p}}_{nm_\mathrm{I}}^n - \boldsymbol{R}_b^n(t)\boldsymbol{r}_{bm_\mathrm{I}}^b \\
\hat{\boldsymbol{v}}_{nb}^b &= \hat{\boldsymbol{v}}_{nm_\mathrm{I}}^b + \boldsymbol{\omega}_{nm_\mathrm{I}}^b \times \boldsymbol{r}_{m_\mathrm{I}b}^b \\
&= \boldsymbol{R}_b^n(t)^\top\hat{\boldsymbol{v}}_{nm_\mathrm{I}}^n - \boldsymbol{\omega}_{nb}^b \times \boldsymbol{r}_{bm_\mathrm{I}}^b
\end{aligned}$$

How to compute the observer gains K_i (i = 1, 2, 3) such that the equilibrium point is exponentially stable?

14.3.1 Fixed-Gain Observer with Attitude Measurements

Estimation error dynamics

$$\begin{bmatrix} \dot{\tilde{\boldsymbol{p}}}^n_{nm_\mathrm{I}} \\ \dot{\tilde{\boldsymbol{v}}}^n_{nm_\mathrm{I}} \\ \dot{\tilde{\boldsymbol{b}}}^b_{\mathrm{acc}} \end{bmatrix} = \begin{bmatrix} -\boldsymbol{K}_1 & \boldsymbol{I}_3 & \boldsymbol{0}_{3\times 3} \\ -\boldsymbol{K}_2 & \boldsymbol{0}_{3\times 3} & -\boldsymbol{R}^n_b(t) \\ -\boldsymbol{K}_3 \boldsymbol{R}^n_b(t)^\top & \boldsymbol{0}_{3\times 3} & \boldsymbol{0}_{3\times 3} \end{bmatrix} \begin{bmatrix} \tilde{\boldsymbol{p}}^n_{nm_\mathrm{I}} \\ \tilde{\boldsymbol{v}}^n_{nm_\mathrm{I}} \\ \tilde{\boldsymbol{b}}^b_{\mathrm{acc}} \end{bmatrix}$$

$$\Updownarrow$$

$$\dot{\boldsymbol{x}} = \boldsymbol{A}(t)\boldsymbol{x}$$

Change of coordinates

$$\boldsymbol{x} = \boldsymbol{M}(t)\boldsymbol{z} \qquad \boldsymbol{M}(t) = \mathrm{diag}\{\boldsymbol{R}^n_b(t),\, \boldsymbol{R}^n_b(t),\, \boldsymbol{I}_3\}$$

$$\boldsymbol{M}^{-1}(t) = \boldsymbol{M}^\top(t)$$

Assume that the angular velocity $\boldsymbol{\omega}^b_{nb}$ is small such that

$$\dot{\boldsymbol{M}}(t) = \boldsymbol{0}$$

Stability analysis

$$\dot{\boldsymbol{z}} = \boldsymbol{F}\boldsymbol{z} \qquad \boldsymbol{F} = \boldsymbol{M}^\top(t)\boldsymbol{A}(t)\boldsymbol{M}(t)$$

$$\boldsymbol{F} = \begin{bmatrix} -\boldsymbol{K}_1 & \boldsymbol{I}_3 & \boldsymbol{0}_{3\times 3} \\ -\boldsymbol{K}_2 & \boldsymbol{0}_{3\times 3} & -\boldsymbol{I}_3 \\ -\boldsymbol{K}_3 & \boldsymbol{0}_{3\times 3} & \boldsymbol{0}_{3\times 3} \end{bmatrix}$$

Property 14.1 (Commuting Matrices)
A matrix $\boldsymbol{K}(t) \in \mathbb{R}^{3\times 3}$ *is said to* commute *with the rotation matrix* $\boldsymbol{R}^n_b(t)$ *if*

$$\boldsymbol{K}(t)\boldsymbol{R}^n_b(t) = \boldsymbol{R}^n_b(t)\boldsymbol{K}(t)$$

We can choose the commuting gain matrix as

$$\boldsymbol{K}_i = \mathrm{diag}\{k_i,\, k_i,\, l_i\}, \quad i = 1, 2, 3$$

where surge and sway have the same gain constants $k_i > 0$ and heave can be tuned independently by the constant $l_i > 0$.

Finally, stability can be checked by computing the eigenvalues of $\boldsymbol{F}$.
A necessary condition for exponential stability is that the eigenvalues of $\boldsymbol{F}$ lie in the left half-plane, that is $\boldsymbol{F}$ must be Hurwitz.

14.3.2 Direct Kalman Filter with Attitude Measurements

Assume that the transformations matrices

$$\boldsymbol{R}(\boldsymbol{\Theta}_{nb}) := \boldsymbol{R}_b^n(t), \qquad \boldsymbol{T}(\boldsymbol{\Theta}_{nb}) := \boldsymbol{T}_b^n(t)$$

are known (the Euler angles are measured using an AHRS)

Strapdown Navigation Equations

$$\begin{aligned}
\dot{\boldsymbol{p}}_{nm_\mathrm{I}}^n &= \boldsymbol{v}_{nm_\mathrm{I}}^n \\
\dot{\boldsymbol{v}}_{nm_\mathrm{I}}^n &= \boldsymbol{R}_b^n(t)(\boldsymbol{f}_{\mathrm{imu}}^b - \boldsymbol{b}_{\mathrm{acc}}^b - \boldsymbol{w}_{\mathrm{acc}}^b) + \boldsymbol{g}^n \\
\dot{\boldsymbol{b}}_{\mathrm{acc}}^b &= \boldsymbol{w}_{b,\,\mathrm{acc}}^b \\
\dot{\boldsymbol{\Theta}}_{nb} &= \boldsymbol{T}_b^n(t)(\boldsymbol{\omega}_{\mathrm{imu}}^b - \boldsymbol{b}_{\mathrm{ars}}^b - \boldsymbol{w}_{\mathrm{ars}}^b) \\
\dot{\boldsymbol{b}}_{\mathrm{ars}}^b &= \boldsymbol{w}_{b,\,\mathrm{ars}}^b
\end{aligned}$$

15-State LTV Kalman Filter Model using AHRS Measurements

$$\begin{aligned}
\dot{\boldsymbol{x}} &= \boldsymbol{A}(t)\boldsymbol{x} + \boldsymbol{B}(t)\boldsymbol{u} + \boldsymbol{E}(t)\boldsymbol{w} \\
\boldsymbol{y} &= \boldsymbol{C}\boldsymbol{x} + \boldsymbol{\varepsilon}
\end{aligned}$$

The attitude measurements are represented as time-varying matrices $\boldsymbol{R}_b^n(t)$ and $\boldsymbol{T}_b^n(t)$ in $\boldsymbol{A}(t)$ and $\boldsymbol{E}(t)$

$$\boldsymbol{x} = [(\boldsymbol{p}_{nm_\mathrm{I}}^n)^\top, (\boldsymbol{v}_{nm_\mathrm{I}}^n)^\top, (\boldsymbol{b}_{\mathrm{acc}}^b)^\top, \boldsymbol{\Theta}_{nb}^\top, (\boldsymbol{b}_{\mathrm{ars}}^b)^\top]^\top$$

$$\boldsymbol{w} = [(\boldsymbol{w}_{\mathrm{acc}}^b)^\top, (\boldsymbol{w}_{\mathrm{b,\,acc}}^b)^\top, (\boldsymbol{w}_{\mathrm{ars}}^b)^\top, (\boldsymbol{w}_{b,\,\mathrm{ars}}^b)^\top]^\top$$

$$\boldsymbol{A}(t) = \begin{bmatrix}
\boldsymbol{0}_{3\times3} & \boldsymbol{I}_3 & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} \\
\boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & -\boldsymbol{R}_b^n(t) & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} \\
\boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} \\
\boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & -\boldsymbol{T}_b^n(t) \\
\boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3}
\end{bmatrix}, \quad \boldsymbol{B}(t) = \begin{bmatrix}
\boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} \\
\boldsymbol{R}_b^n(t) & \boldsymbol{0}_{3\times3} & \boldsymbol{I}_3 \\
\boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} \\
\boldsymbol{0}_{3\times3} & \boldsymbol{T}_b^n(t) & \boldsymbol{0}_{3\times3} \\
\boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3}
\end{bmatrix}$$

$$\boldsymbol{C} = \begin{bmatrix}
\boldsymbol{I}_3 & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} \\
\boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{I}_3 & \boldsymbol{0}_{3\times3}
\end{bmatrix}, \quad \boldsymbol{E}(t) = \begin{bmatrix}
\boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} \\
-\boldsymbol{R}_b^n(t) & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} \\
\boldsymbol{0}_{3\times3} & \boldsymbol{I}_3 & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} \\
\boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & -\boldsymbol{T}_b^n(t) & \boldsymbol{0}_{3\times3} \\
\boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{I}_3
\end{bmatrix}$$

The input vector contains specific force, angular rates and gravity →

Inputs

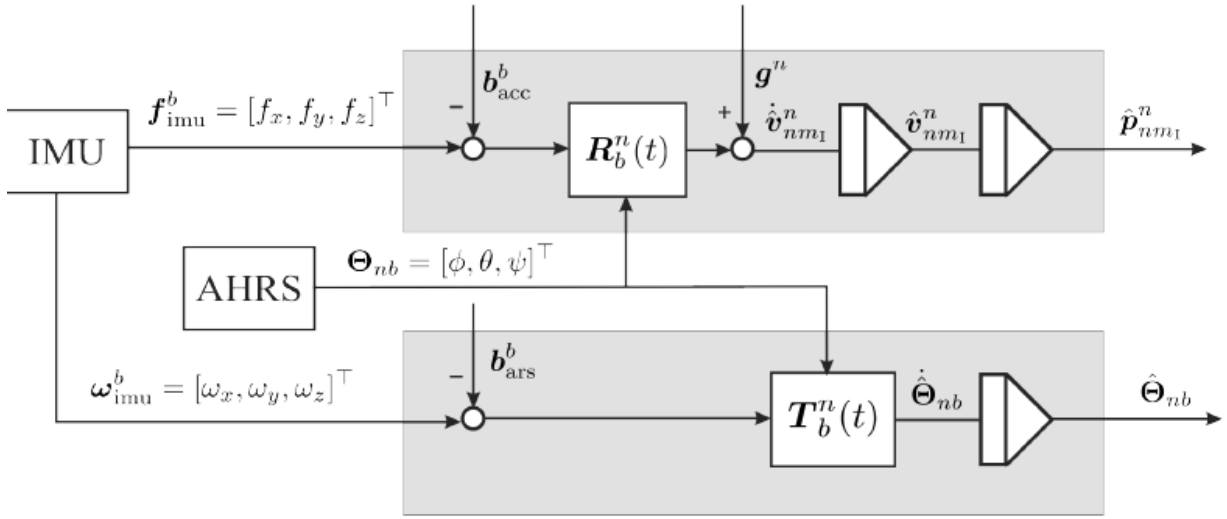
$$\boldsymbol{u} = [(\boldsymbol{f}_{\mathrm{imu}}^b)^\top, (\boldsymbol{\omega}_{\mathrm{imu}}^b)^\top, (\boldsymbol{g}^n)^\top]^\top$$

Measurements

$$\boldsymbol{y} = [(\boldsymbol{p}_{nm_\mathrm{I}}^n)^\top, \boldsymbol{\Theta}_{nb}^\top]^\top$$

It can be shown that the linear discrete-time KF (Table 13.1) will estimate all 15 states thanks to observability (see proof in the textbook).

14.3.2 Observability Analysis of the Direct Kalman Filter/ Fixed-Gain Observer



$$
\begin{aligned}
\dot{p}^n_{nm_I} &= v^n_{nm_I} \\
\dot{v}^n_{nm_I} &= R^n_b(t)(f^b_{\text{imu}} - b^b_{\text{acc}} - w^b_{\text{acc}}) + g^n \\
\dot{b}^b_{\text{acc}} &= w^b_{b,\,\text{acc}} \\
\dot{\Theta}_{nb} &= T^n_b(t)(\omega^b_{\text{imu}} - b^b_{\text{ars}} - w^b_{\text{ars}}) \\
\dot{b}^b_{\text{ars}} &= w^b_{b,\,\text{ars}}
\end{aligned}
$$

Because of the AHRS measurement, we have two decoupled systems for attitude and translational motions

$$
\begin{aligned}
\dot{x} &= A(t)x + B(t)u + E(t)w \\
y &= Cx + \varepsilon
\end{aligned}
$$

$$
A(t) = \begin{bmatrix}
0_{3\times3} & I_3 & 0_{3\times3} & 0_{3\times3} & 0_{3\times3} \\
0_{3\times3} & 0_{3\times3} & -R^n_b(t) & 0_{3\times3} & 0_{3\times3} \\
0_{3\times3} & 0_{3\times3} & 0_{3\times3} & 0_{3\times3} & 0_{3\times3} \\
0_{3\times3} & 0_{3\times3} & 0_{3\times3} & 0_{3\times3} & -T^n_b(t) \\
0_{3\times3} & 0_{3\times3} & 0_{3\times3} & 0_{3\times3} & 0_{3\times3}
\end{bmatrix}, \quad
B(t) = \begin{bmatrix}
0_{3\times3} & 0_{3\times3} & 0_{3\times3} \\
R^n_b(t) & 0_{3\times3} & I_3 \\
0_{3\times3} & 0_{3\times3} & 0_{3\times3} \\
0_{3\times3} & T^n_b(t) & 0_{3\times3} \\
0_{3\times3} & 0_{3\times3} & 0_{3\times3}
\end{bmatrix}
$$

$$
C = \begin{bmatrix}
I_3 & 0_{3\times3} & 0_{3\times3} & 0_{3\times3} & 0_{3\times3} \\
0_{3\times3} & 0_{3\times3} & 0_{3\times3} & I_3 & 0_{3\times3}
\end{bmatrix}, \quad
E(t) = \begin{bmatrix}
0_{3\times3} & 0_{3\times3} & 0_{3\times3} & 0_{3\times3} \\
-R^n_b(t) & 0_{3\times3} & 0_{3\times3} & 0_{3\times3} \\
0_{3\times3} & I_3 & 0_{3\times3} & 0_{3\times3} \\
0_{3\times3} & 0_{3\times3} & -T^n_b(t) & 0_{3\times3} \\
0_{3\times3} & 0_{3\times3} & 0_{3\times3} & I_3
\end{bmatrix}
$$

14.3.2 Observability Analysis of the Direct Kalman Filter/ Fixed-Gain Observer

The pair $(\boldsymbol{A}(t), \boldsymbol{C}(t))$ of a LTV system of dimension n is observable if Chen (2013, Theorem 6.012).

$$\operatorname{rank} \begin{bmatrix} \boldsymbol{N}_0(t) \\ \boldsymbol{N}_1(t) \\ \vdots \\ \boldsymbol{N}_{n-1}(t) \end{bmatrix} = n$$

where

$$\boldsymbol{N}_0(t) = \boldsymbol{C}(t)$$

$$\boldsymbol{N}_{m+1}(t) = \boldsymbol{N}_m(t)\boldsymbol{A}(t) + \frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{N}_m(t), \quad m = 0, 1, ..., n-1$$

The observability proof implies that both the accelerometer and ARS biases will converge to their true values when the attitude and translational motion observers are decoupled (i.e., attitude is known from an external AHRS). In the general coupled case, where attitude is estimated along with position and velocity, the biases are not identifiable during straight, level motion. Convergence is guaranteed only if the vehicle undergoes persistent excitation (PE) in roll and pitch, meaning that it performs sufficient attitude maneuvers so that the rotation and translation dynamics are richly excited. Without such maneuvers, accelerometer biases remain unobservable and will not converge.

The attitude system is observable since

$$\boldsymbol{A}_1 = \begin{bmatrix} \boldsymbol{0}_{3\times3} & -\boldsymbol{T}_b^n(t) \\ \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} \end{bmatrix}, \quad \boldsymbol{C}_1 = \begin{bmatrix} \boldsymbol{I}_3 & \boldsymbol{0}_{3\times3} \end{bmatrix}$$

$$\operatorname{rank} \begin{bmatrix} \boldsymbol{N}_0(t) \\ \boldsymbol{N}_1(t) \end{bmatrix} = \operatorname{rank} \begin{bmatrix} \boldsymbol{I}_3 & \boldsymbol{0}_{3\times3} \\ \boldsymbol{0}_{3\times3} & -\boldsymbol{T}_b^n(t) \end{bmatrix} = 6$$

The Translational motion system is observable since

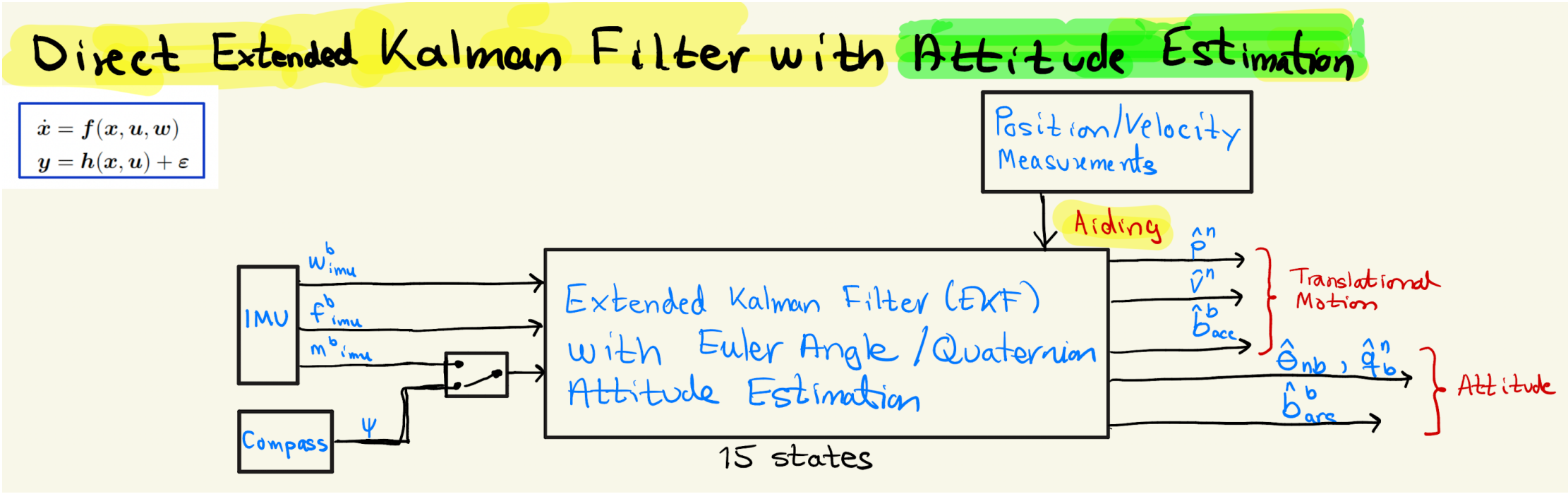
$$\boldsymbol{A}_2 = \begin{bmatrix} \boldsymbol{0}_{3\times3} & \boldsymbol{I}_3 & \boldsymbol{0}_{3\times3} \\ \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & -\boldsymbol{R}_b^n(t) \\ \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} \end{bmatrix}, \quad \boldsymbol{C}_2 = \begin{bmatrix} \boldsymbol{I}_3 & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} \end{bmatrix}$$

$$\operatorname{rank} \begin{bmatrix} \boldsymbol{N}_0(t) \\ \boldsymbol{N}_1(t) \\ \boldsymbol{N}_2(t) \end{bmatrix} = \operatorname{rank} \begin{bmatrix} \boldsymbol{I}_3 & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} \\ \boldsymbol{0}_{3\times3} & \boldsymbol{I}_3 & \boldsymbol{0}_{3\times3} \\ \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & -\boldsymbol{R}_b^n(t) \end{bmatrix} = 9$$

Observability is guaranteed since the rotation matrix has orthogonal columns and therefore $\operatorname{rank}(\boldsymbol{R}_b^n(t)) = 3$.

Chen, C.-T. (2012). Linear System Theory and Design. Oxford University Press.

Direct Kalman Filter with Attitude Estimation



14.3.3 Direct Extended Kalman Filter (EKF) with Euler Angle Attitude Estimation

Strapdown Navigation Equations

$$
\begin{aligned}
\dot{\boldsymbol{p}}^n_{nm_\mathrm{I}} &= \boldsymbol{v}^n_{nm_\mathrm{I}} \\
\dot{\boldsymbol{v}}^n_{nm_\mathrm{I}} &= \boldsymbol{R}(\boldsymbol{\Theta}_{nb})(\boldsymbol{f}^b_{\mathrm{imu}} - \boldsymbol{b}^b_{\mathrm{acc}} - \boldsymbol{w}^b_{\mathrm{acc}}) + \boldsymbol{g}^n \\
\dot{\boldsymbol{b}}^b_{\mathrm{acc}} &= \boldsymbol{w}^b_{b,\,\mathrm{acc}} \\
\dot{\boldsymbol{\Theta}}_{nb} &= \boldsymbol{T}(\boldsymbol{\Theta}_{nb})(\boldsymbol{\omega}^b_{\mathrm{imu}} - \boldsymbol{b}^b_{\mathrm{ars}} - \boldsymbol{w}^b_{\mathrm{ars}}) \\
\dot{\boldsymbol{b}}^b_{\mathrm{ars}} &= \boldsymbol{w}^b_{b,\,\mathrm{ars}}
\end{aligned}
$$

Attitude Θ_{nb} is estimated together with position $\boldsymbol{p}^n_{nm_I}$, linear velocity $\boldsymbol{v}^n_{nm_I}$, and biases. The resulting model $\boldsymbol{f(x,u,w)}$ is nonlinear. Hence, we must use the discrete-time EKF to estimate the Euler angles.

Measurements

$$
\boldsymbol{y} = \begin{bmatrix} \boldsymbol{p}^n_{nm_\mathrm{I}} \\ \boldsymbol{v}^n_{nm_\mathrm{I}}\ (\text{optionally}) \\ \boldsymbol{f}^b_{\mathrm{imu}}\ (\text{optionally}) \\ \boldsymbol{m}^b_{\mathrm{mag}}\ (\text{or compass}) \end{bmatrix}
$$

$$
\boldsymbol{h}(\boldsymbol{x},\boldsymbol{u}) = \begin{bmatrix} \boldsymbol{p}^n_{nm_\mathrm{I}} \\ \boldsymbol{v}^n_{nm_\mathrm{I}}\ (\text{optionally}) \\ \boldsymbol{S}(\cancel{\boldsymbol{\omega}^b_{\mathrm{imu}} - \boldsymbol{b}^b_{\mathrm{ars}}})\boldsymbol{R}^\top(\boldsymbol{\Theta}_{nb})\boldsymbol{v}^n_{nm_\mathrm{I}} - \boldsymbol{R}^\top(\boldsymbol{\Theta}_{nb})\boldsymbol{g}^n + \cancel{\boldsymbol{b}^b_{\mathrm{acc}}}\ (\text{optionally}) \\ \boldsymbol{R}^\top(\boldsymbol{\Theta}_{nb})\boldsymbol{m}^n \end{bmatrix}
$$

We can use specific force as dynamic measurement (not only input). This is not straightforward. We can also use the static gravity term. Usually, we drop the measurement on line #3.

15-State EKF (Euler Angle Representation)

$$
\begin{aligned}
\dot{\boldsymbol{x}} &= \boldsymbol{f}(\boldsymbol{x},\boldsymbol{u},\boldsymbol{w}) \\
\boldsymbol{y} &= \boldsymbol{h}(\boldsymbol{x},\boldsymbol{u}) + \boldsymbol{\varepsilon}
\end{aligned}
$$

$$
\boldsymbol{x} = [(\boldsymbol{p}^n_{nm_\mathrm{I}})^\top,\ (\boldsymbol{v}^n_{nm_\mathrm{I}})^\top,\ (\boldsymbol{b}^b_{\mathrm{acc}})^\top,\ \boldsymbol{\Theta}^\top_{nb},\ (\boldsymbol{b}^b_{\mathrm{ars}})^\top]^\top
$$

$$
\boldsymbol{w} = [(\boldsymbol{w}^b_{\mathrm{acc}})^\top,\ (\boldsymbol{w}^b_{b,\,\mathrm{acc}})^\top,\ (\boldsymbol{w}^b_{\mathrm{ars}})^\top,\ (\boldsymbol{w}^b_{b,\,\mathrm{ars}})^\top]^\top
$$

$$
\boldsymbol{f}(\boldsymbol{x},\boldsymbol{u},\boldsymbol{w}) = \begin{bmatrix} \boldsymbol{v}^n_{nm_\mathrm{I}} \\ \boldsymbol{R}(\boldsymbol{\Theta}_{nb})(\boldsymbol{f}^b_{\mathrm{imu}} - \boldsymbol{b}^b_{\mathrm{acc}} - \boldsymbol{w}_{\mathrm{acc}}) + \boldsymbol{g}^n \\ \boldsymbol{w}_{b,\,\mathrm{acc}} \\ \boldsymbol{T}(\boldsymbol{\Theta}_{nb})(\boldsymbol{\omega}^b_{\mathrm{imu}} - \boldsymbol{b}^b_{\mathrm{ars}} - \boldsymbol{w}_{\mathrm{ars}}) \\ \boldsymbol{w}_{b,\,\mathrm{ars}} \end{bmatrix}
$$

Inputs

$$
\boldsymbol{u} = [(\boldsymbol{f}^b_{\mathrm{imu}})^\top,\ (\boldsymbol{\omega}^b_{\mathrm{imu}})^\top,\ (\boldsymbol{g}^n)^\top]^\top
$$

14.3.3 Direct Extended Kalman Filter (EKF) with Euler Angle Attitude Estimation

Discrete-Time Jacobians

$$\boldsymbol{A}_d[k] = \boldsymbol{I}_{15} + h \left. \frac{\partial \boldsymbol{f}(\boldsymbol{x}[k], \boldsymbol{u}[k], \boldsymbol{w}[k])}{\partial \boldsymbol{x}[k]} \right|_{\boldsymbol{x}[k]=\hat{\boldsymbol{x}}[k],\, \boldsymbol{w}[k]=\boldsymbol{0}}$$

$$\boldsymbol{E}_d[k] = h \left. \frac{\partial \boldsymbol{f}(\boldsymbol{x}[k], \boldsymbol{u}[k], \boldsymbol{w}[k])}{\partial \boldsymbol{w}[k]} \right|_{\boldsymbol{x}[k]=\hat{\boldsymbol{x}}[k],\, \boldsymbol{w}[k]=\boldsymbol{0}}$$

$$\boldsymbol{C}_d[k] = \left. \frac{\partial \boldsymbol{h}(\boldsymbol{x}[k], \boldsymbol{u}[k])}{\partial \boldsymbol{x}[k]} \right|_{\boldsymbol{x}[k]=\hat{\boldsymbol{x}}^-[k]}$$

15-State EKF (Euler Angle Representation)

$$\begin{aligned} \dot{\boldsymbol{x}} &= \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{u}, \boldsymbol{w}) \\ \boldsymbol{y} &= \boldsymbol{h}(\boldsymbol{x}, \boldsymbol{u}) + \boldsymbol{\varepsilon} \end{aligned}$$

$$\boldsymbol{x} = [(\boldsymbol{p}^n_{nm_\mathrm{I}})^\top, (\boldsymbol{v}^n_{nm_\mathrm{I}})^\top, (\boldsymbol{b}^b_{\mathrm{acc}})^\top, \boldsymbol{\Theta}^\top_{nb}, (\boldsymbol{b}^b_{\mathrm{ars}})^\top]^\top$$

$$\boldsymbol{w} = [(\boldsymbol{w}^b_{\mathrm{acc}})^\top, (\boldsymbol{w}^b_{\mathrm{b,\,acc}})^\top, (\boldsymbol{w}^b_{\mathrm{ars}})^\top, (\boldsymbol{w}^b_{\mathrm{b,\,ars}})^\top]^\top$$

Predictor:

$$\hat{\boldsymbol{x}}^-[k+1] = \hat{\boldsymbol{x}}[k] + h\boldsymbol{f}(\hat{\boldsymbol{x}}[k], \boldsymbol{u}[k], \boldsymbol{0})$$
$$\hat{\boldsymbol{P}}^-[k+1] = \boldsymbol{A}_d[k]\hat{\boldsymbol{P}}[k]\boldsymbol{A}_d^\top[k] + \boldsymbol{E}_d[k]\boldsymbol{Q}_d[k]\boldsymbol{E}_d^\top[k]$$

Kalman gain matrix:

$$\boldsymbol{K}[k] = \hat{\boldsymbol{P}}^-[k]\boldsymbol{C}_d^\top[k] \left(\boldsymbol{C}_d[k]\hat{\boldsymbol{P}}^-[k]\boldsymbol{C}_d^\top[k] + \boldsymbol{R}_d[k]\right)^{-1}$$

Corrector:

$$\hat{\boldsymbol{x}}[k] = \hat{\boldsymbol{x}}^-[k] + \boldsymbol{K}[k]\left(\boldsymbol{y}[k] - \boldsymbol{h}(\hat{\boldsymbol{x}}^-[k], \boldsymbol{u}[k])\right)$$
$$\hat{\boldsymbol{P}}[k] = (\boldsymbol{I}_{15} - \boldsymbol{K}[k]\boldsymbol{C}_d[k])\hat{\boldsymbol{P}}^-[k](\boldsymbol{I}_{15} - \boldsymbol{K}[k]\boldsymbol{C}_d[k])^\top + \boldsymbol{K}[k]\boldsymbol{R}_d[k]\boldsymbol{K}^\top[k]$$

Inputs

$$\boldsymbol{u} = [(\boldsymbol{f}^b_{\mathrm{imu}})^\top, (\boldsymbol{\omega}^b_{\mathrm{imu}})^\top, (\boldsymbol{g}^n)^\top]^\top$$

Measurements

$$\boldsymbol{y} = \begin{bmatrix} \boldsymbol{p}^n_{nm_\mathrm{I}} \\ \boldsymbol{v}^n_{nm_\mathrm{I}} \text{ (optionally)} \\ \boldsymbol{f}^b_{\mathrm{imu}} \text{ (optionally)} \\ \boldsymbol{m}^b_{\mathrm{mag}} \text{ (or compass)} \end{bmatrix}$$

There is no global stability/convergence proof for the EKF and covariance blow-up is a problem (consider to use the more stable MEKF instead).
Estimation of the acceleration bias for this case is hard, and it usually requires that the craft is sufficient excited in roll and pitch (agile maneuvers).

14.4 Indirect Filters for Aided INS

14.4 Indirect Filters for Aided INS

The **Error-State Kalman Filter (ESKF)** is the standard method used in Strapdown INS for estimating the state and correcting errors in position, velocity, and orientation.
Why, see next slide?

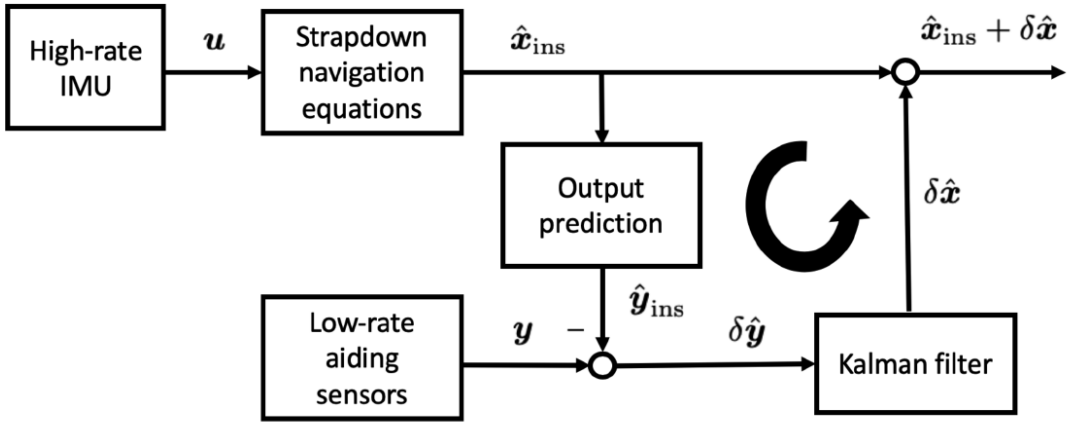
Indirect filtering implies that the KF filter is formulated as an error-state filter. In this context, the INS state $\boldsymbol{x}_{\text{ins}}[k]$ and error state $\delta\boldsymbol{x}[k]$ are related to the true state $\boldsymbol{x}[k]$ by

$$\boldsymbol{x}[k] = \boldsymbol{x}_{\text{ins}}[k] + \delta\boldsymbol{x}[k]$$

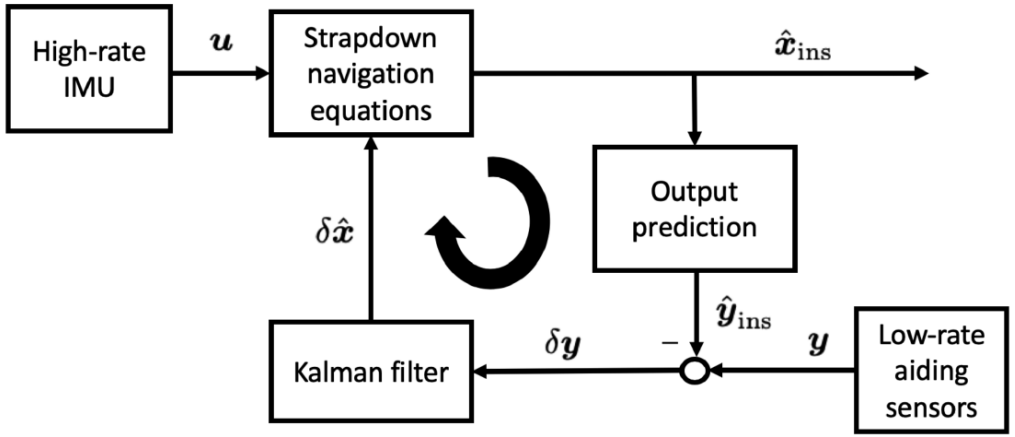
We want to estimate the error state $\delta\boldsymbol{x}[k]$

The indirect filter can be represented as a feedforward or a feedback filter. The reason for the term feedforward filter can be seen in the figure where the error corrections are added to the INS state as a feedforward compensation.

However, for the feedback filter the error estimates are used to update the INS estimates directly in order to prevent the INS errors to grow. This is achieved by regulating the error $\delta\boldsymbol{x}[k]$ to zero such that $\boldsymbol{x}_{\text{ins}}[k] \rightarrow \boldsymbol{x}[k]$ analogues to a feedback control system.



Feedforward filter



Feedback filter (reset mechanism)

14.4 Why is the ESKF the Standard Method used in Strapdown INS

1. **Handling of Nonlinearities:**

INS systems involve the integration of accelerometer and gyroscope data, which results in nonlinear dynamics due to the relationship between orientation (attitude) and angular velocity. The ESKF efficiently handles these nonlinearities by linearizing the error dynamics rather than the full system dynamics, making it more suitable for practical implementations in INS compared to the traditional EKF and UKF.

2. Error-State Representation:

The ESKF focuses on the error state of the system, such as position, velocity, and orientation errors, rather than directly estimating the full state. This allows for more stable estimation of small errors while reducing the impact of accumulating integration errors common in strapdown INS systems. By estimating these smaller error states, the ESKF avoids the divergence problems that could arise in other Kalman Filter implementations.

3. Efficient Update and Correction:

IMU measurements are often combined with external corrections (such as GNSS). The ESKF excels at fusing multiple sensor measurements, allowing for frequent corrections of the error state and keeping the system accurate. It updates the error state efficiently using a linear update step, while the propagation of the main state remains largely independent of these updates.

4. Robustness Against Biases:

INS systems are prone to sensor biases (e.g., accelerometer and gyroscope biases). The ESKF is particularly effective at tracking and estimating these biases as part of the error state, making it more robust for long-term navigation applications. By estimating and correcting sensor biases, the ESKF minimizes their influence on the final position, velocity, and attitude estimates.

5. Suitability for High-Rate Systems:

INS systems often operate at high update rates (e.g., 1000–2000 Hz), and the ESKF is computationally efficient enough to handle these fast updates. The error-state formulation allows for quick corrections based on new sensor data without having to re-linearize the full system, making it an ideal solution for real-time applications in navigation.

14.4 Indirect Filters for Aided INS

Table 14.1: Discrete-time error-state Kalman filter.

Initial values	$\delta\hat{\boldsymbol{x}}^-[0] = \delta\boldsymbol{x}_0$ $\hat{\boldsymbol{P}}^-[0] = \mathrm{E}[(\delta\boldsymbol{x}[0] - \delta\hat{\boldsymbol{x}}^-[0])(\delta\boldsymbol{x}[0] - \delta\hat{\boldsymbol{x}}^-[0])^\top] = \boldsymbol{P}_0$
KF gain	$\boldsymbol{K}[k] = \hat{\boldsymbol{P}}^-[k]\boldsymbol{C}_d^\top[k](\boldsymbol{C}_d[k]\hat{\boldsymbol{P}}^-[k]\boldsymbol{C}_d^\top[k] + \boldsymbol{R}_d[k])^{-1}$
State corrector	$\delta\hat{\boldsymbol{x}}[k] = \delta\hat{\boldsymbol{x}}^-[k] + \boldsymbol{K}[k]\big(\delta\boldsymbol{y}[k] - \boldsymbol{C}_d[k]\delta\hat{\boldsymbol{x}}^-[k] - \boldsymbol{D}_d[k]\boldsymbol{u}[k]\big)$
Covariance corrector	$\hat{\boldsymbol{P}}[k] = (\boldsymbol{I}_n - \boldsymbol{K}[k]\boldsymbol{C}_d[k])\,\hat{\boldsymbol{P}}^-[k]\,(\boldsymbol{I}_n - \boldsymbol{K}[k]\boldsymbol{C}_d[k])^\top + \boldsymbol{K}[k]\boldsymbol{R}_d[k]\boldsymbol{K}^\top[k]$
State predictor	$\delta\hat{\boldsymbol{x}}^-[k+1] = \boldsymbol{A}_d[k]\,\delta\hat{\boldsymbol{x}}[k] + \boldsymbol{B}_d[k]\,\boldsymbol{u}[k]$
Covariance predictor	$\delta\hat{\boldsymbol{P}}^-[k+1] = \boldsymbol{A}_d[k]\hat{\boldsymbol{P}}[k]\boldsymbol{A}_d^\top[k] + \boldsymbol{E}_d[k]\boldsymbol{Q}_d[k]\boldsymbol{E}_d^\top[k]$

where

$\boldsymbol{Q}_d$, $\boldsymbol{R}_d$	Covariance matrices for the process and measurement noises
$\delta\hat{\boldsymbol{x}}^-$, $\hat{\boldsymbol{P}}^-$	*A priori* error state and covariance matrix estimates (before update)
$\delta\hat{\boldsymbol{x}}$, $\hat{\boldsymbol{P}}$	*A posteriori* error state and covariance matrix estimates (after update)

The discrete-time **Error-State Kalman Filter (ESKF)** is specifically designed to integrate high-rate IMU measurements with low-rate position corrections (aiding).

The Error-State KF filter (ESKF) is based on a discrete-time state-space model:

$$\delta\boldsymbol{x}[k+1] = \boldsymbol{A}_d[k]\delta\boldsymbol{x}[k] + \boldsymbol{B}_d[k]\boldsymbol{u}[k] + \boldsymbol{E}_d[k]\boldsymbol{w}[k]$$
$$\delta\boldsymbol{y}[k] = \boldsymbol{C}_d[k]\delta\boldsymbol{x}[k] + \boldsymbol{D}_d[k]\boldsymbol{u}[k] + \boldsymbol{\varepsilon}[k]$$

Feedforward filter:

$$\hat{\boldsymbol{x}}[k] = \hat{\boldsymbol{x}}_{\mathrm{ins}}[k] + \delta\hat{\boldsymbol{x}}[k]$$

Scalar ESKF Example

(Position, Velocity and Angle)

14.4.1 Scalar ESKF Example (Position, Velocity and Angle)

Strapdown Navigation Equations

$$\dot{p} = v \quad \text{Position}$$
$$\dot{v} = a \quad \text{Velocity}$$
$$\dot{\theta} = \omega \quad \text{Angle}$$

IMU Measurements: The inputs to the strapdown navigation equations are the IMU specific force f_{imu} and ARS measurements ω_{imu}:

$$f_{\text{imu}} = a - g + b_{\text{acc}} + w_{\text{acc}} \quad \text{Specific force}$$
$$\dot{b}_{\text{acc}} = w_{b,\,\text{acc}} \quad \text{Accelerometer bias}$$
$$\omega_{\text{imu}} = \omega + b_{\text{ars}} + w_{\text{ars}} \quad \text{Angular rate}$$
$$\dot{b}_{\text{ars}} = w_{b,\,\text{ars}} \quad \text{ARS bias}$$

Aiding: The measurements used for aiding are position and angle

$$y_1 = p + \varepsilon_1 \quad \text{Position}$$
$$y_2 = \theta + \varepsilon_2 \quad \text{Angle}$$

Inertial navigation system (INS):

$$\dot{\hat{p}}_{\text{ins}} = \hat{v}_{\text{ins}}$$
$$\dot{\hat{v}}_{\text{ins}} = f_{\text{imu}} - \hat{b}_{\text{acc,ins}} + g$$
$$\dot{\hat{b}}_{\text{acc,ins}} = 0$$
$$\dot{\hat{\theta}}_{\text{ins}} = \omega_{\text{imu}} - \hat{b}_{\text{ars,ins}}$$
$$\dot{\hat{b}}_{\text{ars,ins}} = 0$$

We want to estimate the 5 error states:

$$\delta p = p - \hat{p}_{\text{ins}}$$
$$\delta v = v - \hat{v}_{\text{ins}}$$
$$\delta b_{\text{acc}} = b_{\text{acc}} - \hat{b}_{\text{acc, ins}}$$
$$\delta\theta = \theta - \hat{\theta}_{\text{ins}}$$
$$\delta b_{\text{ars}} = b_{\text{ars}} - \hat{b}_{\text{ars, ins}}$$

In particular, the two IMU biases δb_{acc} and δb_{ars} are important estimates to avoid drift.

14.4.1 Scalar ESKF Example (Position, Velocity and Angle)

Error State Dynamics

$$
\begin{aligned}
\delta\dot{p} &= \dot{p} - \dot{\hat{p}}_{\text{ins}} \\
&= v - \hat{v}_{\text{ins}} = \delta v \\
\delta\dot{v} &= (f + g) - \dot{\hat{v}}_{\text{ins}} \\
&= (f_{\text{imu}} - b_{\text{acc}} - w_{\text{acc}} + g) - (f_{\text{imu}} - \hat{b}_{\text{acc, ins}} + g) \\
&= -b_{\text{acc}} - w_{\text{acc}} + \hat{b}_{\text{acc, ins}} \\
&= -\delta b_{\text{acc}} - w_{\text{acc}} \\
\delta\dot{\theta} &= \omega - \dot{\hat{\theta}}_{\text{ins}} \\
&= (\omega_{\text{imu}} - b_{\text{ars}} - w_{\text{ars}}) - (\omega_{\text{imu}} - \hat{b}_{\text{ars, ins}}) \\
&= -b_{\text{ars}} - w_{\text{ars}} + \hat{b}_{\text{ars,ins}} \\
&= -\delta b_{\text{ars}} - w_{\text{ars}}
\end{aligned}
$$

Error Measurements

$$
\begin{aligned}
\delta y_1 &= (p + \varepsilon_1) - \hat{p}_{\text{ins}} \\
&= \delta p + \varepsilon_1 \\
\delta y_2 &= (\theta + \varepsilon_2) - \hat{\theta}_{\text{ins}} \\
&= \delta\theta + \varepsilon_2
\end{aligned}
$$

Bias Models: The errors in the INS due to drift and noise on the accelerometer and ARS measurements can be accurately described by first-order bias models (these models are physically motivated)

$$
\begin{aligned}
\delta\dot{b}_{\text{acc}} &:= -\frac{1}{T_{\text{acc}}}\delta b_{\text{acc}} + w_{\text{b,acc}} \\
\delta\dot{b}_{\text{ars}} &:= -\frac{1}{T_{\text{ars}}}\delta b_{\text{ars}} + w_{\text{b,ars}}
\end{aligned}
$$

The time constants ensure that the bias errors go exponentially to zero during dead reckoning, that is situations where there are no aiding.

14.4.1 Scalar ESKF Example (Position, Velocity and Angle)

Error-state model:

$$\delta\dot{\boldsymbol{x}} = \boldsymbol{A}\delta\boldsymbol{x} + \boldsymbol{E}\boldsymbol{w}$$
$$\delta\boldsymbol{y} = \boldsymbol{C}\delta\boldsymbol{x} + \boldsymbol{\varepsilon}$$

where $\delta\boldsymbol{x} = [\delta p,\ \delta v,\ \delta b_{\text{acc}},\ \delta\theta,\ \delta b_{\text{ars}}]^\top$, $\boldsymbol{w} = [w_{\text{acc}},\ w_{\text{b,acc}},\ w_{\text{ars}},\ w_{b,\text{ ars}}]^\top$, $\boldsymbol{\varepsilon} = [\varepsilon_1,\ \varepsilon_2]^\top$ and

$$\boldsymbol{A} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & -\frac{1}{T_{\text{acc}}} & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & -\frac{1}{T_{\text{ars}}} \end{bmatrix}, \quad \boldsymbol{E} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \boldsymbol{C} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

$$\boldsymbol{A}_d \approx \boldsymbol{I}_5 + h\boldsymbol{A}, \qquad \boldsymbol{C}_d = \boldsymbol{C}, \qquad \boldsymbol{E}_d \approx h\boldsymbol{E}$$

Note that the for the **indirect filter**, the error model is not driven by IMU inputs such as specific force and ARS.

Also note that the error model is linear. Hence, a linear discrete-time KF can be used to update the states. This is contrary to direct filters.

Discrete-Time Linear Kalman Filter:

Feedforward filter

Corrector: $\delta\hat{\boldsymbol{x}}[k] = \delta\hat{\boldsymbol{x}}^-[k] + \boldsymbol{K}[k]\left(\delta\boldsymbol{y}[k] - \boldsymbol{C}_d[k]\delta\hat{\boldsymbol{x}}^-[k]\right)$ → $\hat{\boldsymbol{x}}[k] = \hat{\boldsymbol{x}}_{\text{ins}}[k] + \delta\hat{\boldsymbol{x}}[k]$

Predictor: $\delta\hat{\boldsymbol{x}}^-[k+1] = \boldsymbol{A}_d[k]\,\delta\hat{\boldsymbol{x}}[k]$

6-DOF ESKF with Attitude Measurements

(Attitude is Estimated using the MEKF)

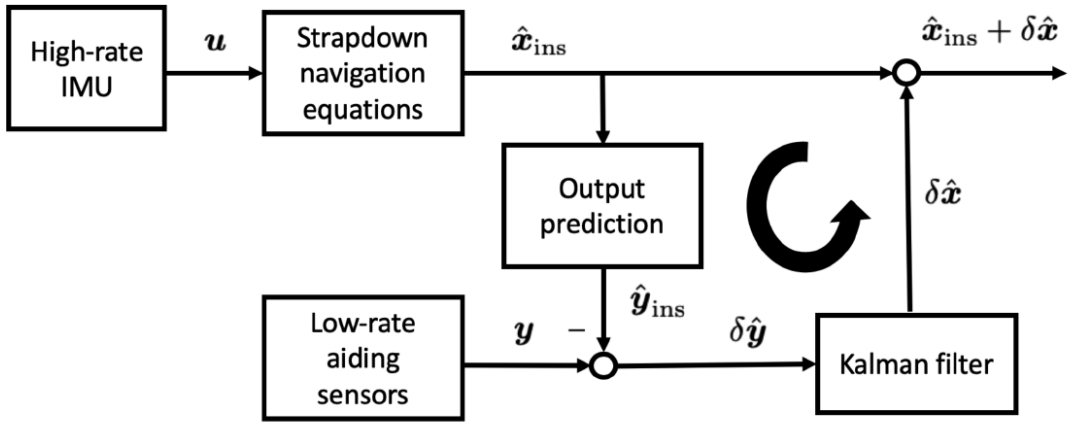
14.4.2 6-DOF ESKF with Attitude Measurements

The scalar introductory example can be generalized to a 15-state error-state filter for 6-DOF motions.
We will discuss both the feedforward and feedback filters.

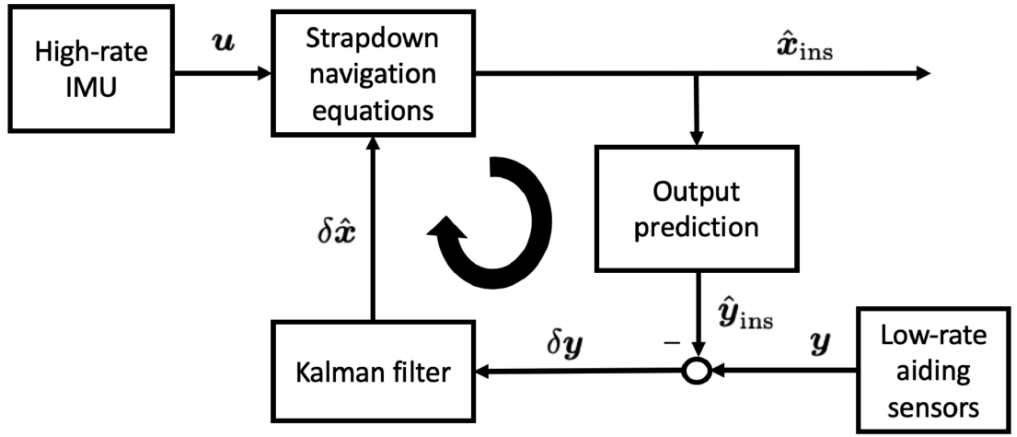
Assumption: The Euler angles are measured using an AHRS such that the attitude transformation matrices are known:

$$\boldsymbol{R}(\boldsymbol{\Theta}_{bn}) := \boldsymbol{R}_b^n(t), \qquad \boldsymbol{T}(\boldsymbol{\Theta}_{bn}) := \boldsymbol{T}_b^n(t)$$

The use of attitude measurements simplifies the estimation problem; this assumption will be relaxed later.



Feedforward filter



Feedback filter (reset mechanism)

14.4.2 6-DOF ESKF with Attitude Measurements

Strapdown Navigation Equations

$$\dot{\boldsymbol{p}}^n_{nm_\mathrm{I}} = \boldsymbol{v}^n_{nm_\mathrm{I}} \quad \text{Position vector}$$
$$\dot{\boldsymbol{v}}^n_{nm_\mathrm{I}} = \boldsymbol{a}^n_{nm_\mathrm{I}} \quad \text{Velocity vector}$$
$$\dot{\boldsymbol{\Theta}}_{nb} = \boldsymbol{T}^n_b(t)\boldsymbol{\omega}^b_{nb} \quad \text{Euler angles}$$

IMU Measurements: The inputs to the strapdown navigation equations are the IMU specific force and ARS measurements

$$\boldsymbol{f}^b_{\mathrm{imu}} = \boldsymbol{R}^n_b(t)^\top(\boldsymbol{a}^n_{nm_\mathrm{I}} - \boldsymbol{g}^n) + \boldsymbol{b}^b_{\mathrm{acc}} + \boldsymbol{w}^b_{\mathrm{acc}} \quad \text{Specific force}$$
$$\dot{\boldsymbol{b}}^b_{\mathrm{acc}} = \boldsymbol{w}^b_{b,\,\mathrm{acc}} \quad \text{Accelerometer bias}$$
$$\boldsymbol{\omega}^b_{\mathrm{imu}} = \boldsymbol{\omega}^b_{nb} + \boldsymbol{b}^b_{\mathrm{ars}} + \boldsymbol{w}^b_{\mathrm{ars}} \quad \text{Angular rate}$$
$$\dot{\boldsymbol{b}}^b_{\mathrm{ars}} = \boldsymbol{w}^b_{b,\,\mathrm{ars}} \quad \text{ARS bias}$$

Aiding: The measurements used for aiding are position and angle

$$\boldsymbol{y}_1 = \boldsymbol{p}^n_{nm_\mathrm{I}} + \boldsymbol{\varepsilon}_1 \quad \text{GNSS position}$$
$$\boldsymbol{y}_2 = \boldsymbol{v}^n_{nm_\mathrm{I}} + \boldsymbol{\varepsilon}_2 \quad \text{(optionally)} \quad \text{GNSS velocity}$$
$$\boldsymbol{y}_3 = \boldsymbol{\Theta}_{nb} + \boldsymbol{\varepsilon}_3 \quad \text{AHRS}$$

INS Strapdown Equations

$$\dot{\hat{\boldsymbol{p}}}^n_{\mathrm{ins}} = \hat{\boldsymbol{v}}^n_{\mathrm{ins}}$$
$$\dot{\hat{\boldsymbol{v}}}^n_{\mathrm{ins}} = \boldsymbol{R}^n_b(t)\boldsymbol{f}^b_{\mathrm{ins}} + \boldsymbol{g}^n$$
$$\dot{\hat{\boldsymbol{b}}}^b_{\mathrm{acc,\,ins}} = \boldsymbol{0}$$
$$\dot{\hat{\boldsymbol{\theta}}}_{\mathrm{ins}} = \boldsymbol{T}^n_b(t)\boldsymbol{\omega}^b_{\mathrm{ins}}$$
$$\dot{\hat{\boldsymbol{b}}}^b_{\mathrm{ars,\,ins}} = \boldsymbol{0}$$

where the INS inputs are bias compensated IMU measurements

$$\boldsymbol{f}^b_{\mathrm{ins}} := \boldsymbol{f}^b_{\mathrm{imu}} - \Delta\boldsymbol{f}^b$$
$$\boldsymbol{\omega}^b_{\mathrm{ins}} := \boldsymbol{\omega}^b_{\mathrm{imu}} - \Delta\boldsymbol{\omega}^b$$

Feedforward filter: $\Delta\boldsymbol{f}^b = \boldsymbol{0}$, $\Delta\boldsymbol{\omega}^b = \boldsymbol{0}$

Feedback filter: $\Delta\boldsymbol{f}^b = \hat{\boldsymbol{b}}^b_{\mathrm{acc,\,ins}}$, $\Delta\boldsymbol{\omega}^b = \hat{\boldsymbol{b}}^b_{\mathrm{ars,\,ins}}$

Reset

14.4.2 6-DOF ESKF with Attitude Measurements

Error-State Model

$$\delta\dot{\boldsymbol{x}} = \boldsymbol{A}(t)\delta\boldsymbol{x} + \boldsymbol{E}(t)\boldsymbol{w}$$
$$\delta\boldsymbol{y} = \boldsymbol{C}\delta\boldsymbol{x} + \boldsymbol{\varepsilon}$$

$$\delta\boldsymbol{x} = [(\delta\boldsymbol{p}^n)^\top, (\delta\boldsymbol{v}^n)^\top, (\delta\boldsymbol{b}^b_{\text{acc}})^\top, \delta\boldsymbol{\Theta}^\top_{nb}, (\delta\boldsymbol{b}^b_{\text{ars}})^\top]^\top$$
$$\boldsymbol{w} = [\boldsymbol{w}^\top_{\text{acc}}, \boldsymbol{w}^\top_{b,\,\text{acc}}, \boldsymbol{w}^\top_{\text{ars}}, \boldsymbol{w}^\top_{b,\,\text{ars}}]^\top$$

The attitude measurements are represented as time-varying matrices $\boldsymbol{R}^n_b(t)$ and $\boldsymbol{T}^n_b(t)$ in $A(t)$ and $E(t)$

$$\boldsymbol{A}(t) = \begin{bmatrix} \boldsymbol{0}_{3\times3} & \boldsymbol{I}_3 & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} \\ \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & -\boldsymbol{R}^n_b(t) & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} \\ \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & -\frac{1}{T_{\text{acc}}}\boldsymbol{I}_3 & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} \\ \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & -\boldsymbol{T}^n_b(t) \\ \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & -\frac{1}{T_{\text{ars}}}\boldsymbol{I}_3 \end{bmatrix}$$

$$\boldsymbol{E}(t) = \begin{bmatrix} \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} \\ -\boldsymbol{R}^n_b(t) & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} \\ \boldsymbol{0}_{3\times3} & \boldsymbol{I}_3 & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} \\ \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & -\boldsymbol{T}^n_b(t) & \boldsymbol{0}_{3\times3} \\ \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{I}_3 \end{bmatrix}$$

$$\boldsymbol{C} = \begin{bmatrix} \boldsymbol{I}_3 & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} \\ \boldsymbol{0}_{3\times3} & \boldsymbol{I}_3 & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} \\ \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{I}_3 & \boldsymbol{0}_{3\times3} \end{bmatrix}$$

Bias Models: The errors in the INS due to drift and noise on the accelerometer and ARS measurements can be accurately described by first-order bias models (these models are physically motivated)

$$\delta\dot{\boldsymbol{b}}^b_{\text{acc}} = -\frac{1}{T_{\text{acc}}}\delta\boldsymbol{b}^b_{\text{acc}} + \boldsymbol{w}^b_{b,\,\text{acc}}$$
$$\delta\dot{\boldsymbol{b}}^b_{\text{ars}} = -\frac{1}{T_{\text{ars}}}\delta\boldsymbol{b}^b_{\text{ars}} + \boldsymbol{w}^b_{b,\,\text{ars}}$$

The time constants ensure that the bias errors go exponentially to zero during dead reckoning, that is situations where there are no aiding.

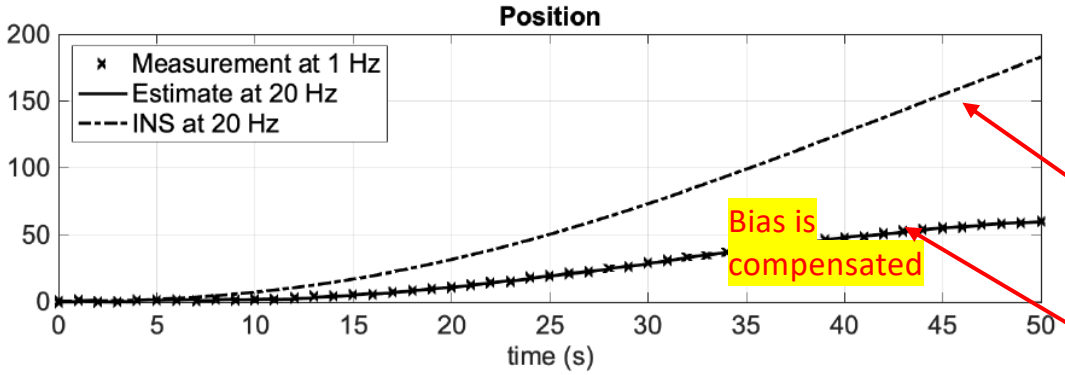
Bias Observability: For the 15-state indirect feedforward filter the estimated state is computed as: $\hat{\boldsymbol{x}} = \hat{\boldsymbol{x}}_{\text{ins}} + \delta\hat{\boldsymbol{x}}$. Since the error dynamics is an LTV system it is possible to show that the pair $(A(t), C)$ is observable (see Section 14.3.2). This implies that the estimated biases converge to their true values:

$$\hat{\boldsymbol{b}}^b_{\text{acc}} = \boldsymbol{0} + \delta\hat{\boldsymbol{b}}^b_{\text{acc}}$$
$$\hat{\boldsymbol{b}}^b_{\text{ars}} = \boldsymbol{0} + \delta\hat{\boldsymbol{b}}^b_{\text{ars}}$$

$$\hat{\boldsymbol{x}}_{\text{ins}} = \begin{bmatrix} \hat{\boldsymbol{p}}^n_{\text{ins}} \\ \hat{\boldsymbol{v}}^n_{\text{ins}} \\ \boldsymbol{0} \\ \hat{\boldsymbol{\Theta}}_{\text{ins}} \\ \boldsymbol{0} \end{bmatrix}$$

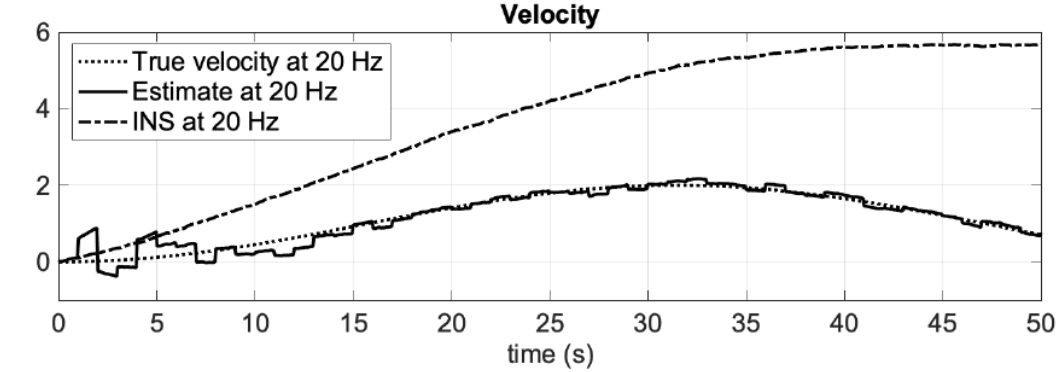
14.4.2 6-DOF ESKF with Attitude Measurements

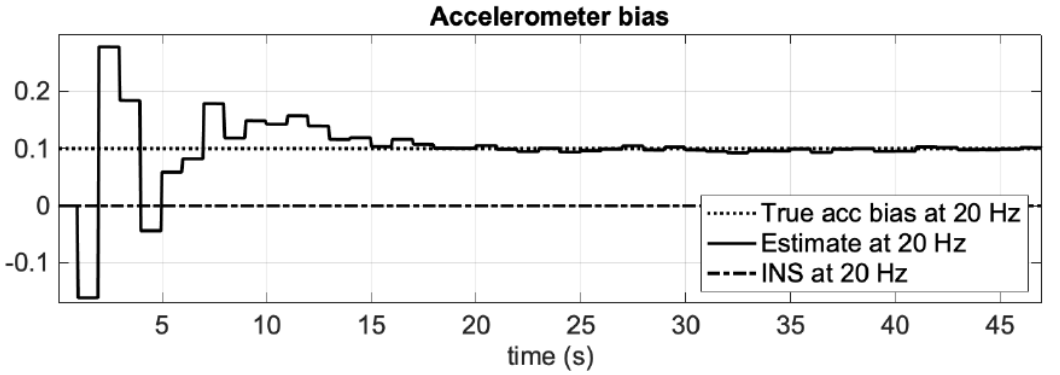
The Indirect Feedback Kalman Filter on the next slide uses reset to avoid this problem



…but the INS drift grows to infinity

Bias is compensated





Summary: Indirect feedforward Kalman filter using attitude measurements

Discrete-time system matrices:

$$\boldsymbol{A}_d \approx \boldsymbol{I}_{15} + h\boldsymbol{A} \qquad \boldsymbol{C}_d = \boldsymbol{C}, \qquad \boldsymbol{E}_d \approx h\boldsymbol{E}$$

State vector:

$$\delta\boldsymbol{x}[k] = [\delta\boldsymbol{p}^n_{nm_I}[k]^\top,\ \delta\boldsymbol{v}^n_{nm_I}[k]^\top,\ \delta\boldsymbol{b}^b_{\text{acc}}[k]^\top,\ \delta\boldsymbol{\Theta}^\top_{nb}[k],\ \delta\boldsymbol{b}^b_{\text{ars}}[k]^\top]^\top$$

Kalman gain:

$$\boldsymbol{K}[k] = \hat{\boldsymbol{P}}^-[k]\boldsymbol{C}_d^\top[k]\left(\boldsymbol{C}_d[k]\hat{\boldsymbol{P}}^-[k]\boldsymbol{C}_d^\top[k] + \boldsymbol{R}_d[k]\right)^{-1}$$

State estimate:

$$\hat{\boldsymbol{x}}_{\text{ins}}[k] = [\hat{\boldsymbol{p}}^n_{\text{ins}}[k]^\top,\ \hat{\boldsymbol{v}}^n_{\text{ins}}[k]^\top,\ \boldsymbol{0}^\top,\ \hat{\boldsymbol{\Theta}}^\top_{\text{ins}}[k],\ \boldsymbol{0}^\top]^\top$$

$$\hat{\boldsymbol{x}}[k] = \hat{\boldsymbol{x}}_{\text{ins}}[k] + \delta\hat{\boldsymbol{x}}[k]$$

Predictor:

$$\delta\hat{\boldsymbol{x}}^-[k+1] = \boldsymbol{A}_d[k]\,\delta\hat{\boldsymbol{x}}[k]$$

$$\hat{\boldsymbol{P}}^-[k+1] = \boldsymbol{A}_d[k]\hat{\boldsymbol{P}}[k]\boldsymbol{A}_d^\top[k] + \boldsymbol{E}_d[k]\boldsymbol{Q}_d[k]\boldsymbol{E}_d^\top[k]$$

Corrector:

$$\delta\hat{\boldsymbol{x}}[k] = \delta\hat{\boldsymbol{x}}^-[k] + \boldsymbol{K}[k]\left(\delta\boldsymbol{y}[k] - \boldsymbol{C}_d\,\delta\hat{\boldsymbol{x}}^-[k]\right)$$

$$\hat{\boldsymbol{P}}[k] = (\boldsymbol{I}_{15} - \boldsymbol{K}[k]\boldsymbol{C}_d[k])\hat{\boldsymbol{P}}^-[k](\boldsymbol{I}_{15} - \boldsymbol{K}[k]\boldsymbol{C}_d[k])^\top + \boldsymbol{K}[k]\boldsymbol{R}_d[k]\boldsymbol{K}^\top[k]$$

INS propagation:

$$\hat{\boldsymbol{p}}^n_{\text{ins}}[k+1] = \hat{\boldsymbol{p}}^n_{\text{ins}}[k] + h\hat{\boldsymbol{v}}^n_{\text{ins}}[k]$$

$$\hat{\boldsymbol{v}}^n_{\text{ins}}[k+1] = \hat{\boldsymbol{v}}^n_{\text{ins}}[k] + h\left(\boldsymbol{R}^n_b[k]\left(\boldsymbol{f}^b_{\text{imu}}[k] - \boldsymbol{0}\right) + \boldsymbol{g}^n\right)$$

$$\hat{\boldsymbol{\Theta}}_{\text{ins}}[k+1] = \hat{\boldsymbol{\Theta}}_{\text{ins}}[k] + h\boldsymbol{T}^n_b[k]\left(\boldsymbol{\omega}^b_{\text{imu}}[k] - \boldsymbol{0}\right)$$

14.4.2 6-DOF ESKF with Attitude Measurements

Indirect Feedback Kalman Filter

For the feedback version of the indirect KF, the error estimates are used to update the INS estimates directly to avoid drift. This prevents the INS errors to grow. This is done by regulating the error $\delta x[k]$ to zero by state reset such that $x_{\text{ins}}[k] \rightarrow x[k]$ analogues to a feedback control system. Hence, the term feedback filter.

Algorithm 14.1 (Feedback Filter Reset Algorithm) *The feedback loop (Figure 14.8) will remove the INS error by regulating $\delta \boldsymbol{x}$ to zero. This can be obtained by modifying the feedforward filter according to:*

1. *Slow measurements: After every slow measurement (position and velocity), the INS states*

$$\hat{\boldsymbol{x}}_{ins} = \left[(\hat{\boldsymbol{p}}^n_{ins})^\top, (\hat{\boldsymbol{v}}^n_{ins})^\top, (\hat{\boldsymbol{b}}^b_{acc,\,ins})^\top, \hat{\boldsymbol{\Theta}}^\top_{ins}, (\hat{\boldsymbol{b}}^b_{ars,\,ins})^\top \right]^\top$$

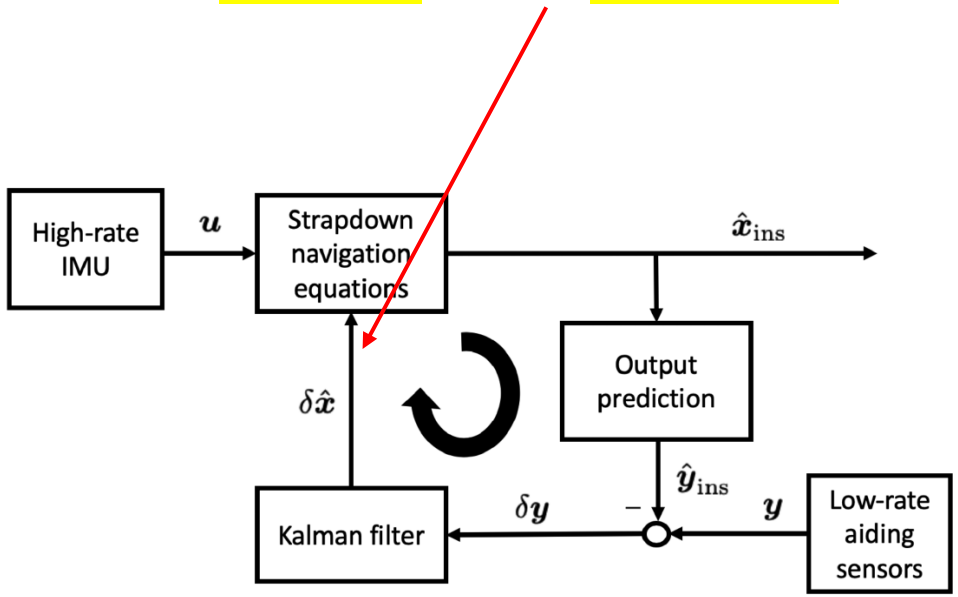
are corrected by setting the error-state vector to zero. This is mathematically equivalent to

$$\hat{\boldsymbol{x}}_{ins}[k] \leftarrow \hat{\boldsymbol{x}}_{ins}[k] + \delta\hat{\boldsymbol{x}}[k]$$

2. *Fast measurements: Step 1 ensures that error-state vector will be zero before the fast measurements (accelerometer and ARS) are applied. Therefore, the state predictor of the feedforward filter* (14.168) *becomes redundant since*

$$\delta\hat{\boldsymbol{x}}^-[k+1] = \boldsymbol{A}_d[k]\boldsymbol{0} \equiv \boldsymbol{0}$$

It should, however, be noted that $\boldsymbol{A}_d[k]$ is still a necessary to compute since $\boldsymbol{A}_d[k]$ is used in the KF covariance predication step.



14.4.2 6-DOF ESKF with Attitude Measurements

```
% MAIN LOOP
for i=1:N

   % IMU measurements
   f_imu = ...
   w_imu = ...

   % AHRS measurements: phi[k], theta[k], psi[k]
   % If only compass measurement psi[k], use static solution
   % [phi, theta] = acc2rollpitch(f_imu)
   R = Rzyx( phi, theta, psi );
   T = Tzyx( phi, theta );

   % Kalman filter matrices
   Ad = ...
   Cd = ...
   Ed = ...

   if (new_measurement)

      y = ...      % GNSS measurement: y[k] = [p[k]; v[k]]'

      % KF gain: K[k]
      K = P_prd * Cd' * inv( Cd * P_prd * Cd' + Rd );
      IKC = eye(15) - K * Cd;

      % Corrector: delta_x_hat[k] and P_hat[k]
      delta_x_hat = K * ( y - Cd * x_ins );
      P_hat = IKC * P_prd * IKC' + K * Rd * K';

      % INS reset: x_ins[k]
      p_ins = p_ins + delta_x_hat(1:3);          % position
      v_ins = v_ins + delta_x_hat(4:6);          % velocity
      b_acc_ins = b_acc_ins + delta_x_hat(7:9);  % acc bias
      theta_ins = theta_ins + delta_x_hat(10:12); % attitude
      b_ars_ins = b_ars_ins + delta_x_hat(13:15); % ars bias

   else

       P_hat = P_prd;              % no measurement

   end
   P_prd = Ad * P_hat * Ad' + Ed * Qd * Ed';

   % INS propagation: x_ins[k+1]
   p_ins = p_ins + h * v_ins;
   v_ins = v_ins + h * ( R * ( f_imu - b_acc_ins) + g_n );
   theta_ins = theta_ins + h * T * ( w_imu - b_ars_ins );

end
```

Matlab:
The pseudocode for the 15-states indirect feedback KF is given below. The stand-alone Matlab function for the INS is included in the MSS toolbox as:

```
[x_ins,P_prd] = ins_ahrs(x_ins, P_prd, mu, h, Qd, Rd, f_imu, ...
    w_imu, y_ahrs, y_pos, y_vel)
```

State reset

Summary: Indirect feedback Kalman filter using attitude measurements

Discrete-time system matrices:

$$\boldsymbol{A}_d \approx \boldsymbol{I}_{15} + h\boldsymbol{A}, \qquad \boldsymbol{C}_d = \boldsymbol{C}, \qquad \boldsymbol{E}_d \approx h\boldsymbol{E}$$

State vectors:

$$\delta\hat{\boldsymbol{x}}[k] = [\delta\hat{\boldsymbol{p}}^n_{nm_\mathrm{I}}[k]^\top, \delta\hat{\boldsymbol{v}}^n_{nm_\mathrm{I}}[k]^\top, \delta\hat{\boldsymbol{b}}^b_{\mathrm{acc}}[k]^\top, \delta\hat{\boldsymbol{\Theta}}_{nb}[k]^\top, \delta\hat{\boldsymbol{b}}^b_{\mathrm{ars}}[k]^\top]^\top$$

$$\hat{\boldsymbol{x}}_{\mathrm{ins}}[k] = [\hat{\boldsymbol{p}}^n_{\mathrm{ins}}[k]^\top, \hat{\boldsymbol{v}}^n_{\mathrm{ins}}[k]^\top, \hat{\boldsymbol{b}}^b_{\mathrm{acc,ins}}[k]^\top, \hat{\boldsymbol{\Theta}}_{\mathrm{ins}}[k]^\top, \hat{\boldsymbol{b}}^b_{\mathrm{ars,ins}}[k]^\top]^\top$$

Kalman gain:

$$\boldsymbol{K}[k] = \hat{\boldsymbol{P}}^-[k]\boldsymbol{C}_d^\top[k]\left(\boldsymbol{C}_d[k]\hat{\boldsymbol{P}}^-[k]\boldsymbol{C}_d^\top[k] + \boldsymbol{R}_d[k]\right)^{-1}$$

INS reset and state estimate:

$$\hat{\boldsymbol{x}}_{\mathrm{ins}}[k] = \hat{\boldsymbol{x}}_{\mathrm{ins}}[k] + \delta\hat{\boldsymbol{x}}[k]$$

Predictor:

$$\hat{\boldsymbol{P}}^-[k+1] = \boldsymbol{A}_d[k]\hat{\boldsymbol{P}}[k]\boldsymbol{A}_d^\top[k] + \boldsymbol{E}_d[k]\boldsymbol{Q}_d[k]\boldsymbol{E}_d^\top[k]$$

Corrector:

$$\delta\hat{\boldsymbol{x}}[k] = \boldsymbol{K}[k]\left(\boldsymbol{y}[k] - \boldsymbol{C}_d[k]\,\hat{\boldsymbol{x}}_{\mathrm{ins}}[k]\right)$$

$$\hat{\boldsymbol{P}}[k] = \left(\boldsymbol{I}_{15} - \boldsymbol{K}[k]\boldsymbol{C}_d[k]\right)\hat{\boldsymbol{P}}^-[k]\left(\boldsymbol{I}_{15} - \boldsymbol{K}[k]\boldsymbol{C}_d[k]\right)^\top + \boldsymbol{K}[k]\boldsymbol{R}_d[k]\boldsymbol{K}^\top[k]$$

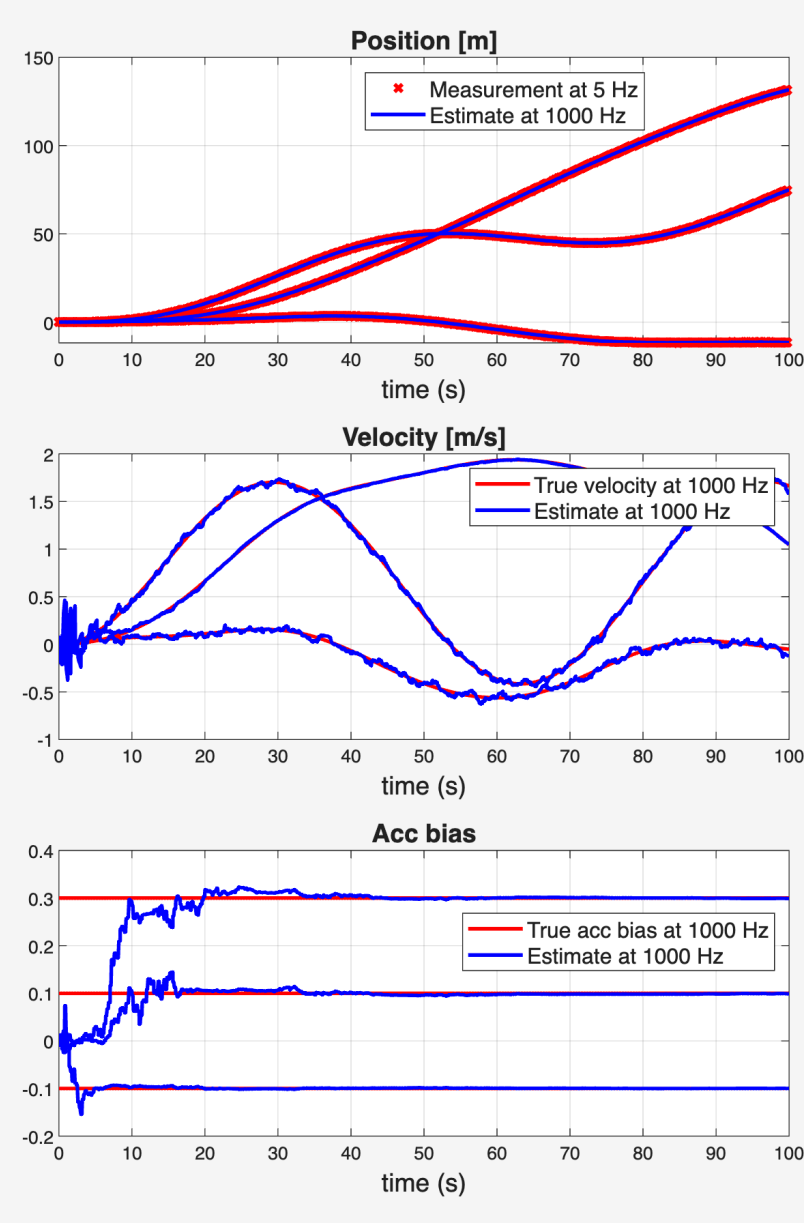
INS propagation:

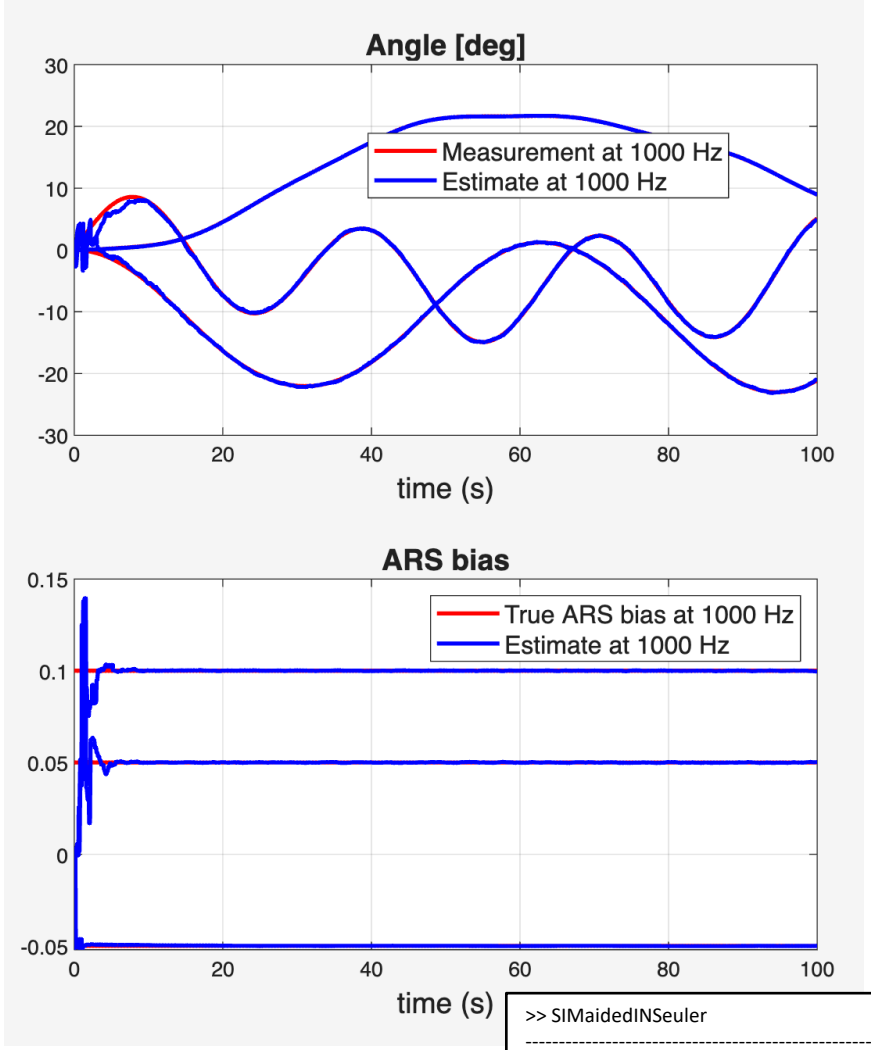
$$\hat{\boldsymbol{p}}^n_{\mathrm{ins}}[k+1] = \hat{\boldsymbol{p}}^n_{\mathrm{ins}}[k] + h\hat{\boldsymbol{v}}^n_{\mathrm{ins}}[k]$$

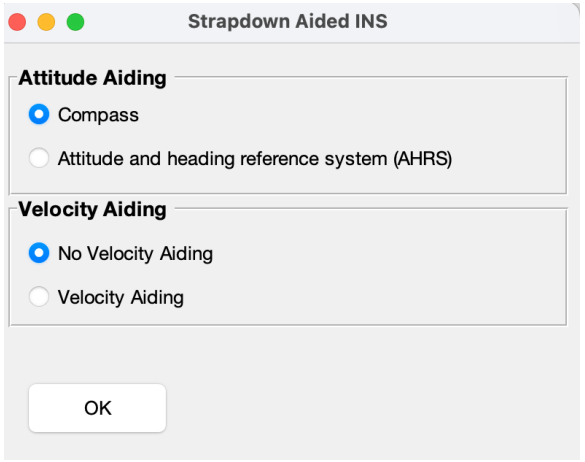
$$\hat{\boldsymbol{v}}^n_{\mathrm{ins}}[k+1] = \hat{\boldsymbol{v}}^n_{\mathrm{ins}}[k] + h\left(\boldsymbol{R}^n_b[k]\left(\boldsymbol{f}^b_{\mathrm{imu}}[k] - \hat{\boldsymbol{b}}^b_{\mathrm{acc,\,ins}}[k]\right) + \boldsymbol{g}^n\right)$$

$$\hat{\boldsymbol{\Theta}}_{\mathrm{ins}}[k+1] = \hat{\boldsymbol{\Theta}}_{\mathrm{ins}}[k] + h\boldsymbol{T}^n_b[k]\left(\boldsymbol{\omega}^b_{\mathrm{imu}}[k] - \hat{\boldsymbol{b}}^b_{\mathrm{ars,\,ins}}[k]\right)$$

MSS Toolbox: SIMaidedINSeuler.m







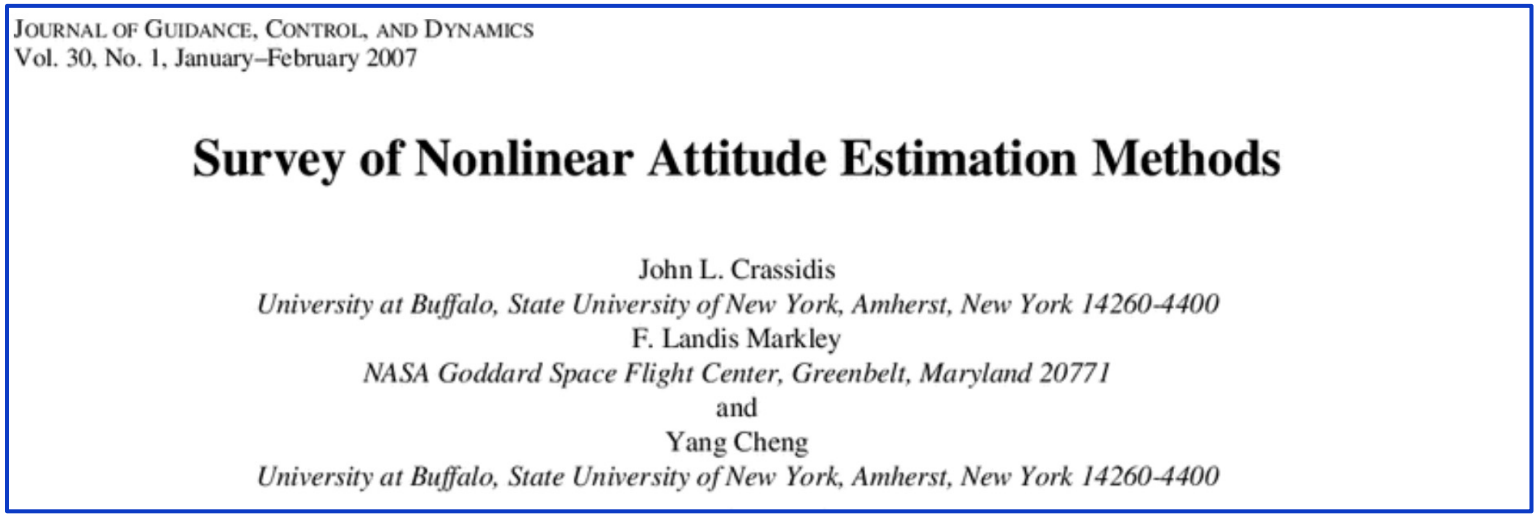

```
>> SIMaidedINSeuler
-------------------------------------------------------------------
MSS toolbox: Error-state (indirect) feedback Kalman filter
Attitude parametrization: Euler angles
INS aided by position at 5 Hz
IMU measurements (specific force and ARS) at 1000 Hz
Compass measurements at 1000 Hz
Magnetic field reference vector for Trondheim (>> type magneticField)
-------------------------------------------------------------------
Simulating...
```

6-DOF ESKF with Attitude Estimation

(Attitude is Estimated using the MEKF)

Multiplicative Extended Kalman Filter (MEKF)

Multiplicative Extended Kalman Filter (MEKF) is an **ESKF** where attitude is parametrized by a four-dimensional unit quaternion. The attitude error, however, is uniquely defined by three parameters, which is the minimal representation for the 3-DOF rotational motion of a rigid body (Crassidis et al. 2007, Markley and Crassidis 2014).


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Survey of Nonlinear Attitude Estimation Methods

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It is well known that is impossible to represent attitude by three parameters globally. Three-parameter representations such as the Euler angles do all have singular points. It is tempting to use the four-parameter unit quaternion in the EKF to avoid singular points. However, the covariance matrix will then be rank deficient because of the unit constraint. Another problem is the EKF injection term. The standard ESKF with additive-error injection cannot be used to create an unbiased estimator with the unit quaternion in the estimated state, as the additive-error injection would violate the quaternion unit constraint as well. To overcome these difficulties the MEKF is used.

Crassidis, J. L., F. L. Markley and Y. Cheng (2007). Survey of Nonlinear Attitude Estimation Methods. Journal of Guidance, Control and Dynamics 30(1), 12-28.
Markley, F. L. and J. L. Crassidis (2014). Fundamentals of Spacecraft Attitude Determination and Control, Volume 33 of Space Technology Library. Springer-Verlag New York,

Multiplicative Extended Kalman Filter (MEKF)

The main idea of the that the unit quaternion error is parametrized using a three-parameter attitude representation, were δη and δε are the real and imaginary components of the error unit quaternion

$$\delta \boldsymbol{q}_b^n = \begin{bmatrix} \delta\eta \\ \delta\boldsymbol{\varepsilon} \end{bmatrix}$$

Scalar-first quaternion

There are many candidates for representation of the unit quaternion error. We will use representation #2.

We multiply by 2 to account for the half-angle representation of rotations in quaternions, ensuring the attitude error corresponds to the full angular error for accurate state updates.

2. Modified Rodrigues parameters

$$\delta \boldsymbol{a}_m = \frac{\delta\boldsymbol{\varepsilon}}{1+\delta\eta}, \qquad \delta \boldsymbol{q}_b^n = \frac{1}{1+\delta\boldsymbol{a}_m^\top \delta\boldsymbol{a}_m} \begin{bmatrix} 1-\delta\boldsymbol{a}_m^\top\delta\boldsymbol{a}_m \\ 2\delta\boldsymbol{a}_m \end{bmatrix}$$

By scaling the modified Rodrigues parameters vector by the factor 4, the covariance estimates will be given in radians squared. This gives

$$\delta\boldsymbol{a} := 4\delta\boldsymbol{a}_m, \qquad \delta \boldsymbol{q}_b^n = \frac{1}{16+\delta\boldsymbol{a}^\top\delta\boldsymbol{a}} \begin{bmatrix} 16-\delta\boldsymbol{a}^\top\delta\boldsymbol{a} \\ 8\delta\boldsymbol{a} \end{bmatrix}$$

1. Rodrigues parameters (Gibbs vector)

$$\delta \boldsymbol{a}_g = \frac{\delta\boldsymbol{\varepsilon}}{\delta\eta}, \qquad \delta \boldsymbol{q}_b^n = \frac{1}{\sqrt{1+\delta\boldsymbol{a}_g^\top\delta\boldsymbol{a}_g}} \begin{bmatrix} 1 \\ \delta\boldsymbol{a}_g \end{bmatrix}$$

It is convenient to scale Gibbs vector by a factor 2 such that the KF covariance estimates will be given in radians squared, which is equivalent to angle errors using a first-order approximation. In other words

$$\delta\boldsymbol{a} := 2\delta\boldsymbol{a}_g, \qquad \delta \boldsymbol{q}_b^n = \frac{1}{\sqrt{4+\delta\boldsymbol{a}^\top\delta\boldsymbol{a}}} \begin{bmatrix} 2 \\ \delta\boldsymbol{a} \end{bmatrix}$$

The Gibbs vector has the property that the elements approach infinity as the rotation approaches 180 degrees. This means that insensible attitude errors of more than 180 degrees cannot exist.

The δ**a** vector replaces the unit quaternion vector $\delta \boldsymbol{q}_b^n$ in error-state Kalman filters.

Multiplicative Extended Kalman Filter (MEKF)

The unit quaternion error $\delta \boldsymbol{q}_b^n$ satisfies (Solà 2016)

$$\boldsymbol{q}_b^n = \hat{\boldsymbol{q}}_{\text{ins}} \otimes \delta \boldsymbol{q}_b^n$$

The nonlinear injection term in the ESKF is represented by the Schur product.

where $\hat{\boldsymbol{q}}_{\text{ins}} \equiv \hat{\boldsymbol{q}}_b^n$. The product of two quaternions (*Hamiltonian product*) is defined as[1]

$$\boldsymbol{q}_1 \otimes \boldsymbol{q}_2 = \begin{bmatrix} \eta_{q_1}\eta_{q_2} - \boldsymbol{\varepsilon}_{q_1}^\top \boldsymbol{\varepsilon}_{q_2} \\ \eta_{q_1}\boldsymbol{\varepsilon}_{q_2} + \eta_{q_2}\boldsymbol{\varepsilon}_{q_1} + \boldsymbol{S}(\boldsymbol{\varepsilon}_{q_1})\boldsymbol{\varepsilon}_{q_2} \end{bmatrix}$$

This is the quaternion (Hamiltonian) product for scalar-first quaternions.

This implies that

$$\begin{aligned} \boldsymbol{R}(\boldsymbol{q}_b^n) &= \boldsymbol{R}(\hat{\boldsymbol{q}}_{\text{ins}} \otimes \delta \boldsymbol{q}_b^n) \\ &= \boldsymbol{R}(\hat{\boldsymbol{q}}_{\text{ins}})\boldsymbol{R}(\delta \boldsymbol{q}_b^n) \\ &\approx \boldsymbol{R}(\hat{\boldsymbol{q}}_{\text{ins}})\,(\boldsymbol{I}_3 + \boldsymbol{S}(\delta \boldsymbol{a})) \end{aligned}$$

Note that for quaternion operations $\boldsymbol{q}_1 \otimes \boldsymbol{q}_2 \neq \boldsymbol{q}_2 \otimes \boldsymbol{q}_1$ but $\boldsymbol{q}_1 + \boldsymbol{q}_2 = \boldsymbol{q}_2 + \boldsymbol{q}_1$.

Rotational Dynamics (Rodriques Parameters)

$$\begin{aligned} \delta \dot{\boldsymbol{a}} &\approx -\boldsymbol{S}(\boldsymbol{\omega}_{\text{ins}}^b)\delta \boldsymbol{a} - \delta \boldsymbol{b}_{\text{ars}}^b - \boldsymbol{w}_{\text{ars}}^b \\ \delta \dot{\boldsymbol{b}}_{\text{ars}}^b &= -\frac{1}{T_{\text{ars}}}\delta \boldsymbol{b}_{\text{ars}}^b + \boldsymbol{w}_{b,\,\text{ars}}^b \end{aligned}$$

The proof for the Rodrigues parameter differential equation is mathematically intricate. However, all equations are derived in Fossen (Ch. 14.4.3, 2021).

$$\boldsymbol{\omega}_{\text{ins}}^b := \boldsymbol{\omega}_{\text{imu}}^b - \hat{\boldsymbol{b}}_{\text{ars}}^b$$

Sola, J. (2016). Quaternion Kinematics for the Error-State Kalman Filter. Technical Report, IRI-TR-16-02, Institut de Robotica i Informitica Industrial, CSIC-UPC.

Multiplicative Extended Kalman Filter (MEKF)

Error Measurement Equation for Reference Vectors

$$
\begin{aligned}
\delta \boldsymbol{y}_i &= (\boldsymbol{v}_i^b + \boldsymbol{\varepsilon}_i) - \boldsymbol{R}^\top(\hat{\boldsymbol{q}}_b^n)\boldsymbol{v}_{0i}^n \\
&= \boldsymbol{R}^\top(\boldsymbol{q}_b^n)\boldsymbol{v}_{0i}^n + \boldsymbol{\varepsilon}_i - \boldsymbol{R}^\top(\hat{\boldsymbol{q}}_b^n)\boldsymbol{v}_{0i}^n \\
&\approx (\boldsymbol{I}_3 - \boldsymbol{S}(\delta \boldsymbol{a}))\, \boldsymbol{R}^\top(\hat{\boldsymbol{q}}_{\text{ins}})\boldsymbol{v}_{0i}^n - \boldsymbol{R}^\top(\hat{\boldsymbol{q}}_{\text{ins}})\boldsymbol{v}_{0i}^n + \boldsymbol{\varepsilon}_i \\
&= -\boldsymbol{S}(\delta \boldsymbol{a})\boldsymbol{R}^\top(\hat{\boldsymbol{q}}_{\text{ins}})\boldsymbol{v}_{0i}^n + \boldsymbol{\varepsilon}_i \\
&= \boldsymbol{S}(\boldsymbol{R}^\top(\hat{\boldsymbol{q}}_{\text{ins}})\boldsymbol{v}_{0i}^n)\delta \boldsymbol{a} + \boldsymbol{\varepsilon}_i
\end{aligned}
$$

Reference Vector #1 – IMU Specific Force Measurements

$$\boldsymbol{v}_{01}^n = [0,\, 0,\, -1]^\top \qquad \boldsymbol{v}_1^b = \frac{\boldsymbol{f}^b}{\left\|\boldsymbol{f}^b\right\|},$$

Reference vector #2 – IMU Magnetic Field Measurements

$$\boldsymbol{v}_{02}^n = \boldsymbol{m}^n \qquad \boldsymbol{v}_2^b = \frac{\boldsymbol{m}^b}{\|\boldsymbol{m}^b\|},$$



The strength and direction of the magnetic field $\boldsymbol{m}^n$ can be assumed to be constant for a given location and period. The numerical values can be found by averaging the magnetometer measurements for a given period while keeping the roll and pitch angles close to zero.

Compass Equation (Yaw angle to Rodrigues Parameters)

Gibbs parameters $\boldsymbol{a}_g = [g_1,\, g_2,\, g_3]^\top$

$$\psi = \tan^{-1}\left(\frac{R_{21}}{R_{11}}\right) = \tan^{-1}\left(\frac{2(g_1 g_2 + g_3)}{1 + g_1^2 - g_2^2 - g_3^2}\right)$$

For Rodrigues (Gibbs) parameters

$$\boldsymbol{a} = 2\boldsymbol{a}_g$$

$$\psi = h(\boldsymbol{a}) = \tan^{-1}\left(\frac{2(a_1 a_2 + 2a_3)}{4 + a_1^2 - a_2^2 - a_3^2}\right)$$

where $\boldsymbol{a} = [a_1,\, a_2,\, a_3]^\top$. Linearization about $\boldsymbol{a} = \hat{\boldsymbol{a}}$ gives

$$\delta y_\psi = \psi - h(\hat{\boldsymbol{a}}) \approx \left.\frac{\mathrm{d}h(\boldsymbol{a})}{\mathrm{d}\boldsymbol{a}}\right|^\top_{\boldsymbol{a}=\hat{\boldsymbol{a}}} \delta\hat{\boldsymbol{a}}$$

$$\boldsymbol{R}(\boldsymbol{q}_b^n) = \begin{bmatrix} 1-2(\varepsilon_2^2+\varepsilon_3^2) & 2(\varepsilon_1\varepsilon_2-\varepsilon_3\eta) & 2(\varepsilon_1\varepsilon_3+\varepsilon_2\eta) \\ 2(\varepsilon_1\varepsilon_2+\varepsilon_3\eta) & 1-2(\varepsilon_1^2+\varepsilon_3^2) & 2(\varepsilon_2\varepsilon_3-\varepsilon_1\eta) \\ 2(\varepsilon_1\varepsilon_3-\varepsilon_2\eta) & 2(\varepsilon_2\varepsilon_3+\varepsilon_1\eta) & 1-2(\varepsilon_1^2+\varepsilon_2^2) \end{bmatrix}$$

$$\psi = \operatorname{atan2}(R_{21}, R_{11})$$

$$\psi = \operatorname{atan2}\left(2(\varepsilon_1\varepsilon_2 + \varepsilon_3\eta),\, 1 - 2(\varepsilon_2^2 + \varepsilon_3^2)\right)$$

Derivation of the linearized compass equation and the formula for c_ψ using Gibbs parameters is mathematically intricate. For details see next slide and Fossen (Ch. 14.4.3, 2021).

$$\delta\boldsymbol{y}_{y_\psi} \approx \boldsymbol{c}_\psi^\top(\hat{\boldsymbol{q}}_{\text{ins}})\delta\boldsymbol{a}$$

$$\boldsymbol{c}_\psi(\hat{\boldsymbol{q}}_{\text{ins}}) := \left.\frac{\mathrm{d}h(\boldsymbol{a})}{\mathrm{d}\boldsymbol{a}}\right|_{\delta\boldsymbol{a}=\delta\hat{\boldsymbol{a}}}$$

$$\hat{\boldsymbol{q}}_{\text{ins}} = \hat{\boldsymbol{q}}_b^n$$

Compass Equation (Yaw angle to Rodrigues Parameters)

Compass Error Measurement

The compass measurement can be included in the error-state filter by parameterizing the rotation matrix in terms of 2 x Gibbs parameters $\boldsymbol{a}_g = [g_1, g_2, g_3]^\top$ Hence, $\boldsymbol{a} = 2\boldsymbol{a}_g$.

$$\psi = \tan^{-1}\left(\frac{R_{21}}{R_{11}}\right) = \tan^{-1}\left(\frac{2(g_1 g_2 + q_3)}{1 + g_1^2 - q_2^2 - q_3^2}\right)$$

$$\psi = h(\boldsymbol{a}) = \tan^{-1}\left(\frac{2(a_1 a_2 + 2a_3)}{4 + a_1^2 - a_2^2 - a_3^2}\right)$$

$$\delta y_\psi = \psi - h(\hat{\boldsymbol{a}}) \approx \left.\frac{\mathrm{d}h(\boldsymbol{a})}{\mathrm{d}\boldsymbol{a}}\right|^\top_{\boldsymbol{a}=\hat{\boldsymbol{a}}} \delta\hat{\boldsymbol{a}}$$

$$\frac{\mathrm{d}h(\boldsymbol{a})}{\mathrm{d}\boldsymbol{a}} = \frac{\partial \tan^{-1}(u)}{\partial u}\frac{\partial u}{\partial \boldsymbol{a}} = \frac{1}{1+u^2}\frac{\partial u}{\partial \boldsymbol{a}} \qquad u := \frac{2(a_1 a_2 + 2a_3)}{4 + a_1^2 - a_2^2 - a_3^2}$$

$$\frac{\partial u}{\partial \boldsymbol{a}} = \left[\frac{-2\left((a_1^2 + a_3^2 - 4)a_2 + a_2^3 + 4a_1 a_3\right)}{(4 + a_1^2 - a_2^2 - a_3^2)^2}, \right.$$

$$\left. \frac{2\left((a_2^2 - a_3^2 + 4)a_1 + a_1^3 + 4a_2 a_3\right)}{(4 + a_1^2 - a_2^2 - a_3^2)^2}, \frac{4\left(a_3^2 + a_1 a_2 a_3 + a_1^2 - a_2^2 + 4\right)}{(4 + a_1^2 - a_2^2 - a_3^2)^2}\right]^\top$$

14.4.3 Feedback Filter Reset Algorithm

Indirect Feedback Kalman Filter

For the feedback version of the indirect KF, the error estimates are used to update the INS estimates directly to avoid drift. This prevents the INS errors to grow. This is done by regulating the error $\delta x[k]$ to zero by state reset such that $x_{\text{ins}}[k] \rightarrow x[k]$ analogues to a feedback control system. Hence, the term feedback filter.

Algorithm 14.1 (Feedback Filter Reset Algorithm) *The feedback loop (Figure 14.8) will remove the INS error by regulating $\delta \boldsymbol{x}$ to zero. This can be obtained by modifying the feedforward filter according to:*

1. *Slow measurements: After every slow measurement (position and velocity), the INS states*

$$\hat{\boldsymbol{x}}_{ins} = \left[(\hat{\boldsymbol{p}}^n_{ins})^\top, (\hat{\boldsymbol{v}}^n_{ins})^\top, (\hat{\boldsymbol{b}}^b_{acc,\,ins})^\top, \hat{\boldsymbol{\Theta}}^\top_{ins}, (\hat{\boldsymbol{b}}^b_{ars,\,ins})^\top \right]^\top$$

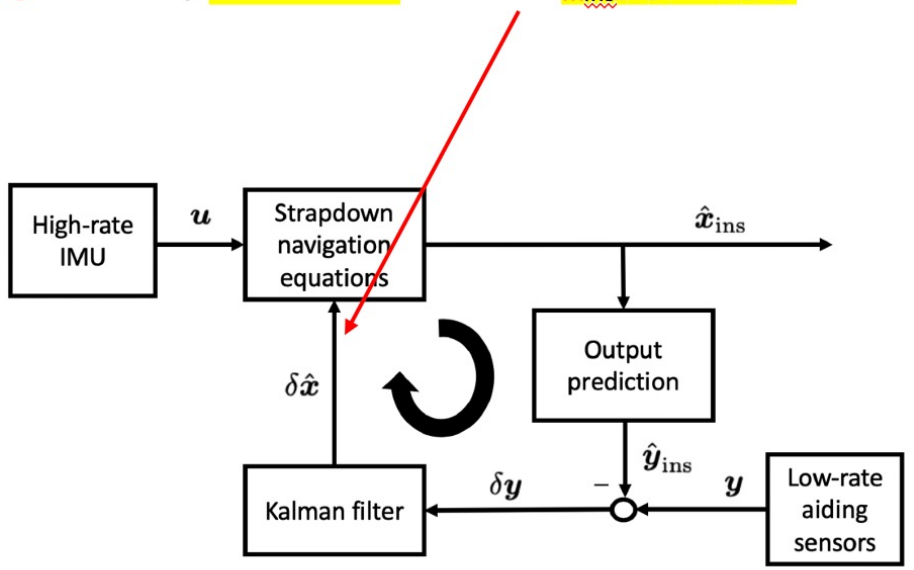
are corrected by setting the error-state vector to zero. This is mathematically equivalent to

$$\hat{\boldsymbol{x}}_{ins}[k] \leftarrow \hat{\boldsymbol{x}}_{ins}[k] + \delta\hat{\boldsymbol{x}}[k]$$

2. *Fast measurements: Step 1 ensures that error-state vector will be zero before the fast measurements (accelerometer and ARS) are applied. Therefore, the state predictor of the feedforward filter* (14.168) *becomes redundant since*

$$\delta\hat{\boldsymbol{x}}^-[k+1] = \boldsymbol{A}_d[k]\boldsymbol{0} \equiv \boldsymbol{0}$$

It should, however, be noted that $\boldsymbol{A}_d[k]$ is still a necessary to compute since $\boldsymbol{A}_d[k]$ is used in the KF covariance predication step.



The **Feedback Filter Reset Algorithm** must be carefully implemented as shown on the next slides.

MSS Toolbox Implementation

Stand-Alone MEKF Attitude Estimator

INS (m-files)

.../MSS/INS/

INS error-state Kalman filter (ESKF) simulators

```
SIMaidedINSeuler     % Simulate the ESKF for aided INS using Euler angles
SIMaidedINSheave     % Simulate the ESKF for aided INS in heave using pressure measurements
SIMaidedINSquat      % Simulate the ESKF for aided INS using unit quaternions (MEKF representation)
SIMquatMEKF          % Simulate the MEKF quaternion-based attitude observer for 9-DOF IMU measurements
SIMquatObserver      % Simulate the nonlinear quaternion-based attitude observer for 9-DOF IMU measurements
```

Functions

```
gravity              % Acceleration of gravity as a function of latitude using the WGS-84 ellipsoid parameters
ins_ahrs             % Error-state Kalman filter (ESKF) for an INS aided by position and AHRS measurements
ins_euler            % Error-state Kalman filter (ESKF) for an INS aided by position and yaw angle measurements
ins_heave            % Error-state Kalman filter (ESKF) for an INS (heave only) aided by pressure measurements
ins_mekf             % Error-state Kalman filter (ESKF) for an INS aided by position and magnetic field measureme
ins_mekf_psi         % Error-state Kalman filter (ESKF) for an INS aided by position and yaw angle measurements
insSignal            % INS signal generator for testing of Kalman filters and observers
magneticField        % NED magnetic field reference vectors 'm_ref' for different cities
quatObserver         % Nonlinear quaternion-based attitude observer for 9-DOF IMU measurements
quatMEKF             % MEKF quaternion-based attitude observer for 9-DOF IMU measurements
```

MSS Stand-Alone MEKF Attitude Estimator Implementation

SIMquatMEKF

Main Loop for simulation of the MEKF attitude estimator quatMEKF.m

```
headingFlag = 2; % 1 for 3-axis magnetometer, 2 for scalar compass

% ==========================================================================
% Simulation parameters
% ==========================================================================
T_final = 60; % Final simulation time (s)
f_fast = 1000; % High-rate IMU measurement frequency (Hz)
f_slow = 100; % Low-rate magnetometer/compass measurement frequency (Hz)
```

MEKF initialization

```
% ------------------------------------------------------------------------
% Initial state estimates
quat_prd = [1 0 0 0]'; % 4x1 initial unit quaternion vector
b_ars_prd = [0 0 0]'; % 3x1 initial ARS bias vector

% MEKF state vector: x = [ax_g ay_g az_g bx_ars by_ars bz_ars]' (Gibbs vector/ARS bias)
Qd = diag([1 1 1 1 1 1]); % 6x6 process noise covariance matrix
P_prd = 1 * eye(6); % % 6x6 initial state covariance matrix

 % Measurement covariance matrix
if headingFlag == 1
    Rd = diag([1 1 1 1 1 1]); % 3x1 gravity vector and 3x1  magnetic field vectors
else
    Rd = diag([1 1 1 1]); % 3x1 gravity vector and 1x1 compass measurement
end
```

```
function [quat, b_ars, P_prd] = quatMEKF(quat, b_ars, P_prd, h, Qd, Rd, m_ref, imu_meas)
% quatMEKF is compatible with MATLAB and GNU Octave (www.octave.org).
% This function computes the updated unit quaternion q[k+1], representing
% the orientation between the BODY and NED frames, as well as the bias
% b_ars[k+1] of the attitude rate sensor (ARS) in a high-performance
% Multiplicative Extended Kalman Filter (MEKF) for quaternion attitude
% estimation (Markley and Crassidis, 2014). The MEKF uses high-rate inertial
% measurements from a 9-DOF inertial measurement unit (IMU). The function
% can be called either as a corrector (with new measurements) or as a
% predictor (without new measurements).
```

MSS Stand-Alone MEKF Attitude Estimator Implementation

```
% Measurements (low rate)
if length(imu_meas) == 6  % No magentic field/compass measurements
    P_hat = P_prd;

else

    if length(imu_meas) == 9        % 9-DOF case: [f_imu' w_imu' m_imu']
        f_imu = imu_meas(1:3)';
        v1  = f_imu / norm(f_imu);
        v01 = [0 0 -1]';

        m_imu = imu_meas(7:9)';
        v2  = m_imu / norm(m_imu);
        v02 = m_ref / norm(m_ref);

        % Measurement matrix
        Cd  = [ Smtrx(R_transposed*v01) O3    % Gravity measurement vector
            Smtrx(R_transposed*v02) O3 ]; % Magentif field measurement vector

        % Innovation vector
        delta_y = [ v1 - R_transposed * v01
            v2 - R_transposed * v02];

 elseif length(imu_meas) == 7     % 7-DOF case: [f_imu' w_imu' psi]

    *** Linearization of compass equation ***

    % Measurement matrix
    Cd = [ Smtrx(R_transposed*v01) O3    % Gravity measurement vector
           c_psi'          zeros(1,3) ]; % Compass measurement

    % Innovation vector
    delta_y = [v1 - R_transposed * v01
               ssa(psi - atan2(u_y, u_x))];

end
```

```
% High-rate IMU measurements: w_imu[k]
w_imu = imu_meas(4:6)';

% Discrete-time ESKF matrices
A = [ -Smtrx(w_imu) -I3
                O3   O3 ];

Ad = expm_taylor(A * h);

Ed = h *[ -I3 O3
          O3 I3  ];
```

$$\delta\dot{\boldsymbol{a}} \approx -\boldsymbol{S}(\boldsymbol{\omega}^b_{\text{ins}})\delta\boldsymbol{a} - \delta\boldsymbol{b}^b_{\text{ars}} - \boldsymbol{w}^b_{\text{ars}}$$

$$\delta\dot{\boldsymbol{b}}^b_{\text{ars}} = -\frac{1}{T_{\text{ars}}}\delta\boldsymbol{b}^b_{\text{ars}} + \boldsymbol{w}^b_{b,\,\text{ars}}$$

Bias is chosen as “random walk” since we do not have dead reckoning (no aiding)

$$\delta\hat{\boldsymbol{x}}[k] = \delta\hat{\boldsymbol{x}}^-[k] + \boldsymbol{K}[k](\delta\boldsymbol{y}[k] - \boldsymbol{C}_d[k]\delta\hat{\boldsymbol{x}}^-[k] - \boldsymbol{D}_d[k]\boldsymbol{u}[k])$$

```
% KF gain: K[k])
K = P_prd * Cd' * invQR(Cd * P_prd * Cd' + Rd);
IKC = eye(size(P_prd)) - K * Cd;

% Corrector: delta_x_hat[k] and P_hat[k]
delta_x_hat = K * delta_y;
P_hat = IKC * P_prd * IKC' + K * Rd * K';
```

$$\delta\hat{\boldsymbol{x}}^-[k+1] = \boldsymbol{A}_d[k]\boldsymbol{0} \equiv \boldsymbol{0}$$

No corrector for state vector, only for covariance matrix.

```
% Form error quaternion from 2 x Gibbs vector: delta_q_hat[k]
delta_a = delta_x_hat(1:3);
delta_q_hat = 1 / sqrt(4 + delta_a' * delta_a) * [2; delta_a];

% ARS bias reset
b_ars = b_ars + delta_x_hat(4:6);
```

$$\hat{\boldsymbol{x}}_{ins}[k] \leftarrow \hat{\boldsymbol{x}}_{ins}[k] + \delta\hat{\boldsymbol{x}}[k]$$

```
% Multiplicative quaternion update (maintains unit norm)
quat = quatprod(quat , delta_q_hat);        % Schur product
quat = quat / norm(quat);                   % Normalization
```

We implement this as a feedback indirect ESKF with reset since the error quaternion behaves similar as the error state in the ESKF.

14.4.3 6-DOF ESKF with Quaternion Attitude Estimation

Strapdown Navigation Equations

$$\dot{\boldsymbol{p}}^n_{nm_\mathrm{I}} = \boldsymbol{v}^n_{nm_\mathrm{I}} \qquad \text{Position vector}$$
$$\dot{\boldsymbol{v}}^n_{nm_\mathrm{I}} = \boldsymbol{a}^n_{nm_\mathrm{I}} \qquad \text{Velocity vector}$$
$$\dot{\boldsymbol{q}}^n_b = \boldsymbol{T}(\boldsymbol{q}^n_b)\boldsymbol{\omega}^b_{nb} \qquad \text{Unit quaternion}$$

IMU Measurements: The inputs to the strapdown navigation equations are the IMU specific force and ARS measurements

$$\boldsymbol{f}^b_{\text{imu}} = \boldsymbol{R}^\top(\boldsymbol{q}^n_b)(\boldsymbol{a}^n_{nm_\mathrm{I}} - \boldsymbol{g}^n) + \boldsymbol{b}^b_{\text{acc}} + \boldsymbol{w}^b_{\text{acc}} \qquad \text{Specific force}$$
$$\dot{\boldsymbol{b}}^b_{\text{acc}} = \boldsymbol{w}^b_{b,\,\text{acc}} \qquad \text{Accelerometer bias}$$
$$\boldsymbol{\omega}^b_{\text{imu}} = \boldsymbol{\omega}^b_{nb} + \boldsymbol{b}^b_{\text{ars}} + \boldsymbol{w}^b_{\text{ars}} \qquad \text{Angular rate}$$
$$\dot{\boldsymbol{b}}^b_{\text{ars}} = \boldsymbol{w}^b_{b,\,\text{ars}} \qquad \text{ARS bias}$$

The nonlinear error state model is derived from these equations.

$$\delta\dot{\boldsymbol{x}} = \boldsymbol{f}(\delta\boldsymbol{x}, \boldsymbol{u}, \boldsymbol{w})$$
$$\delta\boldsymbol{y} = \boldsymbol{h}(\delta\boldsymbol{x}, \boldsymbol{u}) + \boldsymbol{\varepsilon}$$

$$\delta\boldsymbol{x} = [(\delta\boldsymbol{p}^n_{nm_\mathrm{I}})^\top, (\delta\boldsymbol{v}^n_{nm_\mathrm{I}})^\top, (\delta\boldsymbol{b}^b_{\text{acc}})^\top, \delta\boldsymbol{a}^\top, (\delta\boldsymbol{b}^b_{\text{ars}})^\top]^\top$$
$$\boldsymbol{w} = [(\boldsymbol{w}^b_{\text{acc}})^\top, (\boldsymbol{w}^b_{b,\,\text{acc}})^\top, (\boldsymbol{w}^b_{\text{ars}})^\top, (\boldsymbol{w}^b_{b,\,\text{ars}})^\top]^\top$$

INS Navigation Equations (16 States)

$$\dot{\hat{\boldsymbol{p}}}^n_{\text{ins}} = \hat{\boldsymbol{v}}^n_{\text{ins}}$$
$$\dot{\hat{\boldsymbol{v}}}^n_{\text{ins}} = \boldsymbol{R}(\hat{\boldsymbol{q}}_{\text{ins}})\boldsymbol{f}^b_{\text{ins}} + \boldsymbol{g}^n$$
$$\dot{\hat{\boldsymbol{b}}}^b_{\text{ins, acc}} = \boldsymbol{0}$$
$$\dot{\hat{\boldsymbol{q}}}_{\text{ins}} = \boldsymbol{T}(\hat{\boldsymbol{q}}_{\text{ins}})\boldsymbol{\omega}^b_{\text{ins}}$$
$$\dot{\hat{\boldsymbol{b}}}^b_{\text{ins, ars}} = \boldsymbol{0}$$

where the INS inputs are bias compensated IMU measurements

$$\boldsymbol{f}^b_{\text{ins}} := \boldsymbol{f}^b_{\text{imu}} - \Delta\boldsymbol{f}^b$$
$$\boldsymbol{\omega}^b_{\text{ins}} := \boldsymbol{\omega}^b_{\text{imu}} - \Delta\boldsymbol{\omega}^b$$

Feedforward filter: $\Delta\boldsymbol{f}^b = \boldsymbol{0}$, $\Delta\boldsymbol{\omega}^b = \boldsymbol{0}$

Feedback filter: $\Delta\boldsymbol{f}^b = \hat{\boldsymbol{b}}^b_{\text{acc, ins}}$, $\Delta\boldsymbol{\omega}^b = \hat{\boldsymbol{b}}^b_{\text{ars, ins}}$

Reset

14.4.3 6-DOF ESKF with Quaternion Attitude Estimation

Error Measurement Equations

$$\delta \boldsymbol{y} = \begin{bmatrix} \delta \boldsymbol{y}_p & \text{position} \\ \delta \boldsymbol{y}_v & \text{velocity (optionally)} \\ \delta \boldsymbol{v}_1 & \text{gravity reference vector} \\ \delta \boldsymbol{v}_2 & \text{magnetometer reference vector} \\ \delta y_\psi & \text{compass (alternative to } \delta \boldsymbol{v}_2) \end{bmatrix}$$

Gyrocompass and GNSS compass are an alternative to magnetometer reference vector.

Nonlinear Error State-Space Model (15 States)

$$\delta \dot{\boldsymbol{x}} = \boldsymbol{f}(\delta \boldsymbol{x}, \boldsymbol{u}, \boldsymbol{w})$$
$$\delta \boldsymbol{y} = \boldsymbol{h}(\delta \boldsymbol{x}, \boldsymbol{u}) + \boldsymbol{\varepsilon}$$

$$\delta \boldsymbol{x} = [(\delta \boldsymbol{p}^n_{nm_\mathrm{I}})^\top, (\delta \boldsymbol{v}^n_{nm_\mathrm{I}})^\top, (\delta \boldsymbol{b}^b_{\mathrm{acc}})^\top, \delta \boldsymbol{a}^\top, (\delta \boldsymbol{b}^b_{\mathrm{ars}})^\top]^\top$$
$$\boldsymbol{w} = [(\boldsymbol{w}^b_{\mathrm{acc}})^\top, (\boldsymbol{w}^b_{b,\,\mathrm{acc}})^\top, (\boldsymbol{w}^b_{\mathrm{ars}})^\top, (\boldsymbol{w}^b_{b,\,\mathrm{ars}})^\top]^\top$$

Position Error Measurements

$$\begin{aligned} \delta \boldsymbol{y}_p &= (\boldsymbol{p}^n_{nm_\mathrm{I}} + \boldsymbol{\varepsilon}_p) - \hat{\boldsymbol{p}}^n_{nm_\mathrm{I}} \\ &= \delta \boldsymbol{p}^n_{nm_\mathrm{I}} + \boldsymbol{\varepsilon}_p \\ \delta \boldsymbol{y}_v &= (\boldsymbol{v}^n_{nm_\mathrm{I}} + \boldsymbol{\varepsilon}_v) - \hat{\boldsymbol{v}}^n_{nm_\mathrm{I}} \\ &= \delta \boldsymbol{v}^n_{nm_\mathrm{I}} + \boldsymbol{\varepsilon}_v \end{aligned}$$

The NED velocity vector is an optional measurement

Reference Vectors as Error Measurements (e.g. gravity and magnetic field vectors)

$$\begin{aligned} \delta \boldsymbol{y}_i &= (\boldsymbol{v}^b_i + \boldsymbol{\varepsilon}_i) - \boldsymbol{R}^\top(\hat{\boldsymbol{q}}^n_b)\boldsymbol{v}^n_{0i} \\ &= \boldsymbol{R}^\top(\boldsymbol{q}^n_b)\boldsymbol{v}^n_{0i} + \boldsymbol{\varepsilon}_i - \boldsymbol{R}^\top(\hat{\boldsymbol{q}}^n_b)\boldsymbol{v}^n_{0i} \\ &\approx (\boldsymbol{I}_3 - \boldsymbol{S}(\delta \boldsymbol{a}))\,\boldsymbol{R}^\top(\hat{\boldsymbol{q}}_{\mathrm{ins}})\boldsymbol{v}^n_{0i} - \boldsymbol{R}^\top(\hat{\boldsymbol{q}}_{\mathrm{ins}})\boldsymbol{v}^n_{0i} + \boldsymbol{\varepsilon}_i \\ &= -\boldsymbol{S}(\delta \boldsymbol{a})\boldsymbol{R}^\top(\hat{\boldsymbol{q}}_{\mathrm{ins}})\boldsymbol{v}^n_{0i} + \boldsymbol{\varepsilon}_i \\ &= \boldsymbol{S}(\boldsymbol{R}^\top(\hat{\boldsymbol{q}}_{\mathrm{ins}})\boldsymbol{v}^n_{0i})\delta \boldsymbol{a} + \boldsymbol{\varepsilon}_i \end{aligned}$$

$$\delta \dot{\boldsymbol{a}} \approx -\boldsymbol{S}(\boldsymbol{\omega}^b_{\mathrm{ins}})\delta \boldsymbol{a} - \delta \boldsymbol{b}^b_{\mathrm{ars}} - \boldsymbol{w}^b_{\mathrm{ars}}$$
$$\delta \dot{\boldsymbol{b}}^b_{\mathrm{ars}} = -\frac{1}{T_{\mathrm{ars}}}\delta \boldsymbol{b}^b_{\mathrm{ars}} + \boldsymbol{w}^b_{b,\,\mathrm{ars}}$$

$$\boldsymbol{\omega}^b_{\mathrm{ins}} := \boldsymbol{\omega}^b_{\mathrm{imu}} - \hat{\boldsymbol{b}}^b_{\mathrm{ars}}$$

The proof for the Rodrigues parameter differential equation is mathematically intricate. However, all equations are derived in Fossen (Ch. 14.4.3, 2021).

14.4.3 6-DOF ESKF with Quaternion Attitude Estimation

Linearized Error State-Space Model

$$\boldsymbol{A}_d[k] \approx \boldsymbol{I}_{15} + h \left. \frac{\partial \boldsymbol{f}(\delta\boldsymbol{x}[k], \boldsymbol{u}[k], \boldsymbol{w}[k])}{\partial \delta\boldsymbol{x}[k]} \right|_{\delta\boldsymbol{x}[k]=\boldsymbol{0},\, \boldsymbol{w}[k]=\boldsymbol{0}}$$

$$\approx \boldsymbol{I}_{15} + h \begin{bmatrix} \boldsymbol{0}_{3\times3} & \boldsymbol{I}_3 & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} \\ \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & -\boldsymbol{R}(\hat{\boldsymbol{q}}_{\text{ins}}[k]) & -\boldsymbol{R}(\hat{\boldsymbol{q}}_{\text{ins}}[k])\boldsymbol{S}(\boldsymbol{f}^b_{\text{ins}}[k]) & \boldsymbol{0}_{3\times3} \\ \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & -\frac{1}{T_{\text{acc}}}\boldsymbol{I}_3 & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} \\ \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & -\boldsymbol{S}(\boldsymbol{\omega}^b_{\text{ins}}[k]) & -\boldsymbol{I}_3 \\ \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & -\frac{1}{T_{\text{ars}}}\boldsymbol{I}_3 \end{bmatrix}$$

Bias compensated IMU measurements

$$\boldsymbol{f}^b_{\text{ins}} := \boldsymbol{f}^b_{\text{imu}} - \hat{\boldsymbol{b}}^b_{\text{acc ,ins}}$$

$$\boldsymbol{\omega}^b_{\text{ins}} := \boldsymbol{\omega}^b_{\text{imu}} - \hat{\boldsymbol{b}}^b_{\text{ars, ins}}$$

$$\boldsymbol{C}_d[k] \approx \left. \frac{\partial \boldsymbol{h}(\delta\boldsymbol{x}[k], \boldsymbol{u}[k])}{\partial \delta\boldsymbol{x}[k]} \right|_{\delta\boldsymbol{x}[k]=\boldsymbol{0}}$$

$$\approx \begin{bmatrix} \boldsymbol{I}_3 & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} \\ \boldsymbol{0}_{3\times3} & \boldsymbol{I}_3 & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} \\ \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{S}(\boldsymbol{R}^\top(\hat{\boldsymbol{q}}_{\text{ins}}[k])\boldsymbol{v}^n_{01}) & \boldsymbol{0}_{3\times3} \\ \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{S}(\boldsymbol{R}^\top(\hat{\boldsymbol{q}}_{\text{ins}}[k])\boldsymbol{v}^n_{02}) & \boldsymbol{0}_{3\times3} \\ \boldsymbol{0}_{1\times3} & \boldsymbol{0}_{1\times3} & \boldsymbol{0}_{1\times3} & \boldsymbol{c}^\top_\psi(\hat{\boldsymbol{q}}_{\text{ins}}[k]) & \boldsymbol{0}_{1\times3} \end{bmatrix}$$

GNSS position
GNSS velocity
Reference vector #1
Reference vector #2
Compass (linearized using Gibbs vector)

$$\boldsymbol{c}_\psi(\hat{\boldsymbol{q}}_{\text{ins}}) := \left. \frac{\mathrm{d}h(\boldsymbol{a})}{\mathrm{d}\boldsymbol{a}} \right|_{\delta\boldsymbol{a}=\delta\hat{\boldsymbol{a}}}$$

$$\boldsymbol{E}_d[k] \approx h \left. \frac{\partial \boldsymbol{f}(\delta\boldsymbol{x}[k], \boldsymbol{u}[k], \boldsymbol{w}[k])}{\partial \boldsymbol{w}[k]} \right|_{\delta\boldsymbol{x}[k]=\boldsymbol{0},\, \boldsymbol{w}[k]=\boldsymbol{0}}$$

$$\approx h \begin{bmatrix} \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} \\ -\boldsymbol{R}(\hat{\boldsymbol{q}}_{\text{ins}}[k]) & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} \\ \boldsymbol{0}_{3\times3} & \boldsymbol{I}_3 & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} \\ \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & -\boldsymbol{I}_3 & \boldsymbol{0}_{3\times3} \\ \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{I}_3 \end{bmatrix}$$

INS model

$$\begin{aligned} \dot{\hat{\boldsymbol{p}}}^n_{\text{ins}} &= \hat{\boldsymbol{v}}^n_{\text{ins}} \\ \dot{\hat{\boldsymbol{v}}}^n_{\text{ins}} &= \boldsymbol{R}(\hat{\boldsymbol{q}}_{\text{ins}})\boldsymbol{f}^b_{\text{ins}} + \boldsymbol{g}^n \\ \dot{\hat{\boldsymbol{b}}}^b_{\text{ins, acc}} &= \boldsymbol{0} \\ \dot{\hat{\boldsymbol{q}}}_{\text{ins}} &= \boldsymbol{T}(\hat{\boldsymbol{q}}_{\text{ins}})\boldsymbol{\omega}^b_{\text{ins}} \\ \dot{\hat{\boldsymbol{b}}}^b_{\text{ins, ars}} &= \boldsymbol{0} \end{aligned}$$

Nonlinear error-state model

$$\begin{aligned} \delta\dot{\boldsymbol{x}} &= \boldsymbol{f}(\delta\boldsymbol{x}, \boldsymbol{u}, \boldsymbol{w}) \\ \delta\boldsymbol{y} &= \boldsymbol{h}(\delta\boldsymbol{x}, \boldsymbol{u}) + \boldsymbol{\varepsilon} \end{aligned}$$

MSS Toolbox Implementation

6-DOF ESKF with Quaternion Attitude Estimation

INS (m-files)

.../MSS/INS/

INS error-state Kalman filter (ESKF) simulators

```
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```

Functions

```
gravity              % Acceleration of gravity as a function of latitude using the WGS-84 ellipsoid parameters
ins_ahrs             % Error-state Kalman filter (ESKF) for an INS aided by position and AHRS measurements
ins_euler            % Error-state Kalman filter (ESKF) for an INS aided by position and yaw angle measurements
ins_heave            % Error-state Kalman filter (ESKF) for an INS (heave only) aided by pressure measurements
ins_mekf             % Error-state Kalman filter (ESKF) for an INS aided by position and magnetic field measureme
ins_mekf_psi         % Error-state Kalman filter (ESKF) for an INS aided by position and yaw angle measurements
insSignal            % INS signal generator for testing of Kalman filters and observers
magneticField        % NED magnetic field reference vectors 'm_ref' for different cities
quatObserver         % Nonlinear quaternion-based attitude observer for 9-DOF IMU measurements
quatMEKF             % MEKF quaternion-based attitude observer for 9-DOF IMU measurements
```

High-Rate INS Propagation and Reset Algorithm (16 States)

$$
\begin{aligned}
\dot{\hat{\boldsymbol{p}}}^n_{\text{ins}} &= \hat{\boldsymbol{v}}^n_{\text{ins}} \\
\dot{\hat{\boldsymbol{v}}}^n_{\text{ins}} &= \boldsymbol{R}(\hat{\boldsymbol{q}}_{\text{ins}})\boldsymbol{f}^b_{\text{ins}} + \boldsymbol{g}^n \\
\dot{\hat{\boldsymbol{b}}}^b_{\text{ins, acc}} &= \boldsymbol{0} \\
\dot{\hat{\boldsymbol{q}}}_{\text{ins}} &= \boldsymbol{T}(\hat{\boldsymbol{q}}_{\text{ins}})\boldsymbol{\omega}^b_{\text{ins}} \\
\dot{\hat{\boldsymbol{b}}}^b_{\text{ins, ars}} &= \boldsymbol{0}
\end{aligned}
$$

In an INS, the navigation computer typically runs a synchronized loop at 1000–2000 Hz, to handle the high-rate IMU measurements. Accurate numerical integration is crucial to properly compensate for bias drift and to minimize errors caused by sculling and coning effects.

$$\delta\hat{\boldsymbol{x}}^-[k+1] = \boldsymbol{A}_d[k]\boldsymbol{0} \equiv \boldsymbol{0}$$

No corrector for state vector, only for covariance matrix.

$$\delta\hat{\boldsymbol{x}}[k] = \delta\hat{\boldsymbol{x}}^-[k] + \boldsymbol{K}[k](\delta\boldsymbol{y}[k] - \boldsymbol{C}_d[k]\delta\hat{\boldsymbol{x}}^-[k] - \boldsymbol{D}_d[k]\boldsymbol{u}[k])$$

$$\hat{\boldsymbol{x}}_{ins}[k] \leftarrow \hat{\boldsymbol{x}}_{ins}[k] + \delta\hat{\boldsymbol{x}}[k]$$

High-Rate INS Propagation

```
% INS propagation: p_ins[k] and v_ins[k]
a_ins = R * f_ins + g_n;                    % Linear acceleration
p_ins = p_ins + h * v_ins + h^2/2 * a_ins;  % Exact discretization
v_ins = v_ins + h * a_ins;                  % Exact discretization

% q_ins[k+1] is computed using the matrix exponential, which serves as the
% exponential map for matrix Lie groups, ensuring an exact discretization
% of the quaternion differential equation:
%    q_ins_dot = Tquat(w_ins) * q_ins
% You can replace the build-in Matlab function expm.m with the custom-made
% MSS function expm_squaresPade.m for this computation.
q_ins = expm( Tquat(w_ins) * h ) * q_ins;   % Exact discretization
q_ins = q_ins / norm(q_ins);                % Normalization

% INS state vector: x_ins[k+1]
x_ins = [p_ins; v_ins; b_acc_ins; q_ins; b_ars_ins];
```

Low-Rate Corrector with Reset Algorithm

```
% Corrector: delta_x_hat[k] and P_hat[k]
delta_x_hat = K * eps;
P_hat = IKC * P_prd * IKC' + K * Rd * K';

% Error quaternion (2 x Gibbs vector): delta_q_hat[k]
delta_a = delta_x_hat(10:12);
delta_q_hat = 1 / sqrt(4 + delta_a' * delta_a) * [2 delta_a']';

% INS reset: x_ins[k]
p_ins = p_ins + delta_x_hat(1:3);           % Reset position
v_ins = v_ins + delta_x_hat(4:6);           % Reset velocity
b_acc_ins = b_acc_ins + delta_x_hat(7:9);   % Reset ACC bias
b_ars_ins = b_ars_ins + delta_x_hat(13:15); % Reset ARS bias

q_ins = quatprod(q_ins, delta_q_hat);       % Schur product
q_ins = q_ins / norm(q_ins);                % Normalization
```

Low-Rate ESKF Propagation (15 States)

```
% KF gain: K[k]
K = P_prd * Cd' * invQR(Cd * P_prd * Cd' + Rd);
IKC = eye(15) - K * Cd;

% Estimation errors (injection terms)
eps_pos = y_pos - p_ins;
eps_g   = v1 - R' * v01;

if (nargin == 9)
    % No velocity measurements
    eps = [eps_pos; eps_g];
else
    % Velocity measurements
    eps_vel = y_vel - v_ins;
    eps = [eps_pos; eps_vel; eps_g];
end

if length(imu_meas) == 9
    % Magnetic field IMU measurements
    eps_mag = v2 - R' * v02;
    eps = [eps; eps_mag];
end

% Corrector: delta_x_hat[k] and P_hat[k]
delta_x_hat = K * eps;
P_hat = IKC * P_prd * IKC' + K * Rd * K';

% Error quaternion (2 x Gibbs vector): delta_q_hat[k]
delta_a = delta_x_hat(10:12);
delta_q_hat = 1 / sqrt(4 + delta_a' * delta_a) * [2 delta_a']';
```

```
Ed = h *[  03 03     03 03
          -R  03     03 03
           03 I3     03 03
           03 03    -I3 03
           03 03     03 I3  ];
```

```
% Bias compensated IMU measurements
f_ins = f_imu - b_acc_ins;
w_ins = w_imu - b_ars_ins;

v01 = [0 0 -1]'; % Normalized NED reference vector (measuring -g at rest)
v1 = f_imu / norm(f_imu); % Normalized specific force measurement

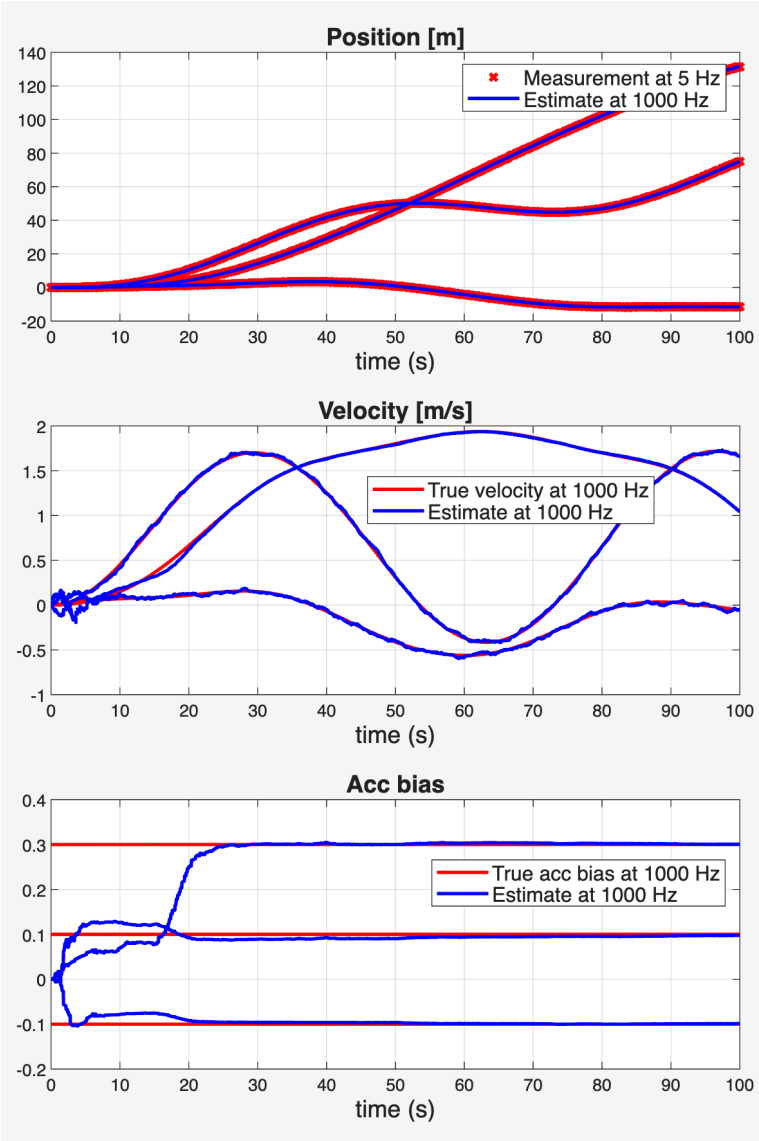
% Discrte-time ESKF matrices
A = [ 03 I3  03            03               03
      03 03 -R            -R*Smtrx(f_ins)   03
      03 03 -(1/T_acc)*I3  03               03
      03 03  03           -Smtrx(w_ins)     -I3
      03 03  03            03               -(1/T_ars)*I3 ];

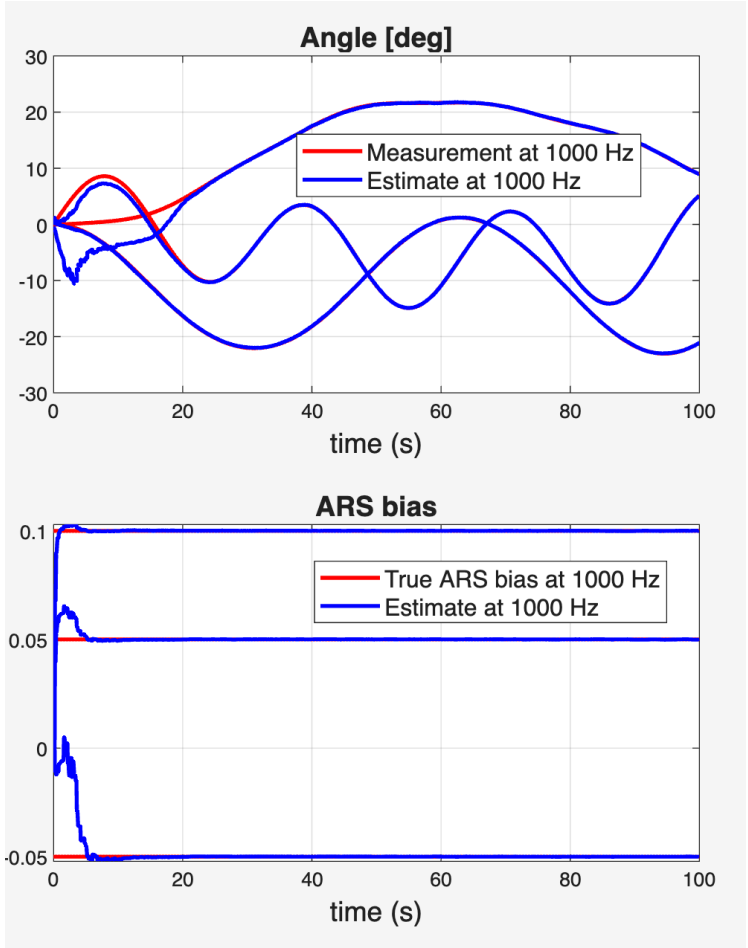
% Ad = eye(15) + h * A + 0.5 * (h * A)^2 + ...
Ad = expm_taylor(A * h);
```

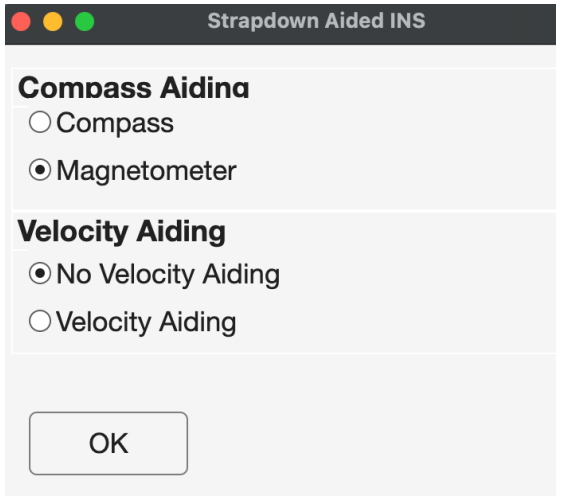
```
if (nargin == 9)
    % No velocity measurements
    Cd = [ I3 03 03 03 03                % NED positions
           03 03 03 Smtrx(R'*v01) 03];   % Gravity
else
    % Velocity measurements
    Cd = [ I3 03 03 03 03                % NED positions
           03 I3 03 03 03                % NED velocities
           03 03 03 Smtrx(R'*v01) 03 ]; % Gravity
end
```

```
% Magnetic field IMU measurements: m_imu[k]
if length(imu_meas) == 9
    m_imu = imu_meas(7:9)';
    v2 =  m_imu / norm(m_imu); % BODY-fixed magnetic field measurement
    v02 = m_ref / norm(m_ref); % NED magnetic field reference vector
    Cd = [ Cd
           03 03 03 Smtrx(R'*v02) 03 ]; % Augment the compass measurement to Cd
end
```

MSS Toolbox: SIMaidedINSquat.m








```
>> SIMaidedINSquat
-------------------------------------------------------------------
MSS toolbox: Error-state (indirect) feedback Kalman filter
Unit quaternion attitude parametrization (MEKF): 2 x Gibbs vector
INS aided by position at 5 Hz
IMU inertial measurements (specific force and ARS) at 1000 Hz
IMU magnetometer measurements at 100 Hz
Magnetic field reference vector for Trondheim (>> type magneticField)
-------------------------------------------------------------------
Simulating...
```

6-DOF ESKF with Quaternion Attitude Estimation

The pseudocode for the 15-state indirect feedback KF where attitude is parametrized using unit quaternions/Gibbs vector is given below. The unit quaternion error-state representation is like the MEKF for quaternion attitude estimation. The stand-alone Matlab/GNU Octave functions for the INS are included in the MSS toolbox as:

IMU magnetometer:

```
>> [x_ins, P_prd] = ins_mekf(x_ins, P_prd, mu, h, Qd, Rd, imu_meas, m_ref, y_pos, y_vel)

imu_meas = [f_imu', w_imu', m_imu']
```

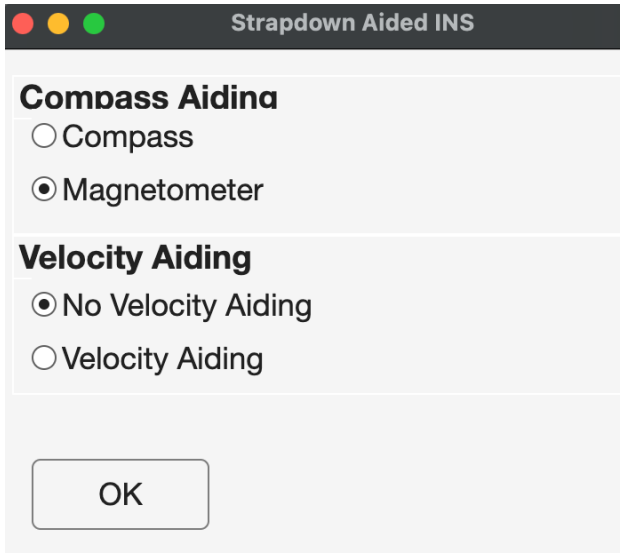
is 1x9 vector with 9-DOF IMU measurements [fx, fy, fz, wx, wy, wz, mx, my, mz].

Compass measurement:

```
>> [x_ins, P_prd] = ins_mekf_psi, x_ins, P_prd, mu, h, Qd, Rd, f_imu, w_imu, y_psi, y_pos, y_vel)
```

Both the magnetometer and the compass implementations can be tested by specifying the configuration parameters in the main loop test script SIMaidedINSquat.m.

```
>> SIMaidedINSquat
-------------------------------------------------------------------
MSS toolbox: Error-state (indirect) feedback Kalman filter
Unit quaternion attitude parametrization (MEKF): 2 x Gibbs vector
INS aided by position at 5 Hz
IMU inertial measurements (specific force and ARS) at 1000 Hz
IMU magnetometer measurements at 100 Hz
Magnetic field reference vector for Trondheim (>> type magneticField)
-------------------------------------------------------------------
Simulating...
```



14.4.3 6-DOF ESKF with Quaternion Attitude Estimation

Concluding remarks:

- The function ins_mekf.m is the implementation of a 9-DOF high-rate IMU with 3-axis magnetometer. This is the standard sensor suite for small vehicles.
- Another possibility is to use compass and/or GNSS heading measurements replacing the magnetometer measurements to improve reliability and resilience in autonomous systems. The MSS toolbox implementation for compass aiding is ins_mekf_psi.m.
- Both the magnetometer and the compass implementations can be tested in main loop by specifying the configuration parameters in the test script SIMaidedINSquat.m.
- It is also possible to test GNSS and DVL velocity aiding by changing the configuration parameters in the script.
- The experiments confirm that both the acceleration and ARS biases converge to their true values. However, there is no observability proof for the 15-state nonlinear case. Hence, care should be taken. From a practical point of view, we know that the acceleration bias will not converge to its true value unless the vehicle is sufficient excited in roll and pitch.
- Computer simulations show that the position, velocity and attitude states converge to their true values.
- Finally, for human interpretation of the results, the attitude plot is obtained by transforming the estimated unit quaternions to Euler angles using q2euler.m.

Chapter Goals — Revisited

Inertial measurement unit (IMU):

- Understand the primary functionality of a MEMS-based IMU. Explain specific force, gyro and ARS.
- Distinguish between 6, 9 and 10 DOFs IMUs
- Be able to write down the IMU measurement equations including the model for white-noise driven biases (drift) terms.
- Be able to transform IMU measurements to different BODY origins using lever arms

Strapdown inertial navigation system (INS):

- Understand the principles for strapdown and gimballed-mounted INS.
- Understand what we mean with aided inertial navigation and why it is needed.
- Understand the static relationship between specific force and roll-pitch angles.
- Understand how an AHRS works, from a mathematical point of view.
- Be able to design attitude estimators using EKF and nonlinear observers (reference vectors).
- Be able to design direct extended Kalman filters (EKFs) for strapdown INS using Euler angles.
- Be able to design indirect (error-state) Kalman filters (ESKFs) for strapdown INS.
  Both the Euler angle and the unit quaternion representation (MEKF).
- Understand the differences of bias models in direct and indirect Kalman filters.
- Understand the problems/limitations of ARS and specific force bias estimation, and how observability is related to vehicle maneuverability.